SIEMENS

SINUMERIK

Industrial Edge for Machine Tools Protect MyMachine /3D Twin Create MyVirtual Machine /3D 3D machine model

Configuration Manual

Valid for
Protect MyMachine /3D Twin V1.3.2 or higher
Create MyVirtual Machine /3D V1.3.2 or higher

09/2023
A5E50684179B AJ

Legal information

Warning notice system

This manual contains notices you have to observe in order to ensure your personal safety, as well as to prevent damage to property. The notices referring to your personal safety are highlighted in the manual by a safety alert symbol, notices referring only to property damage have no safety alert symbol. These notices shown below are graded according to the degree of danger.

DANGER
indicates that death or severe personal injury **will** result if proper precautions are not taken.

WARNING
indicates that death or severe personal injury **may** result if proper precautions are not taken.

CAUTION
indicates that minor personal injury can result if proper precautions are not taken.

NOTICE
indicates that property damage can result if proper precautions are not taken.

If more than one degree of danger is present, the warning notice representing the highest degree of danger will be used. A notice warning of injury to persons with a safety alert symbol may also include a warning relating to property damage.

Qualified Personnel

The product/system described in this documentation may be operated only by **personnel qualified** for the specific task in accordance with the relevant documentation, in particular its warning notices and safety instructions. Qualified personnel are those who, based on their training and experience, are capable of identifying risks and avoiding potential hazards when working with these products/systems.

Proper use of Siemens products

Note the following:

WARNING
Siemens products may only be used for the applications described in the catalog and in the relevant technical documentation. If products and components from other manufacturers are used, these must be recommended or approved by Siemens. Proper transport, storage, installation, assembly, commissioning, operation and maintenance are required to ensure that the products operate safely and without any problems. The permissible ambient conditions must be complied with. The information in the relevant documentation must be observed.

Trademarks

All names identified by ® are registered trademarks of Siemens AG. The remaining trademarks in this publication may be trademarks whose use by third parties for their own purposes could violate the rights of the owner.

Disclaimer of Liability

We have reviewed the contents of this publication to ensure consistency with the hardware and software described. Since variance cannot be precluded entirely, we cannot guarantee full consistency. However, the information in this publication is reviewed regularly and any necessary corrections are included in subsequent editions.

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A5E50684179B AJ
Ⓟ 09/2023 Subject to change

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Introduction 1

1.1 About SINUMERIK

From simple, standardized CNC machines to premium modular machine designs – the SINUMERIK CNCs offer the right solution for all machine concepts. Whether for individual parts or mass production, simple or complex workpieces – SINUMERIK is the highly dynamic automation solution, integrated for all areas of production. From prototype construction and tool design to mold making, all the way to large-scale series production.

Visit our website for more information SINUMERIK (https://www.siemens.com/sinumerik).

1.2 About this documentation

Target group

This document is intended for commissioning engineers and machine tool manufacturers. The documentation provides a detailed description of creating a 3D model for Protect MyMachine /3D Twin or Create MyVirtual Machine /3D software.

Benefits

The software installation manual instructs the target group on how to create a machine model using the example of a milling machine.

Standard scope

This documentation only describes the functionality of the standard version. This may differ from the scope of the functionality of the system that is actually supplied. Please refer to the ordering documentation only for the functionality of the supplied drive system.

It may be possible to execute other functions in the system which are not described in this documentation. This does not, however, represent an obligation to supply such functions with a new control or when servicing.

For reasons of clarity, this documentation cannot include all of the detailed information on all product types. Further, this documentation cannot take into consideration every conceivable type of installation, operation and service/maintenance.

The machine manufacturer must document any additions or modifications they make to the product themselves.

Websites of third-party companies

This document may contain hyperlinks to third-party websites. Siemens is not responsible for and shall not be liable for these websites and their content. Siemens has no control over the information which appears on these websites and is not responsible for the content and information provided there. The user bears the risk for their use.

1.3 Documentation on the internet

1.3.1 Documentation overview SINUMERIK ONE

Comprehensive documentation about the functions provided in SINUMERIK ONE Version 6.13 and higher is provided in the Documentation overview SINUMERIK ONE (https://support.industry.siemens.com/cs/ww/en/view/109768483).

You can display documents or download them in PDF and HTML5 format.

The documentation is divided into the following categories:

- User: Operating
- User: Programming
- Manufacturer/Service: Functions
- Manufacturer/Service: Hardware
- Manufacturer/Service: Configuration/Setup
- Manufacturer/Service: Safety Integrated
- Information and training
- Manufacturer/Service: SINAMICS

1.3.2 Documentation overview SINUMERIK 840D sl

Comprehensive documentation about the functions provided in SINUMERIK 840D sl Version 4.8 SP4 and higher is provided in the Documentation overview SINUMERIK 840D sl (https://support.industry.siemens.com/cs/ww/en/view/109766213).

You can display the documents or download them in PDF and HTML5 format.

The documentation is divided into the following categories:

- User: Operating
- User: Programming
- Manufacturer/Service: Functions
- Manufacturer/Service: Hardware
- Manufacturer/Service: Configuration/Setup
- Manufacturer/Service: Safety Integrated
- Manufacturer/Service: SINUMERIK Integrate/MindApp
- Information and training
- Manufacturer/Service: SINAMICS

1.3.3 Documentation overview SINUMERIK operator components

Comprehensive documentation about the SINUMERIK operator components is provided in the Documentation overview SINUMERIK operator components (https://support.industry.siemens.com/cs/document/109783841/technische-dokumentation-zu-sinumerik-bedienkomponenten?dti=0&lc=en-WW).

You can display documents or download them in PDF and HTML5 format.

The documentation is divided into the following categories:

- Operator Panels
- Machine control panels
- Machine Pushbutton Panel

- Handheld Unit / Mini handheld devices
- Further operator components

An overview of the most important documents, entries and links to SINUMERIK is provided at SINUMERIK Overview - Topic Page (https://support.industry.siemens.com/cs/document/109766201/sinumerik-an-overview-of-the-most-important-documents-and-links?dti=0&lc=en-WW).

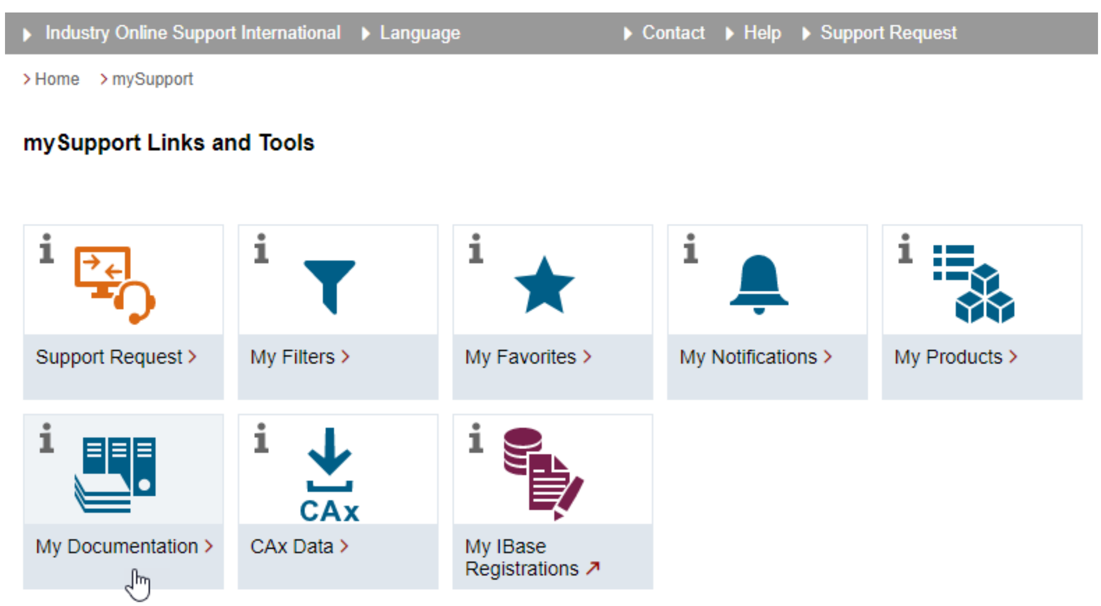
1.4 Feedback on the technical documentation

If you have any questions, suggestions or corrections regarding the technical documentation which is published in the Siemens Industry Online Support, use the link "Provide feedback" which appears at the end of the entry.

1.5 mySupport documentation

With the "mySupport documentation" web-based system you can compile your own individual documentation based on Siemens content, and adapt it for your own machine documentation.

To start the application, click on the "My Documentation" tile on the mySupport homepage (https://support.industry.siemens.com/cs/ww/en/my):



The configured manual can be exported in RTF, PDF or XML format.

Note

Siemens content that supports the mySupport documentation application can be identified by the presence of the "Configure" link.

1.6 Service and Support

Product support

You can find more information about products on the internet:

Product support (https://support.industry.siemens.com/cs/ww/en/)

The following is provided at this address:

- Up-to-date product information (product announcements)
- FAQs
- Manuals
- Downloads
- Newsletters with the latest information about your products
- Global forum for information and best practice sharing between users and specialists
- Local contact persons via our Contacts at Siemens database (→ "Contact")
- Information about field services, repairs, spare parts, and much more (→ "Field Service")

Technical support

Country-specific telephone numbers for technical support are provided on the internet at address (https://support.industry.siemens.com/cs/ww/en/sc/4868) in the "Contact" area.

If you have any technical questions, please use the online form in the "Support Request" area.

Training

You can find information on SITRAIN at the following address (https://www.siemens.com/sitrain).
SITRAIN offers training courses for automation and drives products, systems and solutions from Siemens.

Siemens support on the go

With the award-winning "Siemens Industry Online Support" app, you can access more than 300,000 documents for Siemens Industry products – any time and from anywhere. The app can support you in areas including:

- Resolving problems when implementing a project
- Troubleshooting when faults develop
- Expanding a system or planning a new system

Furthermore, you have access to the Technical Forum and other articles from our experts:

- FAQs
- Application examples
- Manuals
- Certificates
- Product announcements and much more

The "Siemens Industry Online Support" app is available for Apple iOS and Android.

Data matrix code on the nameplate

The data matrix code on the nameplate contains the specific device data. This code can be read with a smartphone and technical information about the device displayed via the "Industry Online Support" mobile app.

1.7 Important product information

Using OpenSSL

This product can contain the following software:

- Software developed by the OpenSSL project for use in the OpenSSL toolkit
- Cryptographic software created by Eric Young.
- Software developed by Eric Young

You can find more information on the internet:

- OpenSSL (https://www.openssl.org)
- Cryptsoft (https://www.cryptsoft.com)

Compliance with the General Data Protection Regulation

Siemens observes standard data protection principles, in particular the data minimization rules (privacy by design).

For this product, this means:

The product does not process or store any personal data, only technical function data (e.g. time stamps). If the user links this data with other data (e.g. shift plans) or if he/she stores person-related data on the same data medium (e.g. hard disk), thus personalizing this data, he/she must ensure compliance with the applicable data protection stipulations.

Safety notes 2

2.1 Fundamental safety instructions

2.1.1 General safety instructions

⚠ WARNING

Danger to life if the safety instructions and residual risks are not observed

If the safety instructions and residual risks in the associated hardware documentation are not observed, accidents involving severe injuries or death can occur.

- Observe the safety instructions given in the hardware documentation.
- Consider the residual risks for the risk evaluation.

⚠ WARNING

Malfunctions of the machine as a result of incorrect or changed parameter settings

As a result of incorrect or changed parameterization, machines can malfunction, which in turn can lead to injuries or death.

- Protect the parameterization against unauthorized access.
- Handle possible malfunctions by taking suitable measures, e.g. emergency stop or emergency off.

2.1.2 Warranty and liability for application examples

Application examples are not binding and do not claim to be complete regarding configuration, equipment or any eventuality which may arise. Application examples do not represent specific customer solutions, but are only intended to provide support for typical tasks.

As the user you yourself are responsible for ensuring that the products described are operated correctly. Application examples do not relieve you of your responsibility for safe handling when using, installing, operating and maintaining the equipment.

2.1.3 Security information

Siemens provides products and solutions with industrial security functions that support the secure operation of plants, systems, machines and networks.

In order to protect plants, systems, machines and networks against cyber threats, it is necessary to implement – and continuously maintain – a holistic, state-of-the-art industrial security concept. Siemens' products and solutions constitute one element of such a concept.

Customers are responsible for preventing unauthorized access to their plants, systems, machines and networks. Such systems, machines and components should only be connected to an enterprise network or the internet if and to the extent such a connection is necessary and only when appropriate security measures (e.g. firewalls and/or network segmentation) are in place.

For additional information on industrial security measures that may be implemented, please visit
https://www.siemens.com/industrialsecurity (https://www.siemens.com/industrialsecurity).

Siemens' products and solutions undergo continuous development to make them more secure. Siemens strongly recommends that product updates are applied as soon as they are available and that the latest product versions are used. Use of product versions that are no longer supported, and failure to apply the latest updates may increase customer's exposure to cyber threats.

To stay informed about product updates, subscribe to the Siemens Industrial Security RSS Feed under
https://www.siemens.com/industrialsecurity
(https://new.siemens.com/global/en/products/services/cert.html#Subscriptions).

Further information is provided on the Internet:

Industrial Security Configuration Manual
(https://support.industry.siemens.com/cs/ww/en/view/108862708)

WARNING

Unsafe operating states resulting from software manipulation

Software manipulations, e.g. viruses, Trojans, or worms, can cause unsafe operating states in your system that may lead to death, serious injury, and property damage.

- Keep the software up to date.
- Incorporate the automation and drive components into a holistic, state-of-the-art industrial security concept for the installation or machine.
- Make sure that you include all installed products into the holistic industrial security concept.
- Protect files stored on exchangeable storage media from malicious software by with suitable protection measures, e.g. virus scanners.
- On completion of commissioning, check all security-related settings.

2.2 Security Disclaimer

Transportation of exported data files have to be secured by technical means like encrypted/signed emails, encrypted/signed USB-Sticks, etc., especially in the public/Internet.

Exported data files have to be stored access restricted within the OEM/end customer area (e.g. access restriction to share points, databases, etc.) by user management with e.g. credentials.

The secure communication between SINUMERIK and the Edge Box is the customer's responsibility. Possibilities for maintaining a secure connection include:

- Point-to-point connection between SINUMERIK and Edge Box. Use of a short communication line and device placement within the same cabinet as SINUMERIK.
- Protection of logical and physical access points of the SINUMERIK system.

Overview 3

3.1 Create 3D machine model

Module Description:

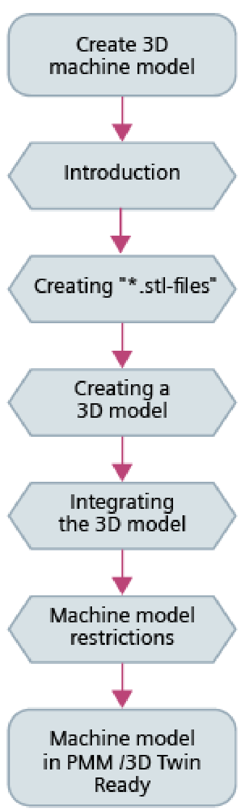
A 3D machine model for Protect MyMachine /3D Twin or Create MyVirtual Machine /3D based on the SinuMill 500 training machine should be created in this documentation.

Module objective:

Create a 3D machine model using the appropriate tool.

Content:

- Create STL files from NX
- Build a 3D model in "MachineBuilder"
- Integrate the 3D model in "Protect MyMachine /3D Twin" or "Create MyVirtual Machine /3D"



3.2 Introduction

The 3D machine is created with the following programs:

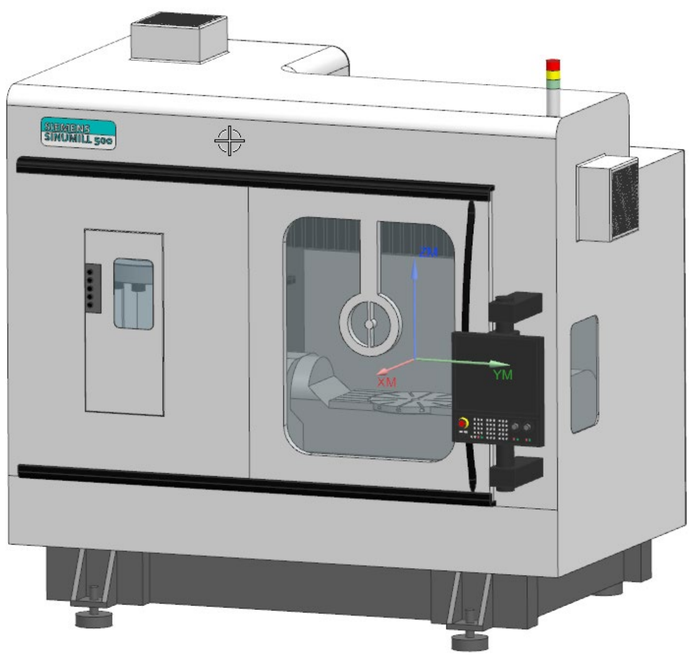
NX

Protect MyMachine /3D Twin or Create MyVirtual Machine /3D

Machine Builder

The 3D model is based on the SinuMill 500 with AC kinematics.

This is also the basis of the CAD/CAM and the 5-axis training documentation from Siemens.



3.3 Constraints for the machine model

The 3D machine is created with the following programs:

- The "lookaheadJog" field of the "protection_area" node should be set to "hidden" in the Machine Builder.
- Other geometry nodes below the milling spindle, except for the tool, are not allowed.
- The machine model for 3D simulation and PMM /3D Twin must be created in metric units (mm).

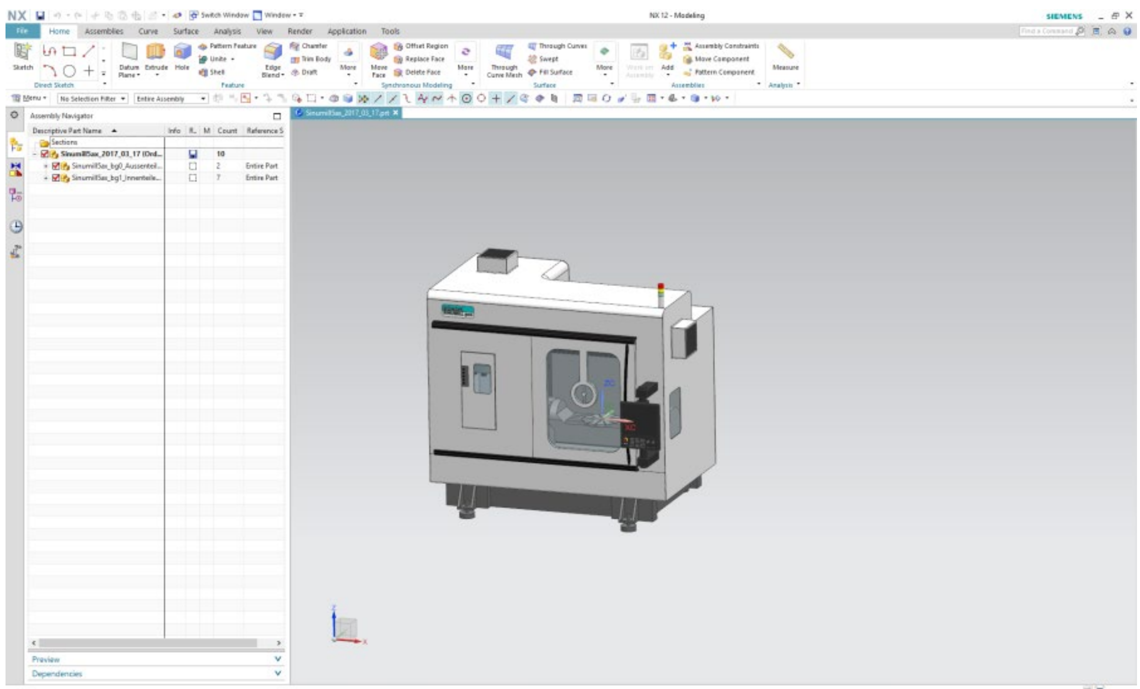
Creating "*.stl-files"

4

4.1 Creating STL-Files

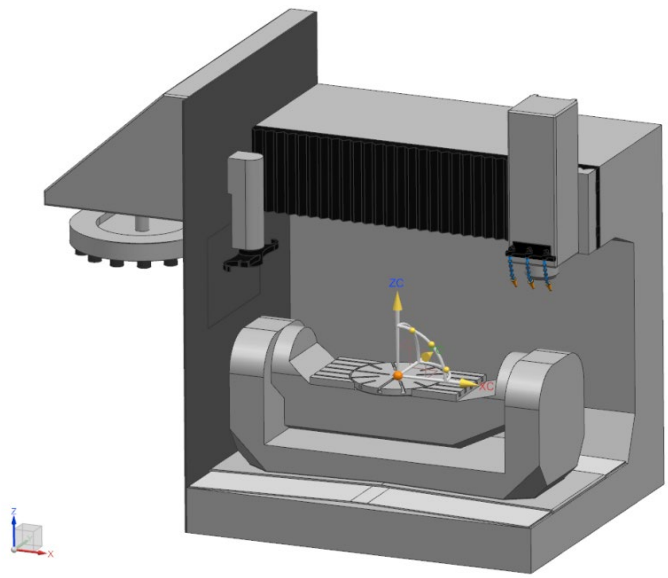
STL (Standard Triangulation Language) files are a standard interface for many CAD systems. These provide geometric information for three-dimensional data models. (Source: Wikipedia).

These files are generated in the NX CAD system and apply to machine components / elements required later for the 3D machine model.

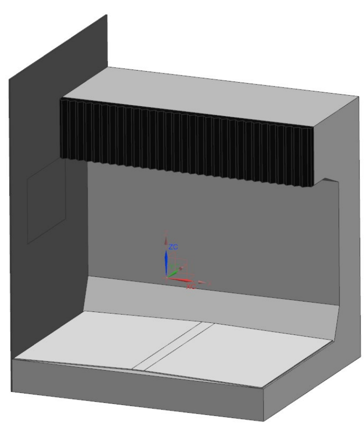


The position of the exported "STL files" refers to the zero point in the "CAD system". The zero point of this machine should lie in the center of the machine table.

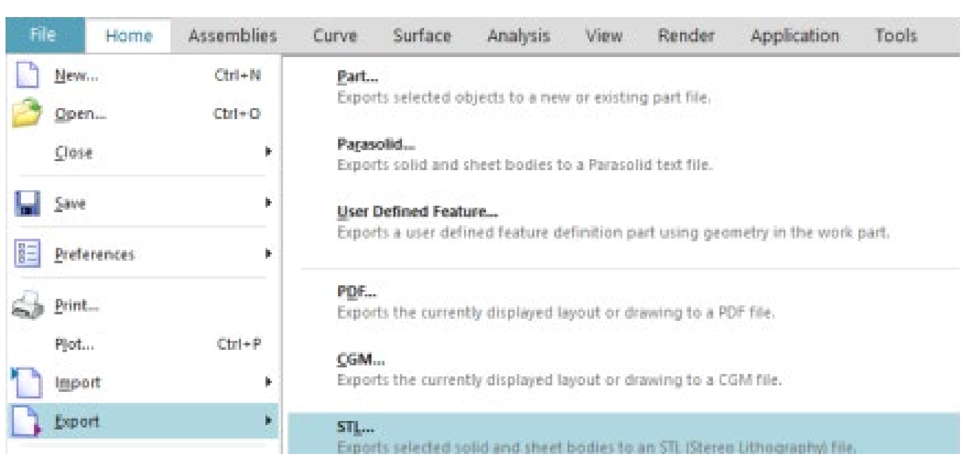
It is recommended that the montage can be made on the "CAD System" directly to zero point. So it will be easier to define offset values.

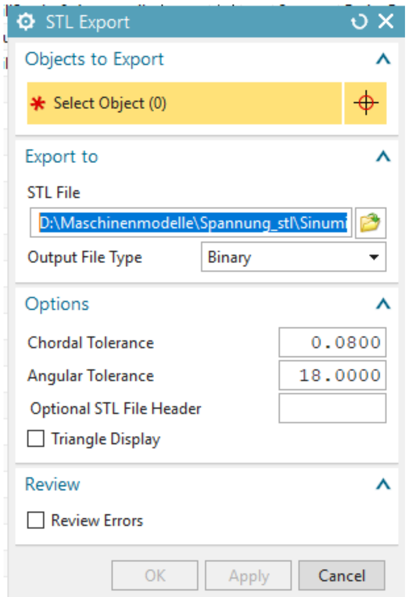


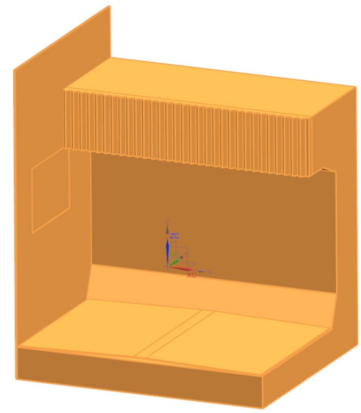
In the first step, the machine bed should be exported as "STL file". For this purpose, all superfluous components are hidden.



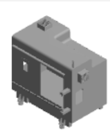
A window opens after selecting "File" "Export" "STL...".





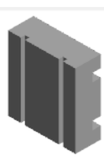


The elements of the machine model (SinuMill 500) should be exported with the following names:

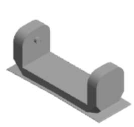


→ housing.stl

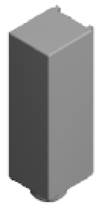
→ door.stl



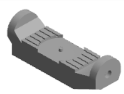
→ xslide.stl



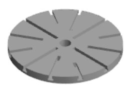
→ yslide.stl



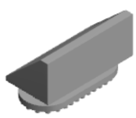
→ zslide.stl



→ aswing.stl



→ Ctable.stl



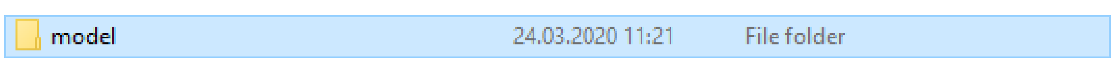
→ magazine.stl

→ controler.stl

WARNING

All "STL Files" must be named in lowercase letters.

These "STL files" must be stored in the same folder.



All file/directory names are written in lowercase.

Name	Date modified	Type	Size
aswing.stl	24.03.2020 08:25	3D Object	566 KB
base.stl	24.03.2020 08:25	3D Object	151 KB
controler.stl	24.03.2020 08:46	3D Object	297 KB
ctable.stl	24.03.2020 08:25	3D Object	575 KB
door.stl	24.03.2020 08:25	3D Object	359 KB
housing.stl	24.03.2020 08:25	3D Object	1.715 KB
magazine.stl	24.03.2020 08:25	3D Object	22.043 KB
xslide.stl	24.03.2020 08:25	3D Object	22 KB
yslide.stl	24.03.2020 08:25	3D Object	420 KB
zslide.stl	24.03.2020 08:25	3D Object	518 KB

WARNING

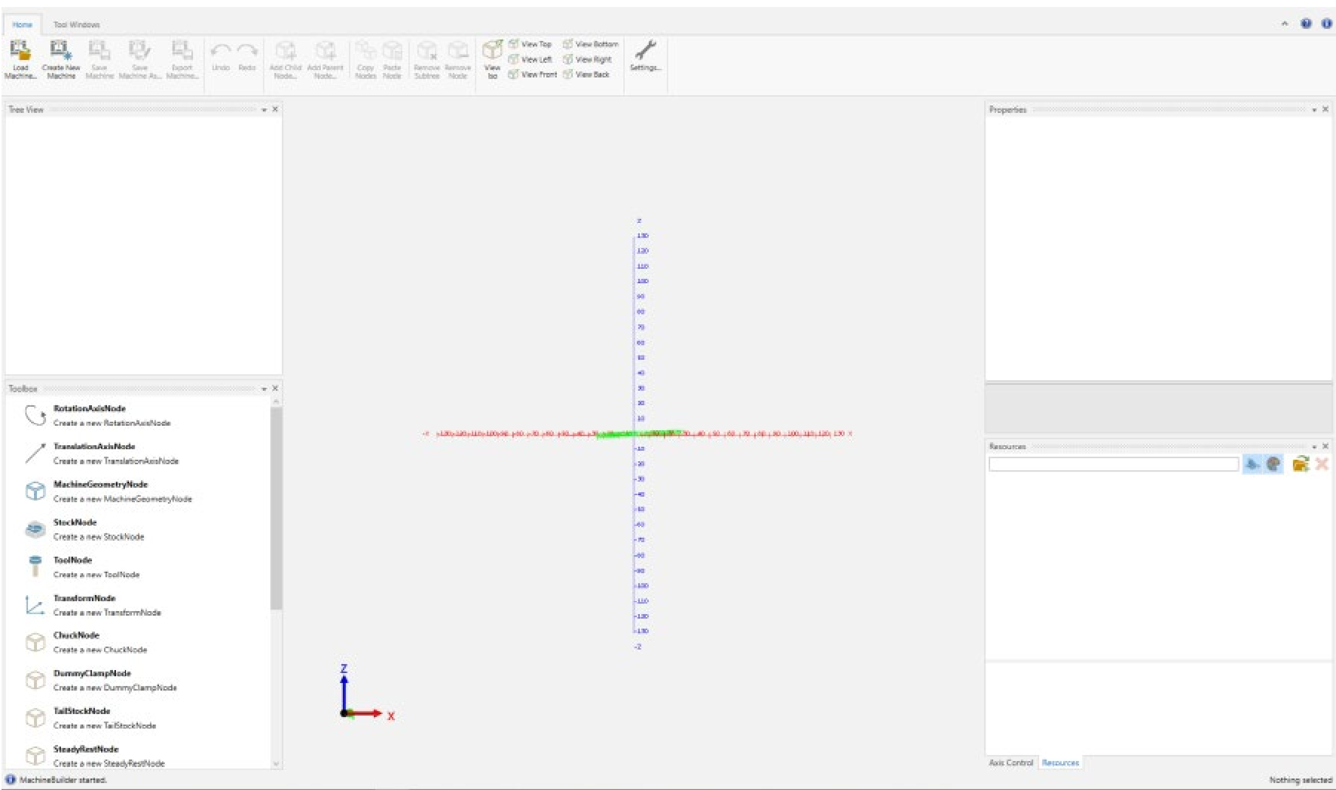
File name and letters

The naming of the Nodes in MachineBuilder should be followed for the next steps. The names should be written in lowercase letters.

Creating a 3D model 5

5.1 Creating a 3D model

A 3D model is created by the "MachineBuilder" tool.

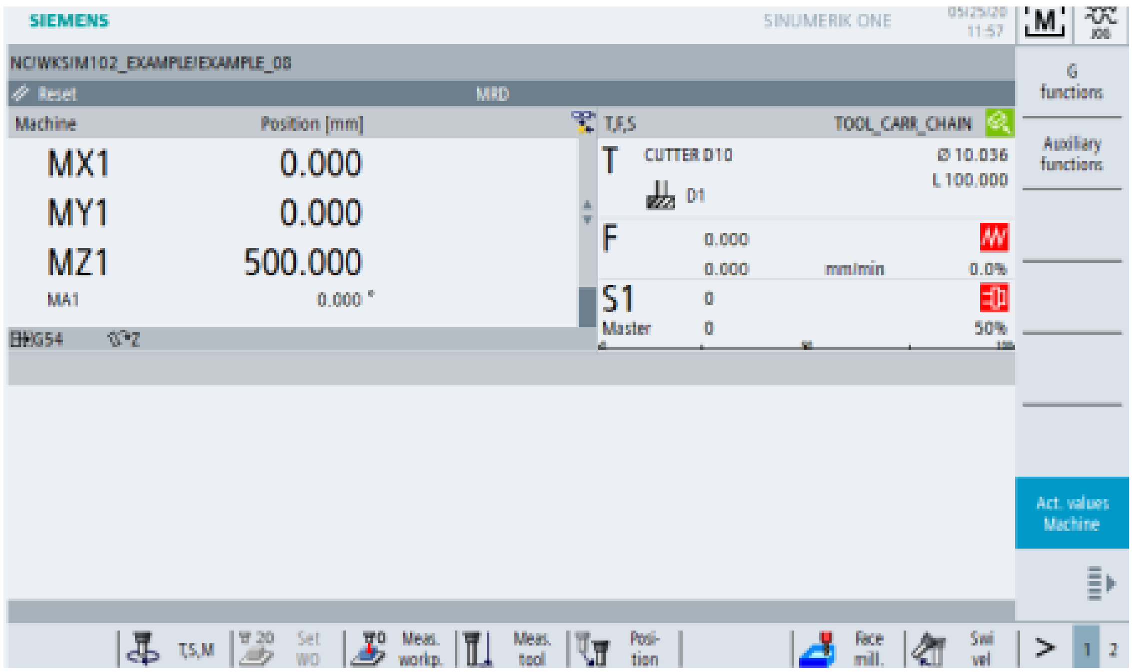


The machine model used later for the simulation is simulated in the "MachineBuilder".

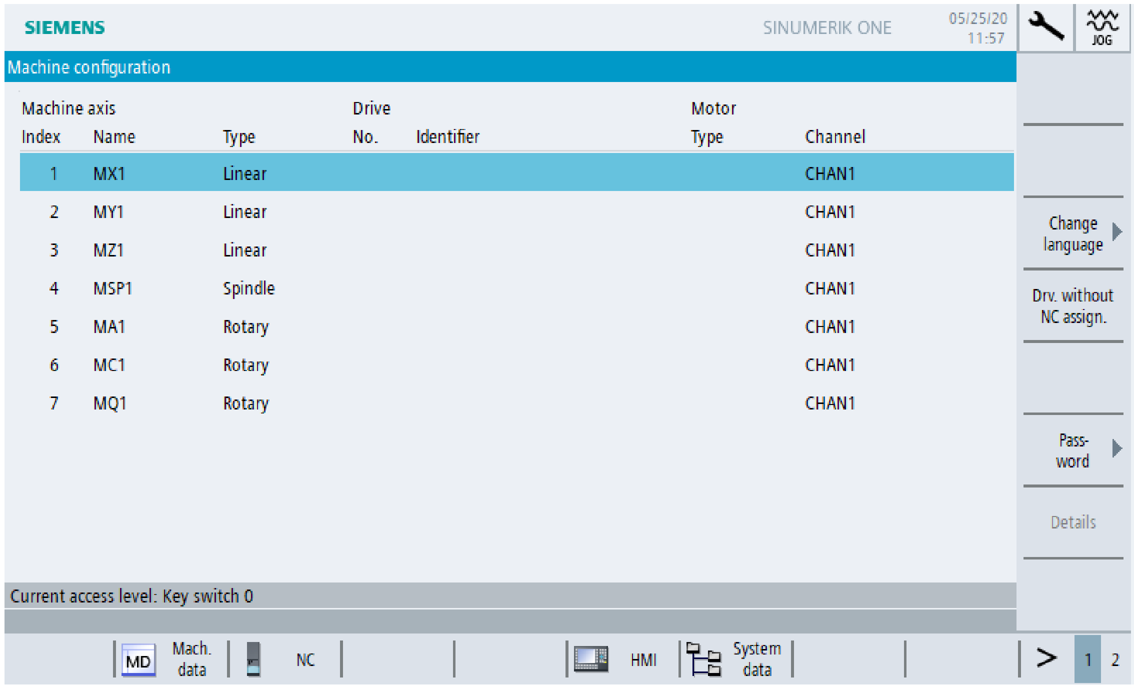
In this example, the machine with name "SinuMill 500" in "Protect MyMachine /3D Twin" or "Create MyVirtual Machine /3D" serves as basis of the following work steps.

The machine kinematics must be known.

- What are the limits of the axes?
- Where are the machine zero points of the axes?
- What are the distances of the tool point from the machine zero point in all axes?
- Which axes does the machine have?



Also, the axis and spindle designations. To ensure the reference between the model and "Protect MyMachine /3D Twin" or "Create MyVirtual Machine /3D", they must be programmed with the same name later in the "MachineBuilder" tool.

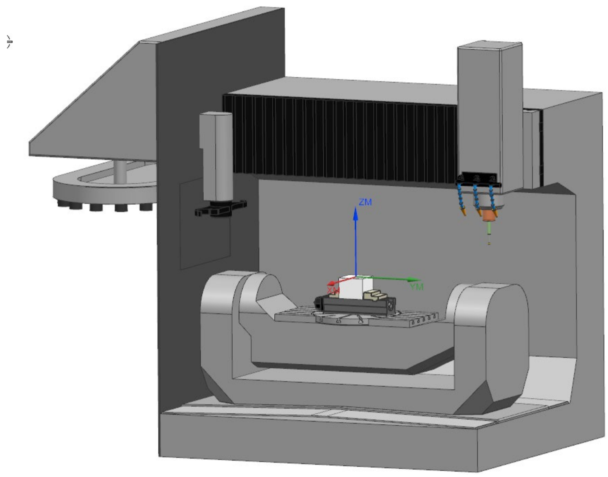


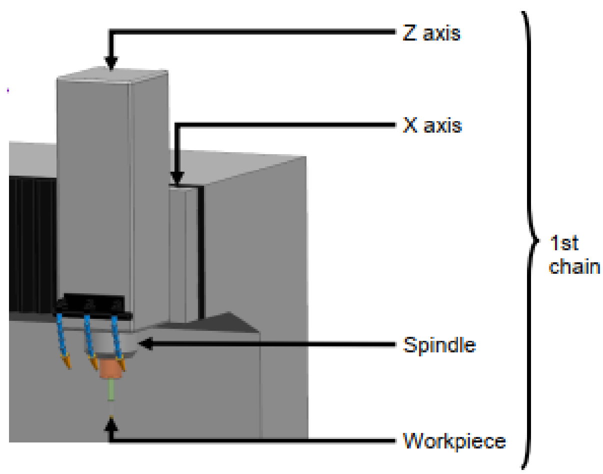
The 3D machine model can be built when all this information is known and the associated "STL files" are available.

A machine tool consists of two chains.

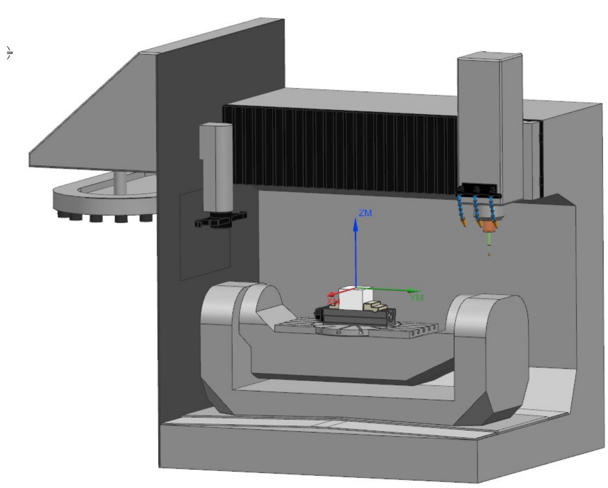
- The first chain describes the machine axes / spindles to the tool tip
- The second chain describes the machine axes / spindles to the workpiece
  - The structure of the chains, the technical data, such as traversal path and axis designations, of the machine are known.
  - These chains are programmed identically in the "MachineBuilder" and added later in the"Protect MyMachine /3D Twin" or "Create MyVirtual Machine /3D" software.

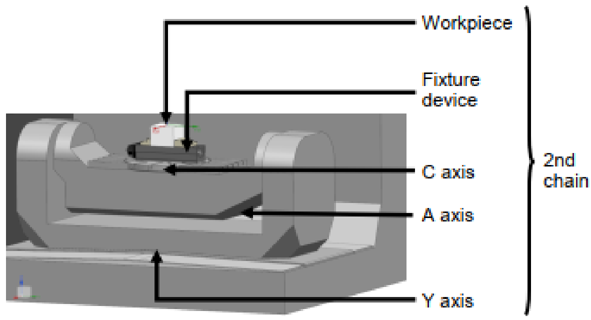
The first chain to the tool tip for SinuMill 500:





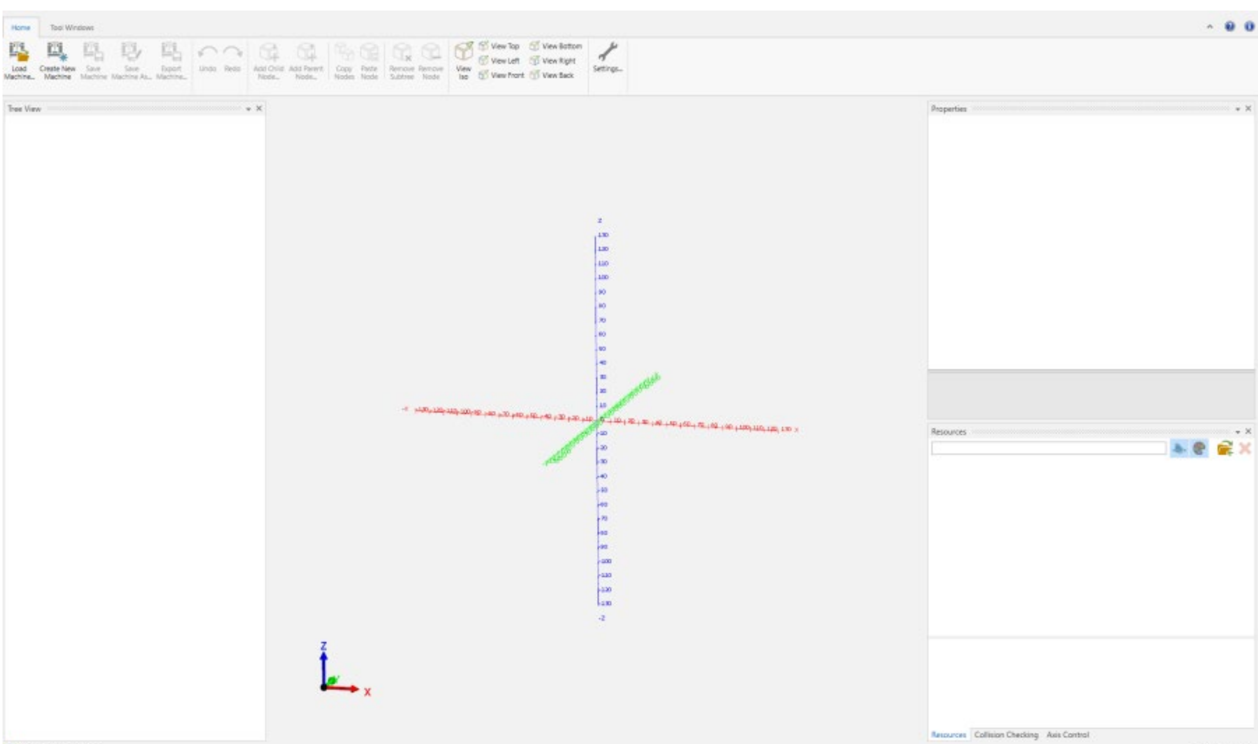
The second chain to the workpiece for SinuMill 500:



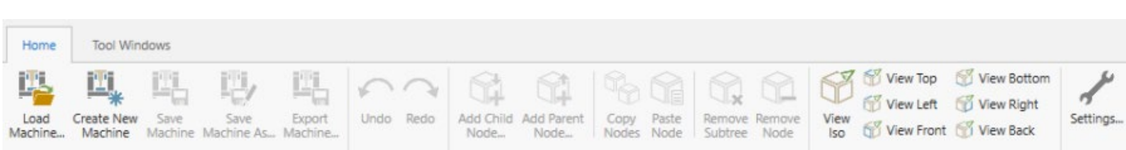


5.2 MachineBuilder

After starting the "MachineBuilder" program, a spatial coordinate system is displayed.

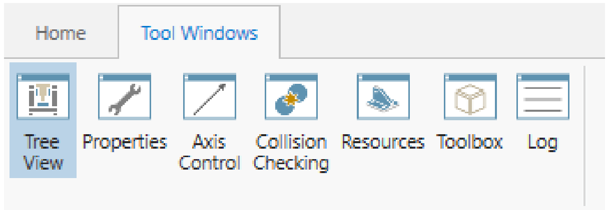


In the "Home" tab, machines can be created or existing machines loaded.



The machine is built using so-called "nodes".

The implementation is performed at the "Tool Windows" tab.



All functions required for building the "SinuMill 500" are explained based on this example.

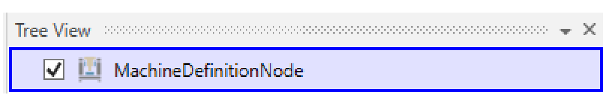
Note

The machine model for PMM /3D Twin and Create MyVirtual Machine /3D must be created in metric units (mm).

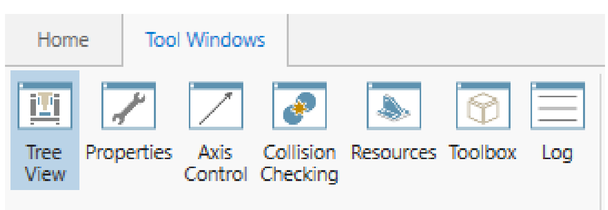
5.2.1 New Machine

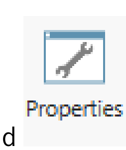
In the "MachineBuilder" tool, pressing 

creates a new machine.

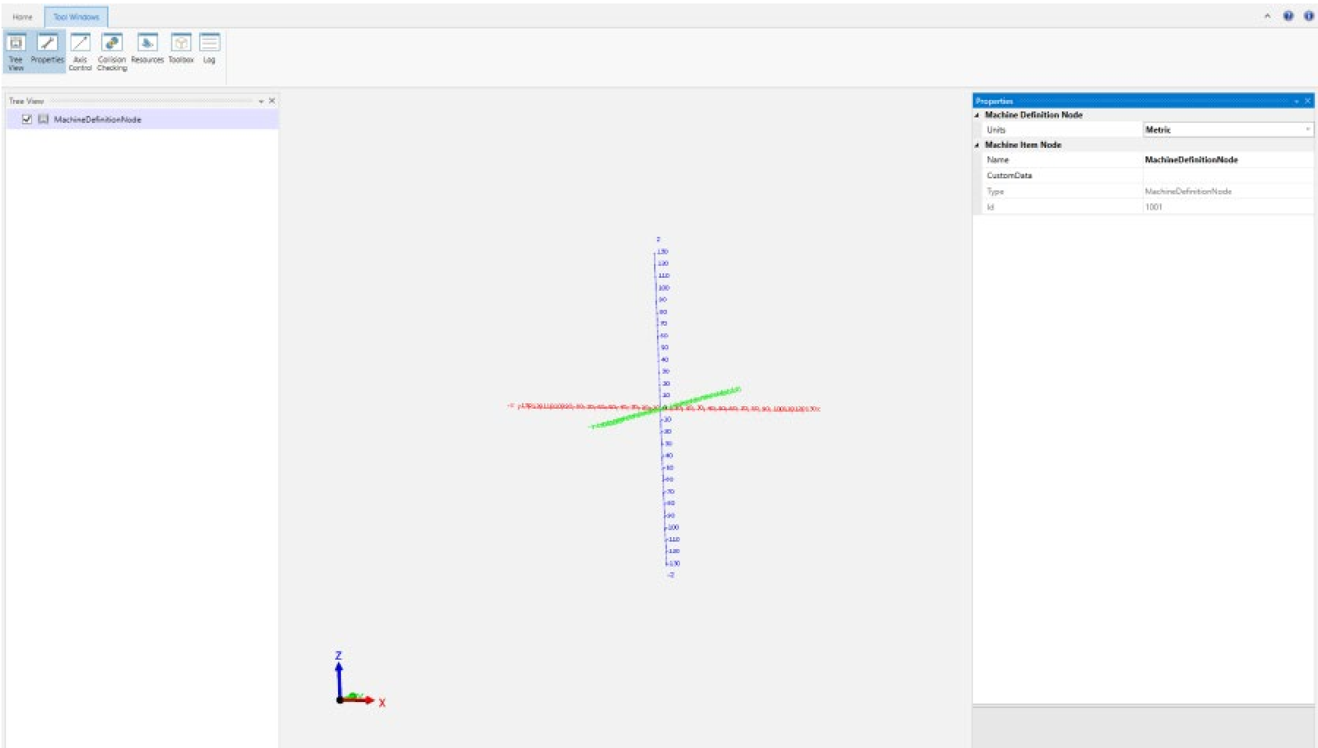


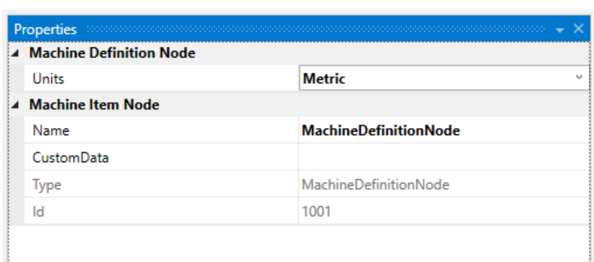
Selecting the "Tool Windows"

tab 

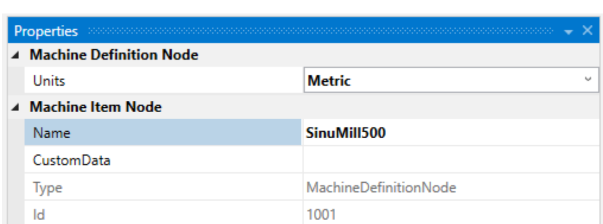
and 

opens an input field at the right.

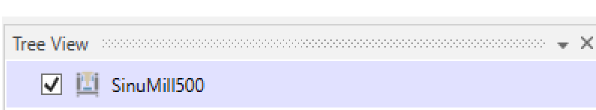




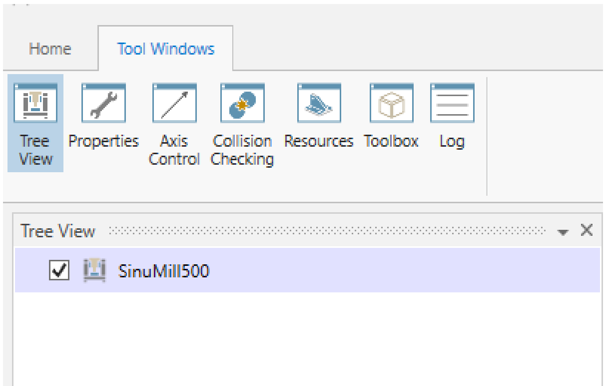
The name of the "SinuMill 500" machine model is entered in this field.



The name of the machine now appears in the "Tree View".

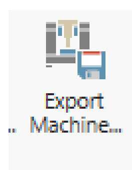


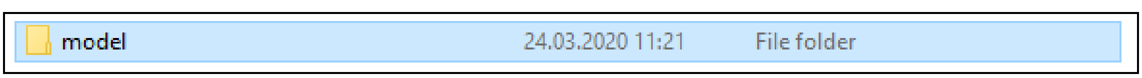
The "Tree View" function should always be activated, because the links, i.e. the "nodes", are represented here.



Icon	Explanation
	The programmed model structure (nodes) is shown.

To use the new machine later in "Protect MyMachine /3D Twin" or "Create MyVirtual Machine /3D", it must be stored as follows:

At the "Home" tab, pressing  stores the machine.



Note

"Protect MyMachine /3D Twin" and "Create MyVirtual Machine /3D" is case sensitive. The name should not include uppercase letters and special symbols.

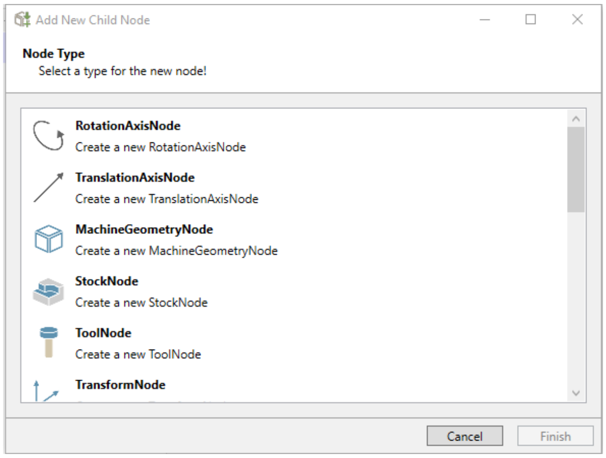
5.2.2 Integration machine axes

The machine axes are integrated in the next step

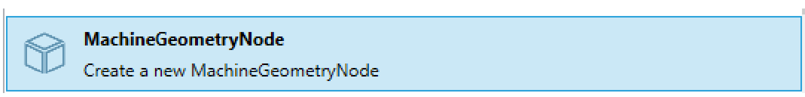
To make this visible in its traversal movement, a coordinate system is loaded as "STL file" with the name "frame". This file is not part of the machine but is integrated to ease understanding.

At the "**Home**" tab and pressing 

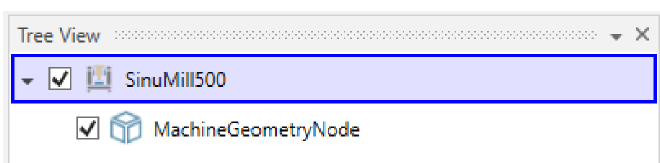
opens a selection field.



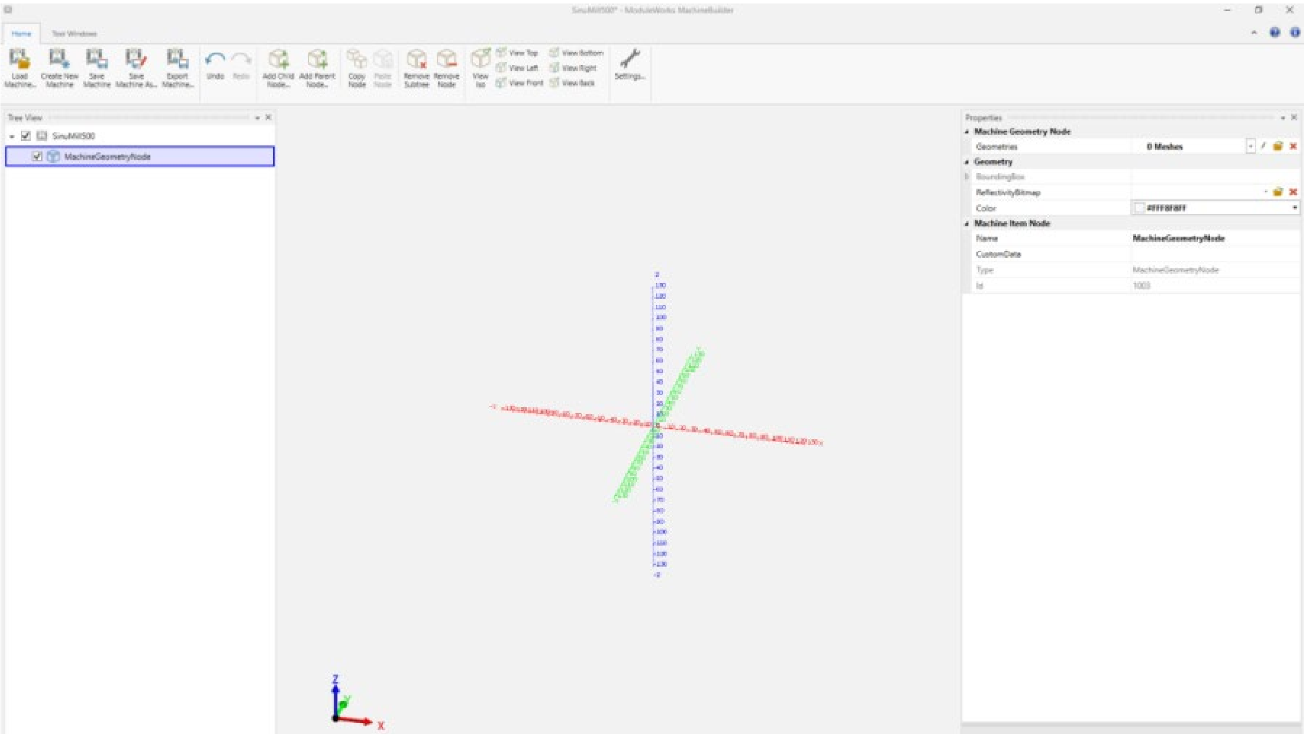
An "STL file" is called "MachineGeometryNode".

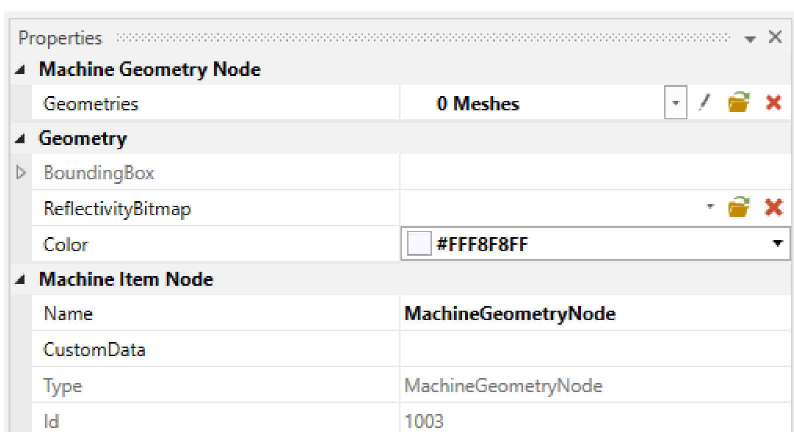


A "MachineGeometryNode" has been added in the "Tree View". An "STL file" is integrated via this node.



When selected, the "Properties" field opens at the right on the screen.

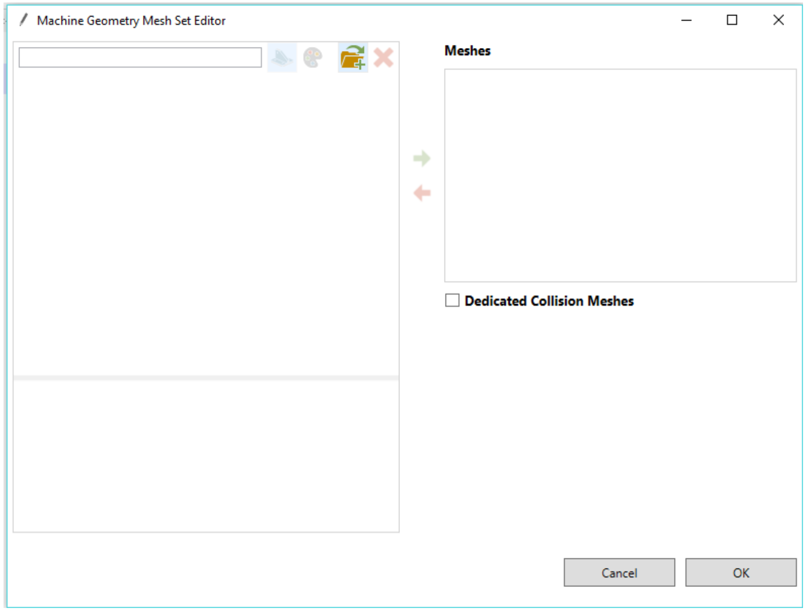




Selecting the pencil symbol next to "Geometries"

opens an input field.

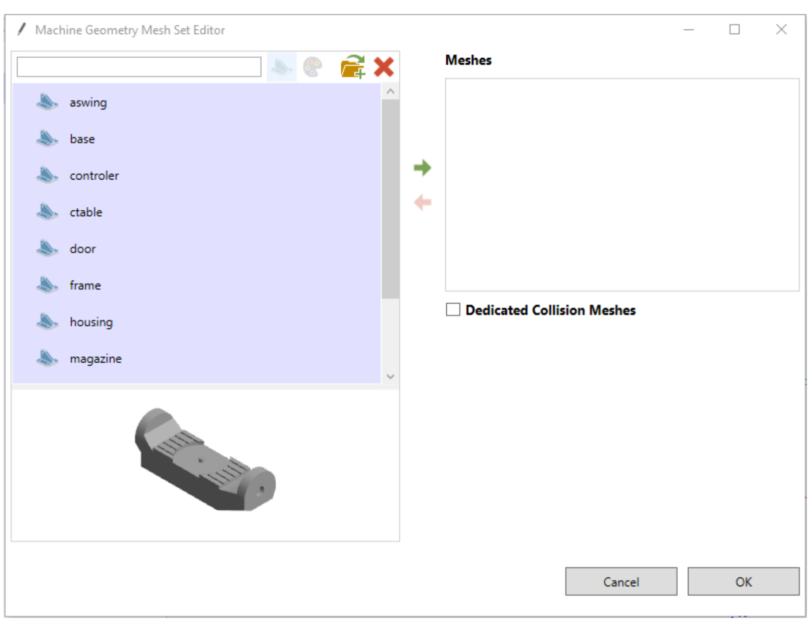
The word "Meshes" means the individual "STL files".

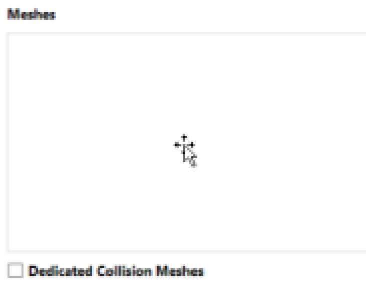


The "model" folder with the "STL files" is called via the symbol and all files can be selected from the opened popup.

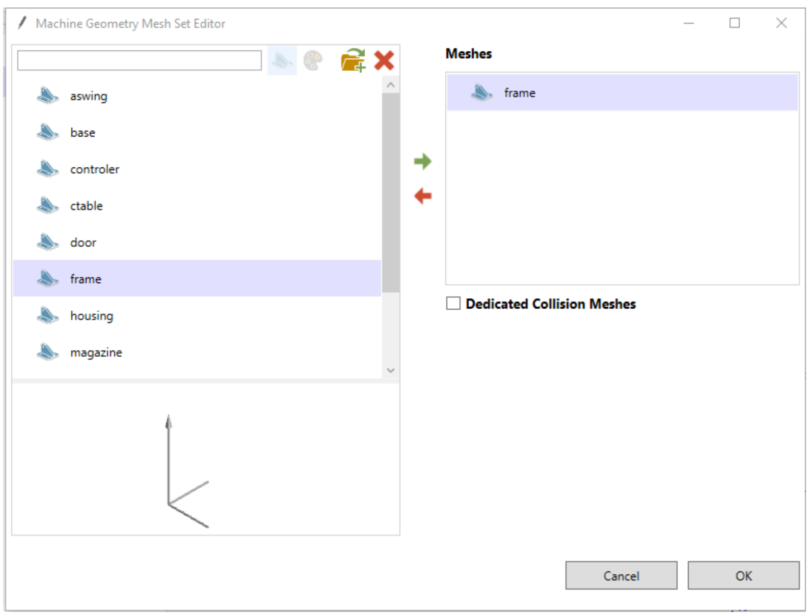
They are now available as "Meshes" with the appropriate graphic.

A specific selection for this "MachineGeometryNode" can be made from "Meshes" on the left side by selecting an element and pressing to transfer it to the Meshes area.

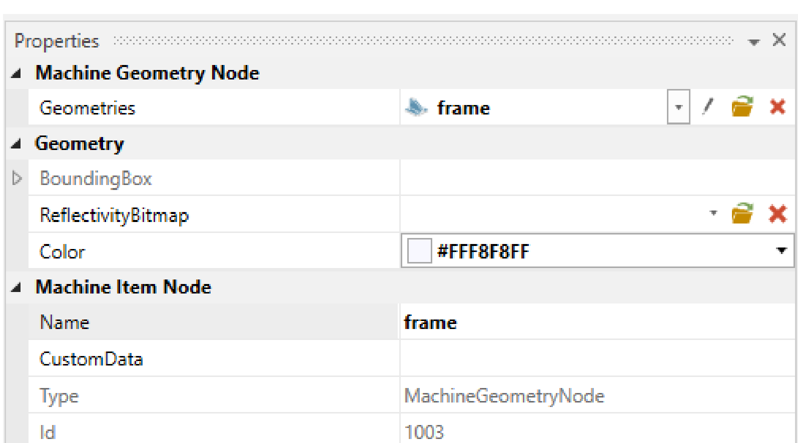




After selecting and transferring the desired "frame" file, it is integrated in the "Meshes".



"frame" is selected as name.



The "frame" is now visible in the coordinate system.

In the first step, the axes with the traversal movements to each other are defined.

Note

Note that, this step depends on the machine that is built by the OEM. Your axes can be in different order, it depends on the kinematics of the machine.

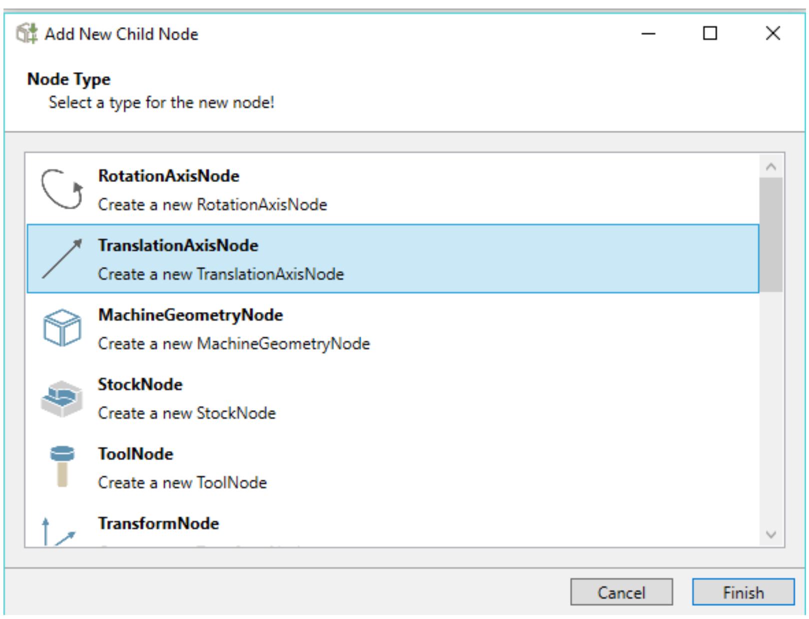
Machine Builder program has a parent child relationship. The axes that moves together should be added as child nodes.

In accordance with the machine ("SinuMill 500") kinematics, the "Y axis" is part of the "A axis" and the "C axis".

This is the "chain" that is programmed with the "MachineBuilder" tool.

Pressing 

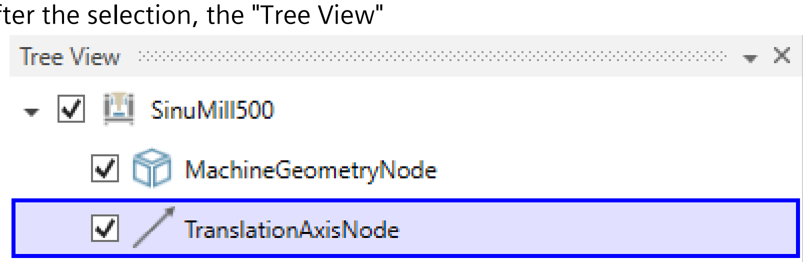
opens a popup, that can be used to select the needed Node.



A "TranslationAxisNode" is used to add the axis of the machine.

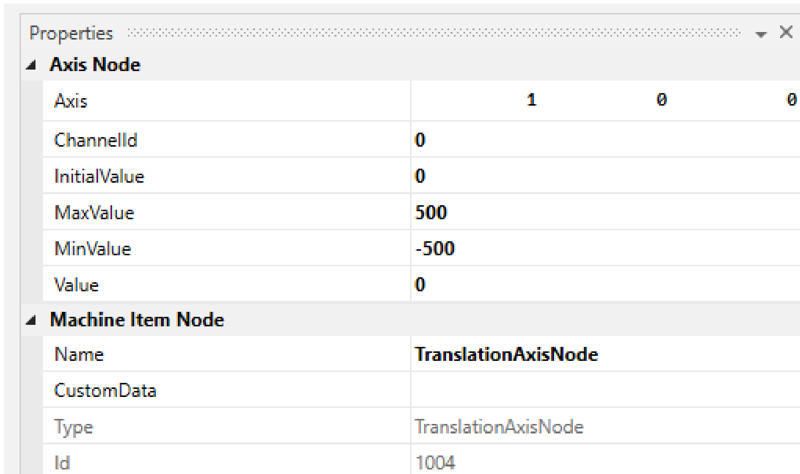
Process for the SinuMill 500 starts selection of a "TranslationAxisNode" in the selection field for the "Y axis".

After the selection, the "Tree View"



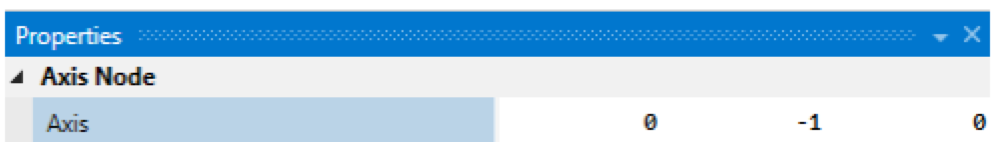
→

By pressing the "TranslationAxisNode", the input fields of the "Properties" opens on the right side of the application.



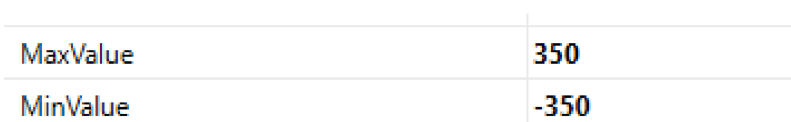
The axis with name and traversal path is described in this field.

The axis and the direction are described in the first field.

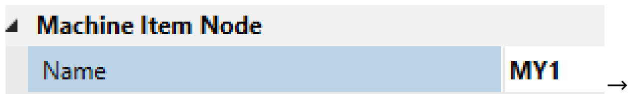


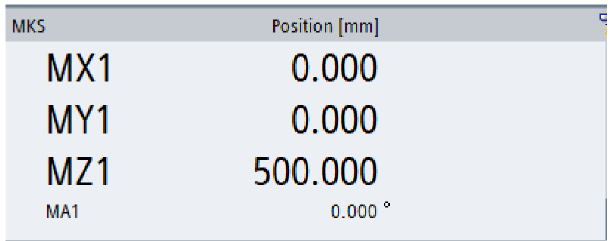
X axis Y axis Z axis

The min. and max. traversal paths are defined in these fields. These values originate at the table center point (machine zero) and correspond to the real traversal movements of the machine.



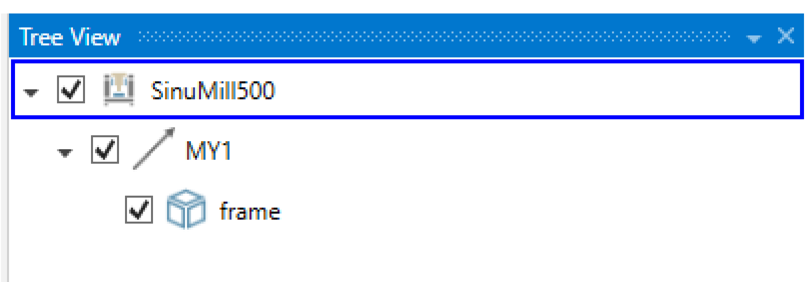
The axis name must match the axis designation of the machine.



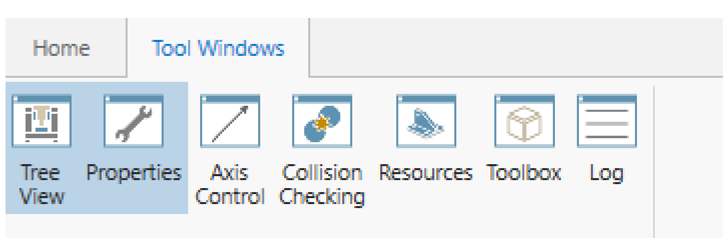


If this is not the case, the model in "Protect MyMachine /3D Twin" or "Create MyVirtual Machine /3D" does not work correctly.

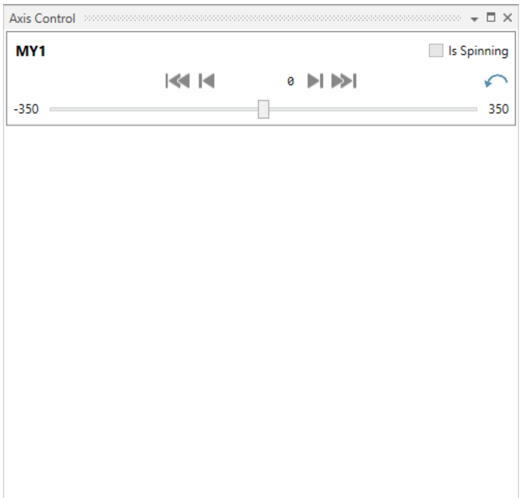
In the "Tree View",  the "frame" is attached to the "Y axis" per drag-and-drop.

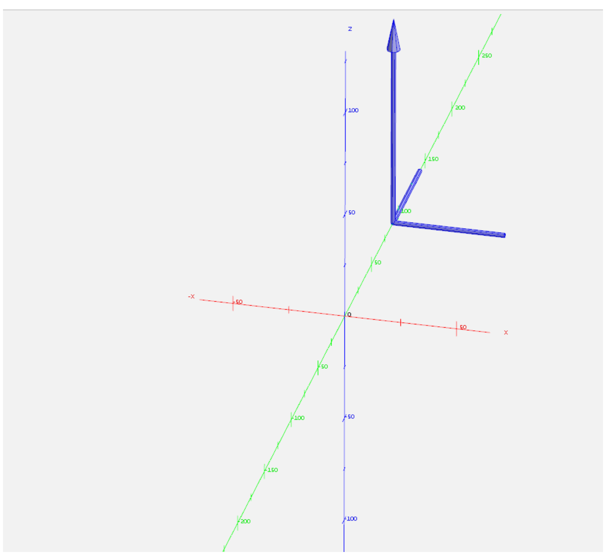


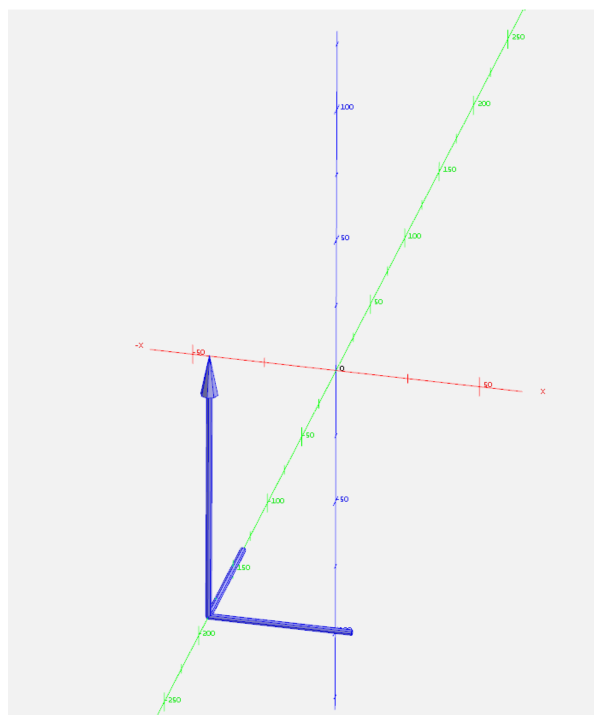
When the "Axis Control"  is activated in the "Tool Windows" tab.



The described "Y axis" appears.



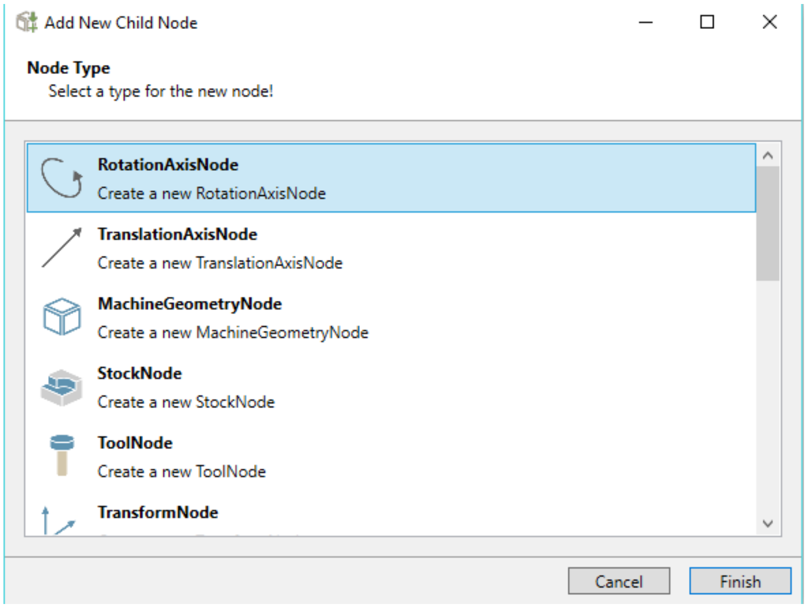




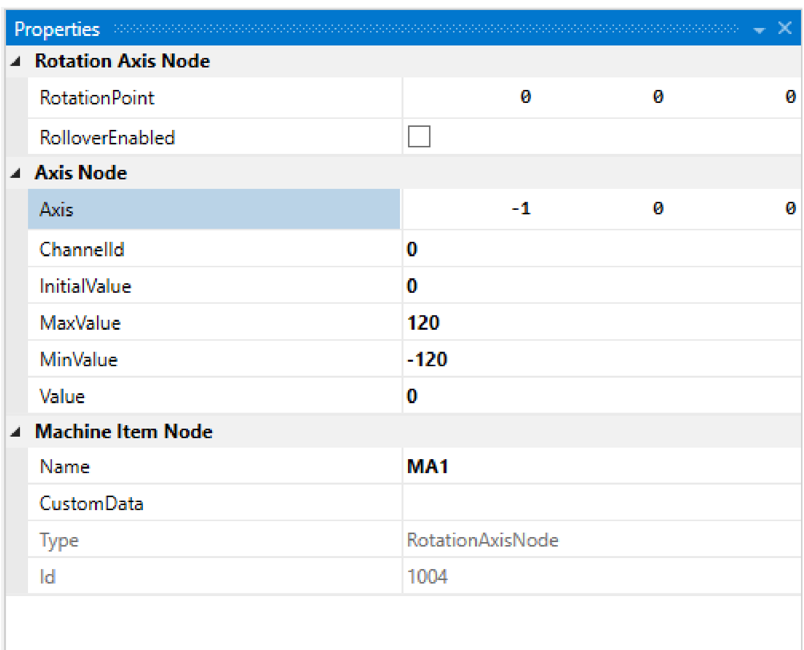
The "A axis" is programmed in the next step. It is connected with the "Y axis".

After selection of 

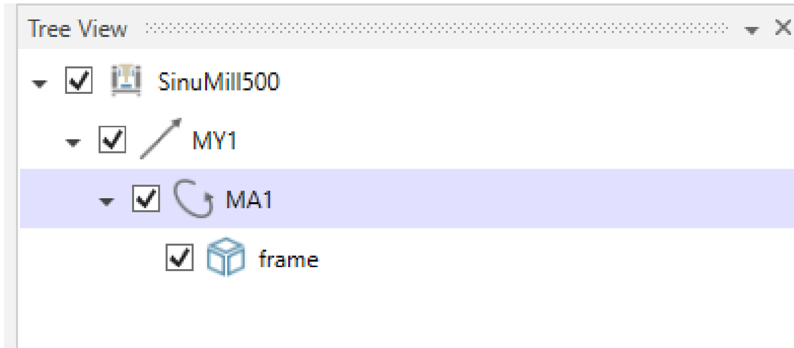
the "RotationAxisNode" is selected in the selection field, because the "A axis" is a rotary axis.



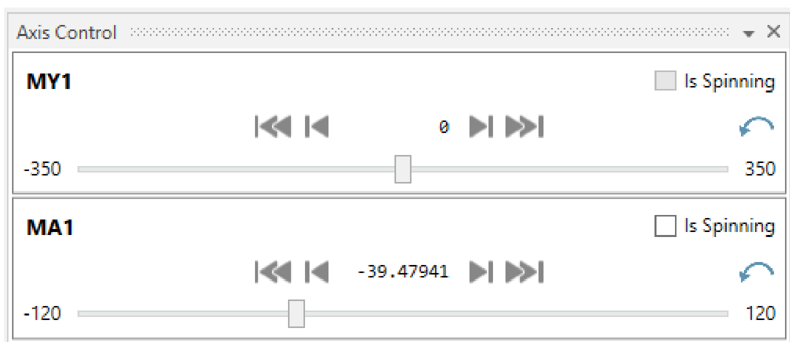
The axis direction, axis path and axis designation are entered in the "Properties" field.

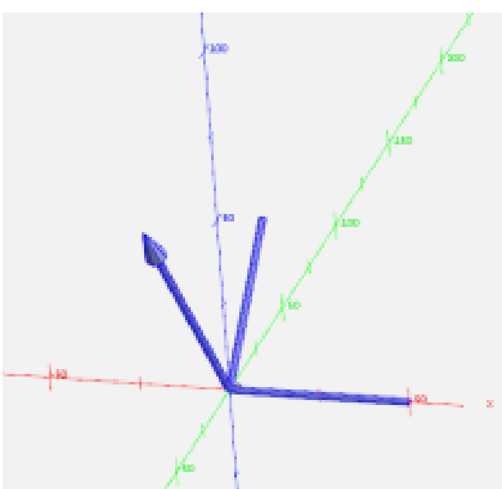


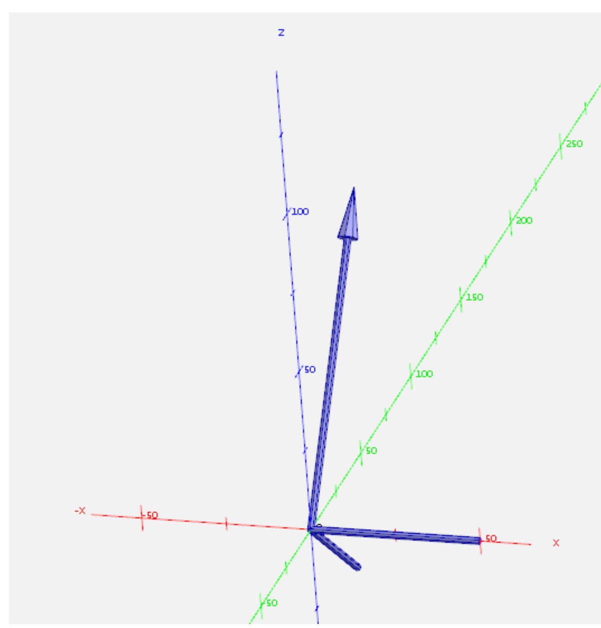
The axis movement must be appended in the "Tree View" at the"Y axis", because the "A axis" is attached to this.



The new axis can then be checked and traversed via the "Axis Control" display.



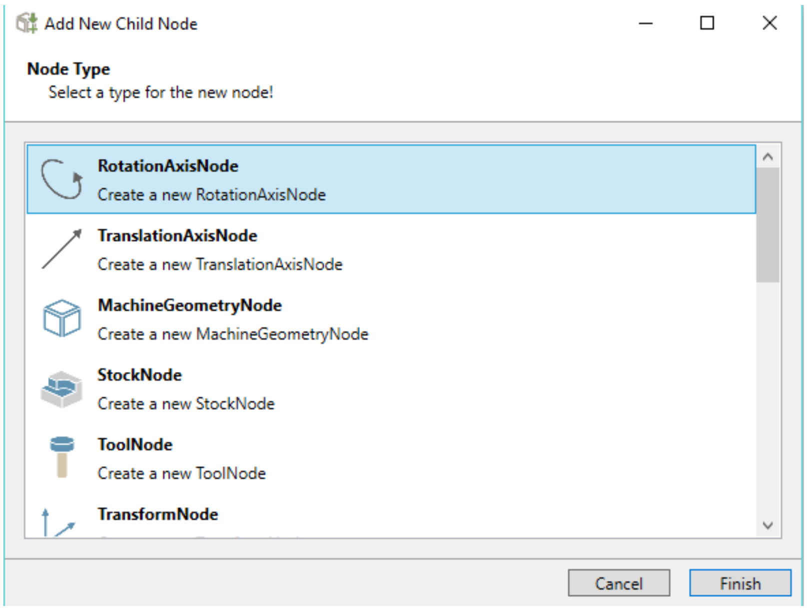




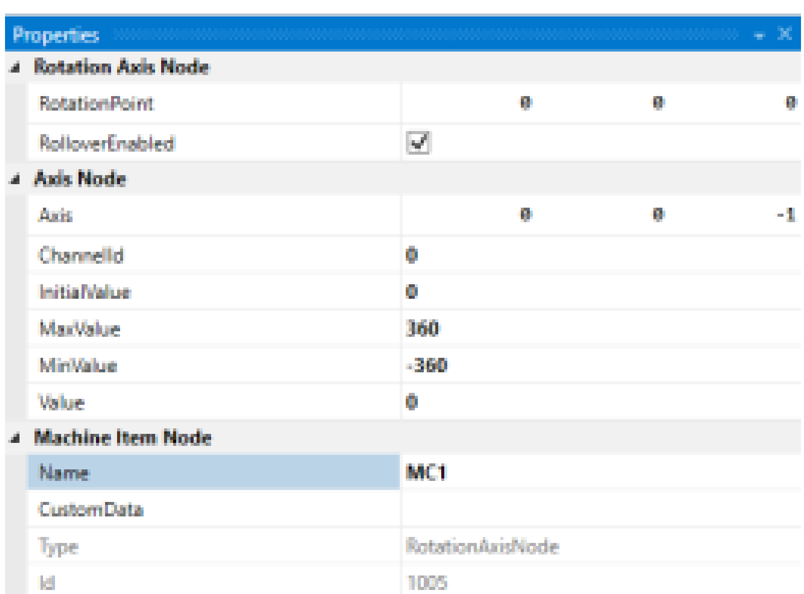
The "C axis" is programmed in the next step. It is connected with the "Y axis" and the "A axis".

After selection of 

the "RotationAxisNode" is selected in the selection field, because the "C axis" is a rotary axis.

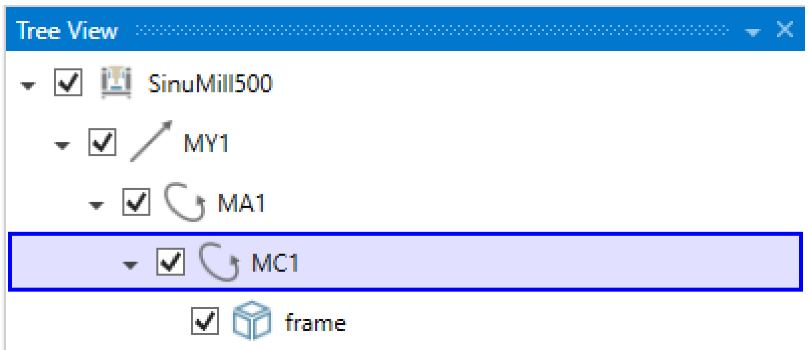


The axis direction, axis path and axis designation are entered in the "Properties" field.

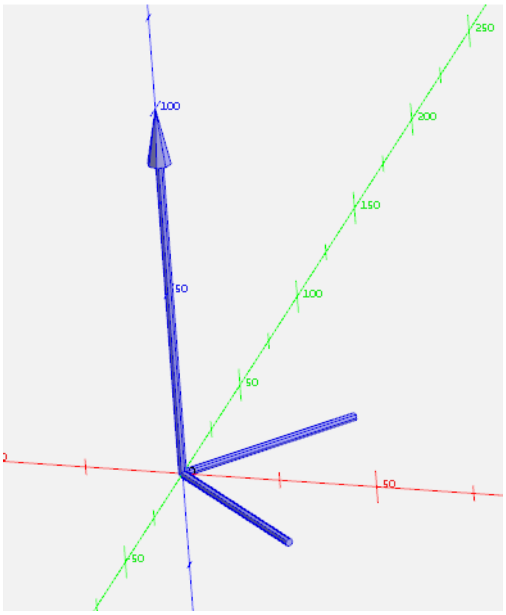


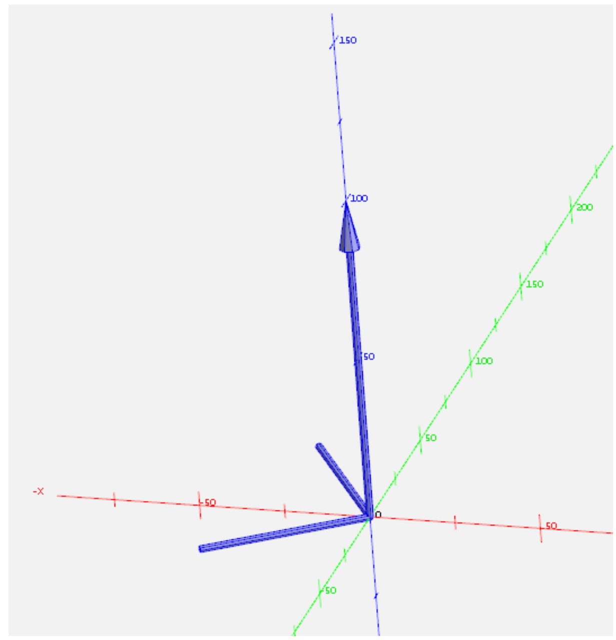
Because the "C axis" is a rotary axis that can turn permanently around itself, the following checkmark must be set:

The axis movement must be appended in the "Tree View" at the "Y axis" and the "A axis", because the "C axis" is attached to this.



The new axis can then be checked and traversed via the "Axis Control" display.



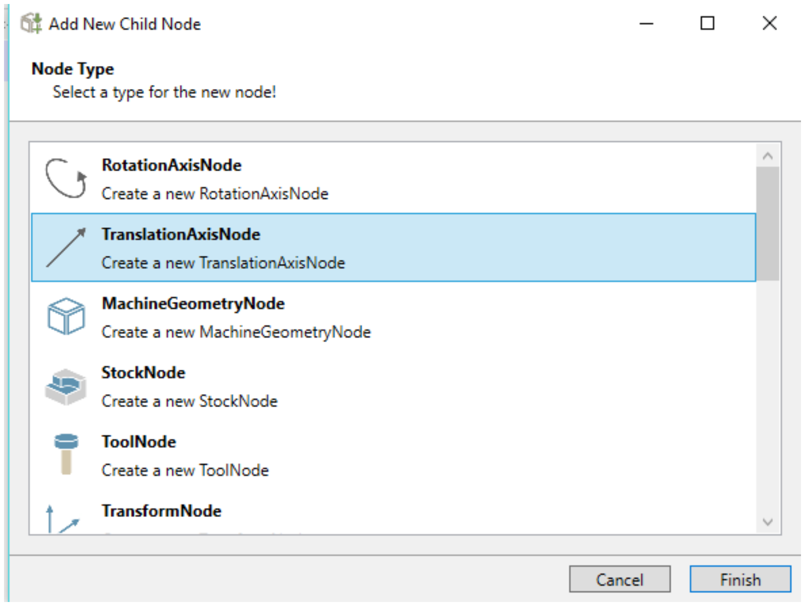


The "X axis" is programmed in the next step.

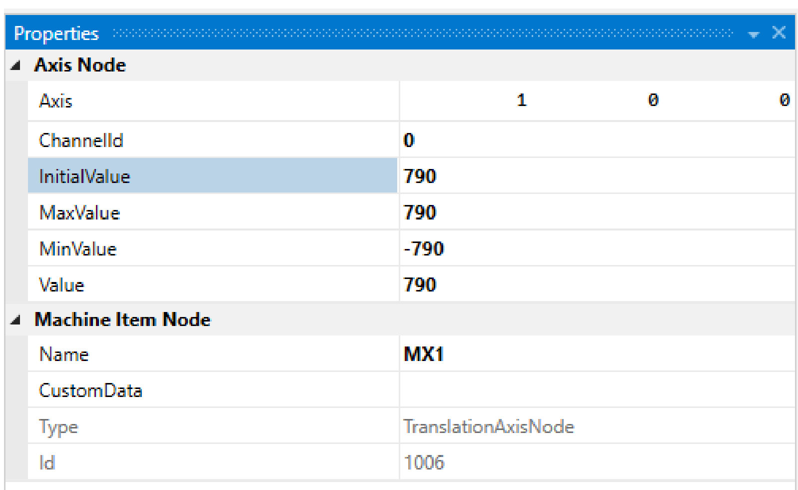
After selection of 

the "TranslationAxisNode" is selected in the selection field, because the "X axis" is a linear axis.

The "X axis" has no dependency with the other axes. Because of that it will be a child node for the machine model and it has no attach to other axes.

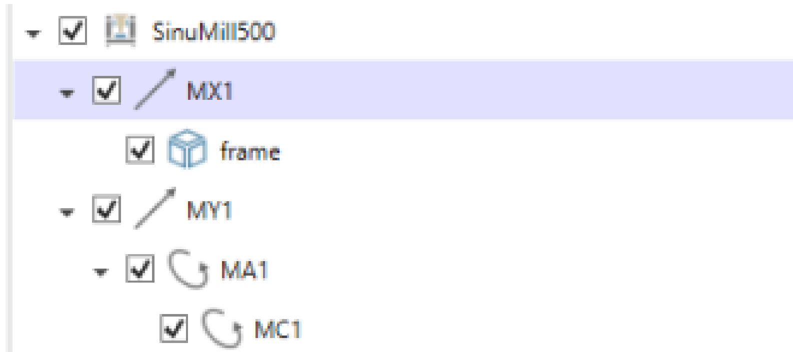


The axis direction, axis path and axis designation are entered in the "Properties" field.

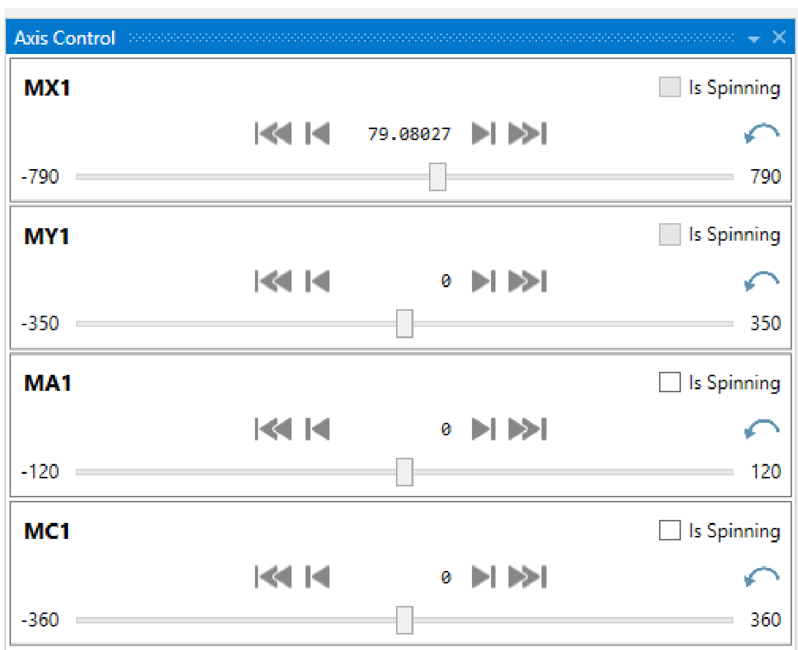


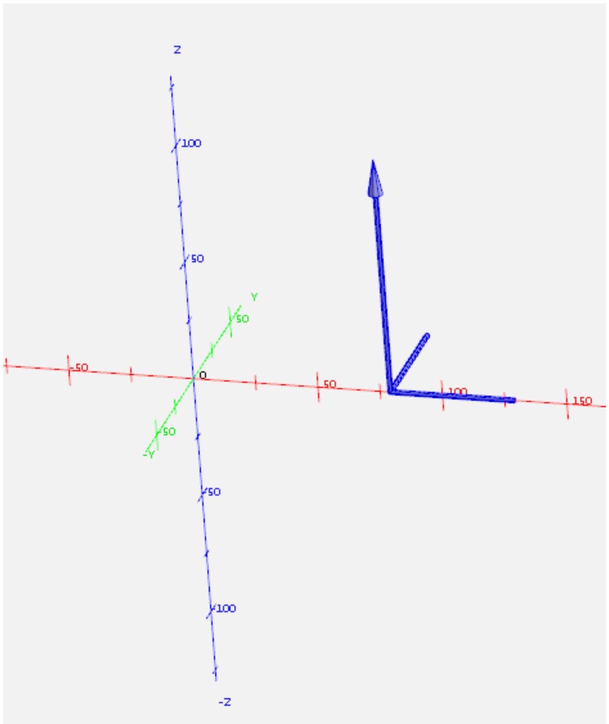
The "Parkposition" of the axis is programmed in the "InitialValue" field.

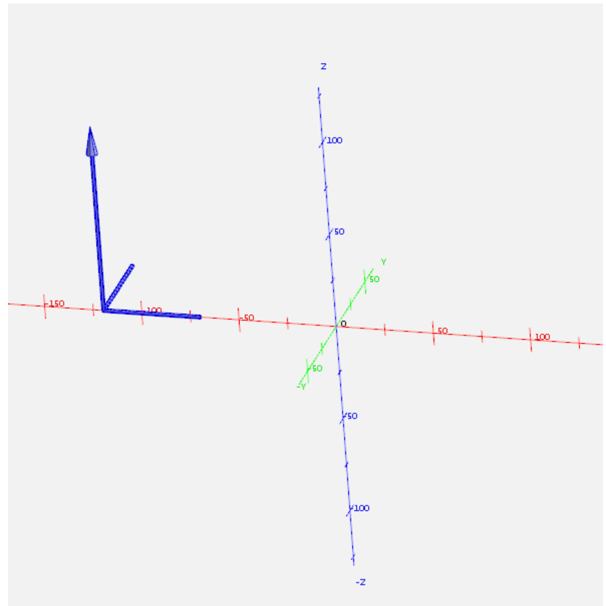
The axis movement must be appended in the "Tree View" directly at the machine.



The new axis can then be checked and traversed via the "**Axis Control**" display.




Z
100
50
Y
50
50
0
50
100
150
50
-Y
50
100
-Z

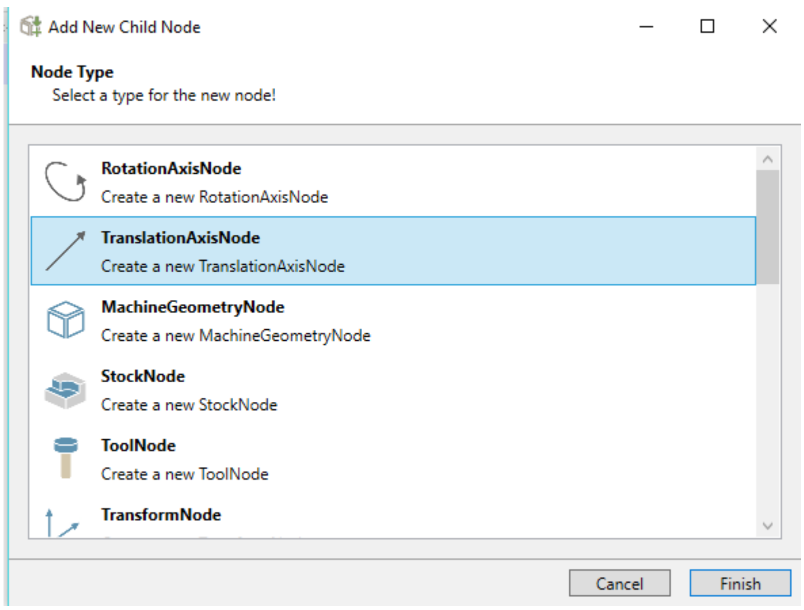

Z
100
50
Y
50
150
100
50
0
50
100
50
-Y
50
100
-Z

The "Z axis" is programmed in the next step.

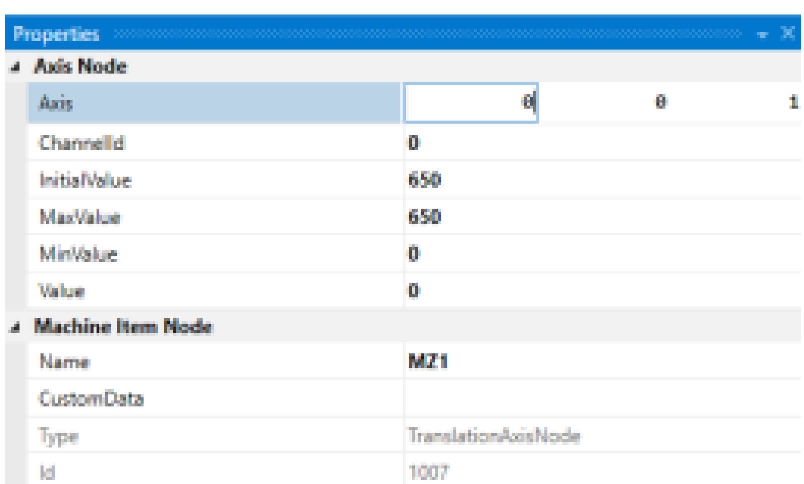
The "X axis" and the "Z axis" are connected with each other.

After selection of 

the "TranslationAxisNode" is selected in the selection field, because the "Z axis" is a linear axis.



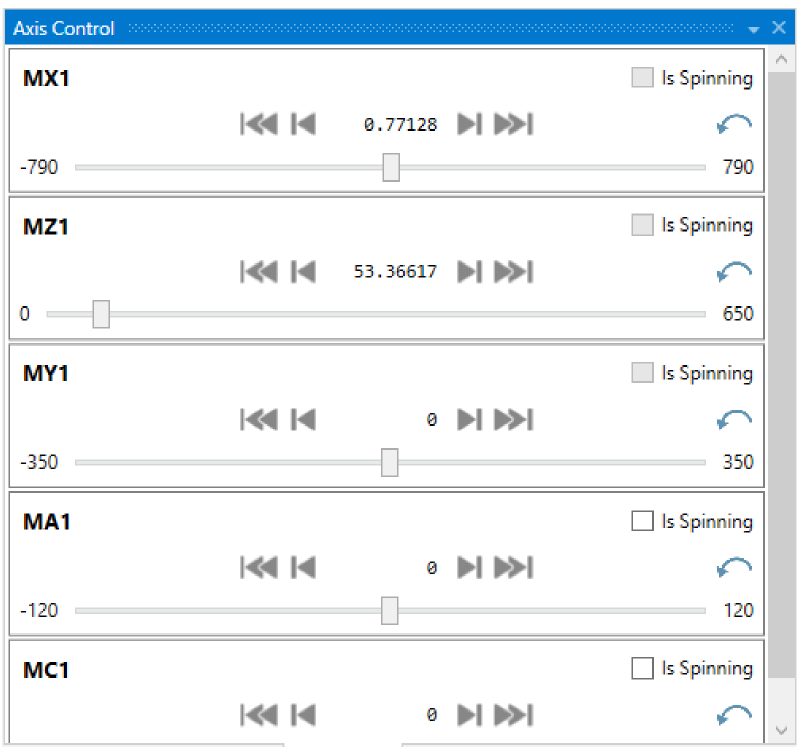
The axis direction, axis path and axis designation are entered in the "Properties" field.

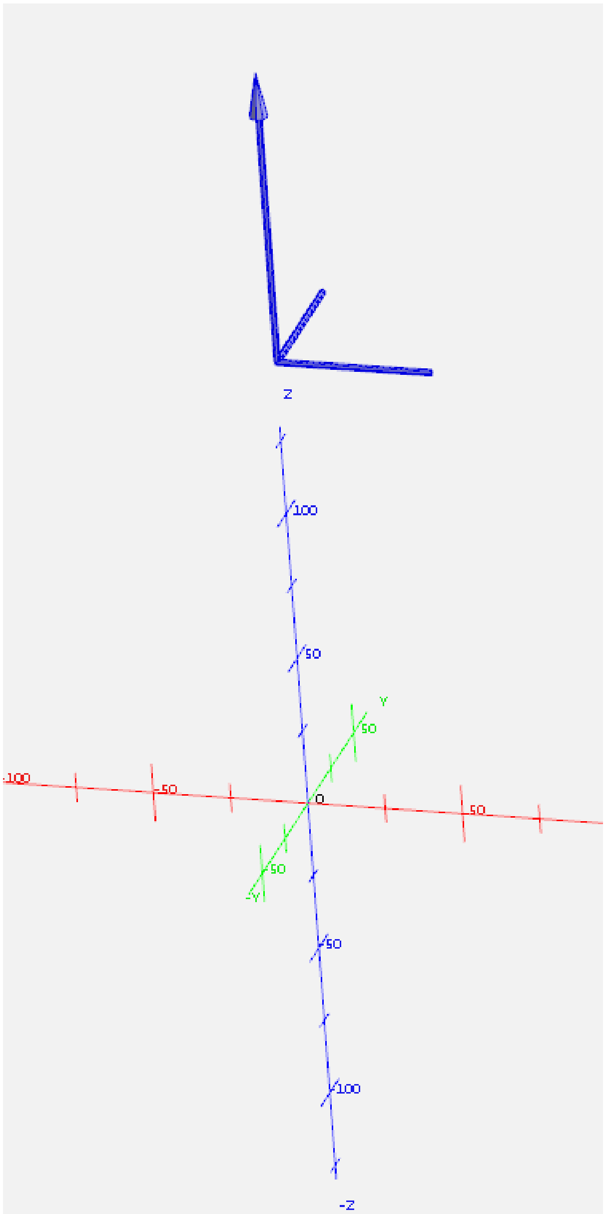


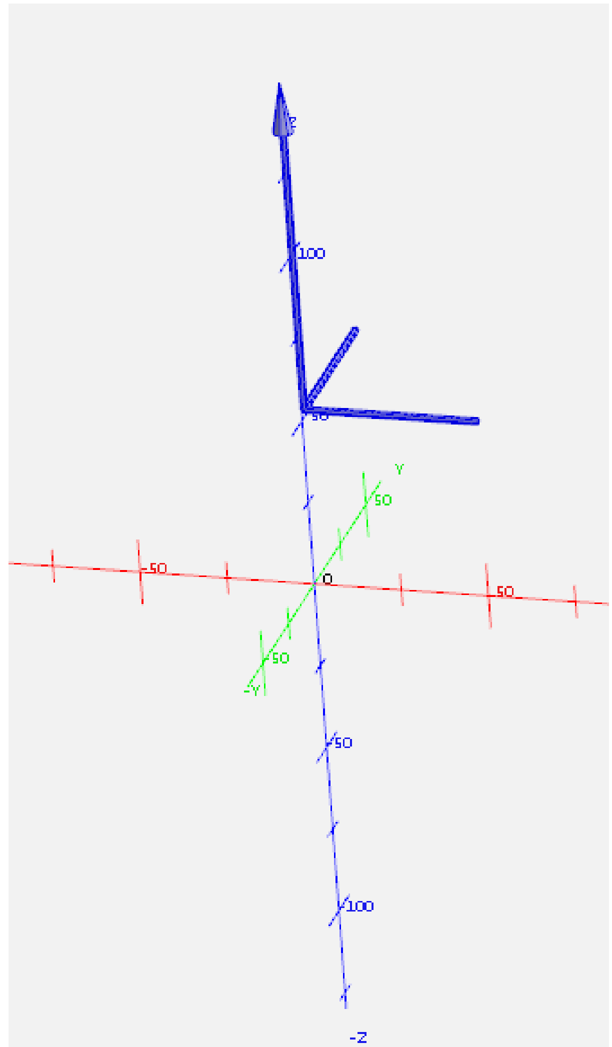
The "Parkposition" of the axis is programmed in the "InitialValue" field.

The axis movement must be appended in the "Tree View" directly at the "Z axis".

The new axis can then be checked and traversed via the "Axis Control" display.



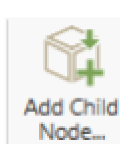


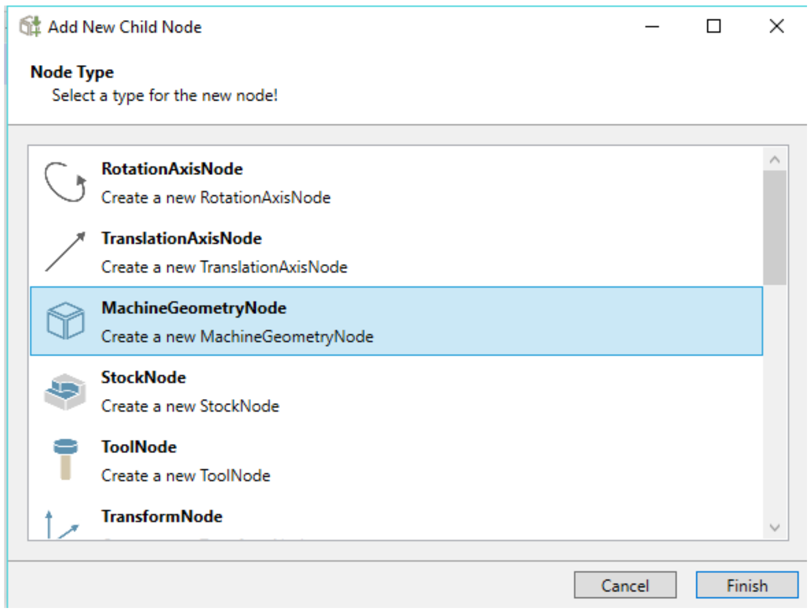


The axes are always configured.

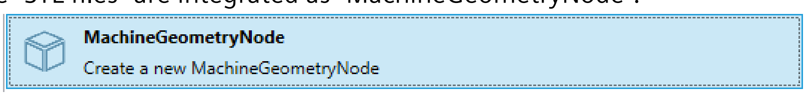
5.2.3 Integration components as *.stl-files

The components are now integrated as "STL files".

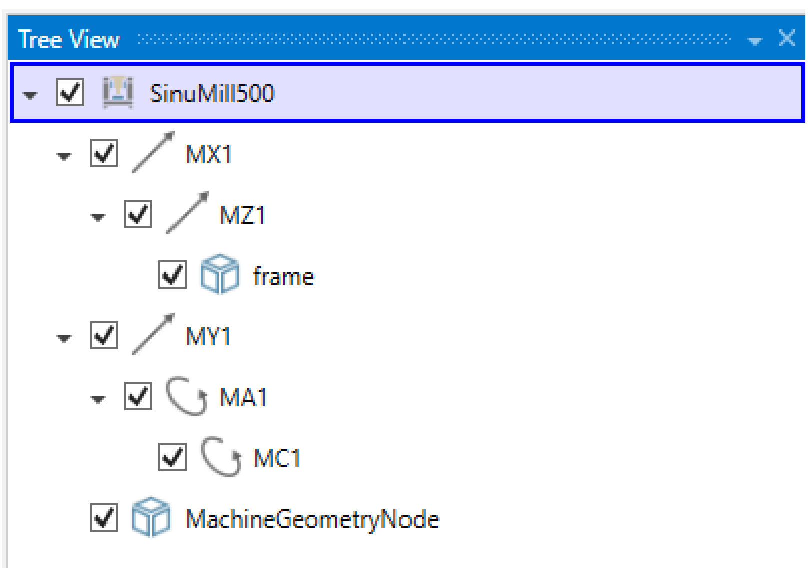
After selection of 



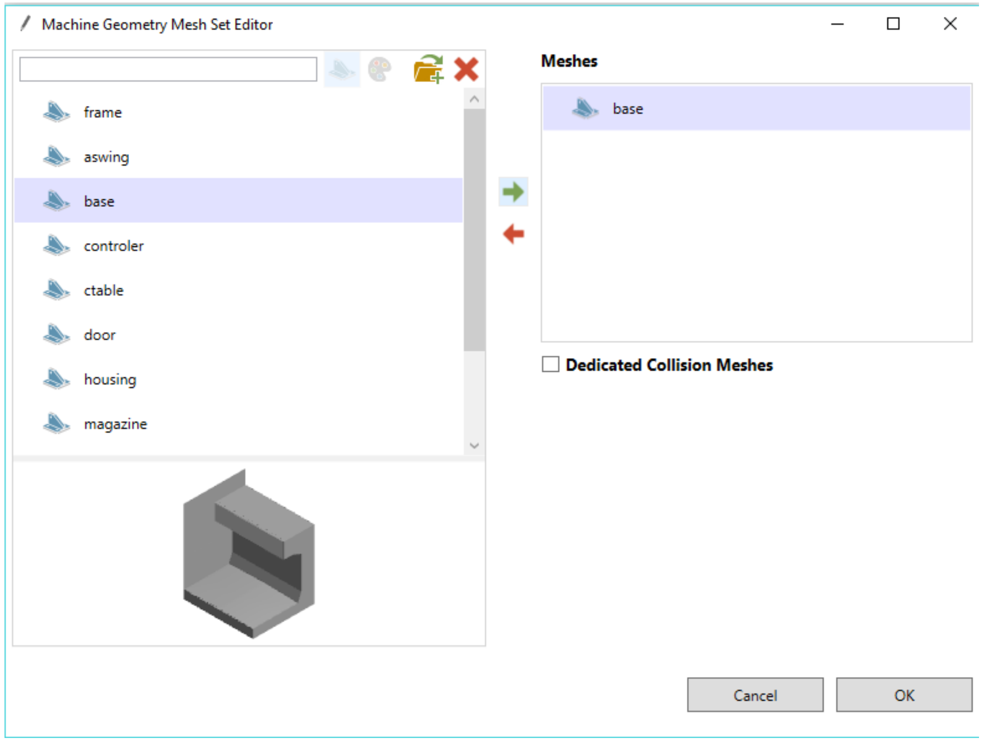
the "STL files" are integrated as "MachineGeometryNode".

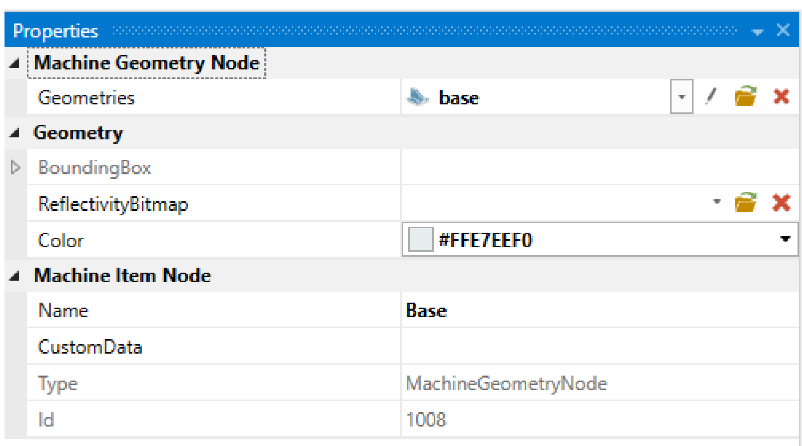
→ 

The First machine STL that will be imported is going to be machine bed, because of that "MachineGeometryNode" is not attached to an axis.

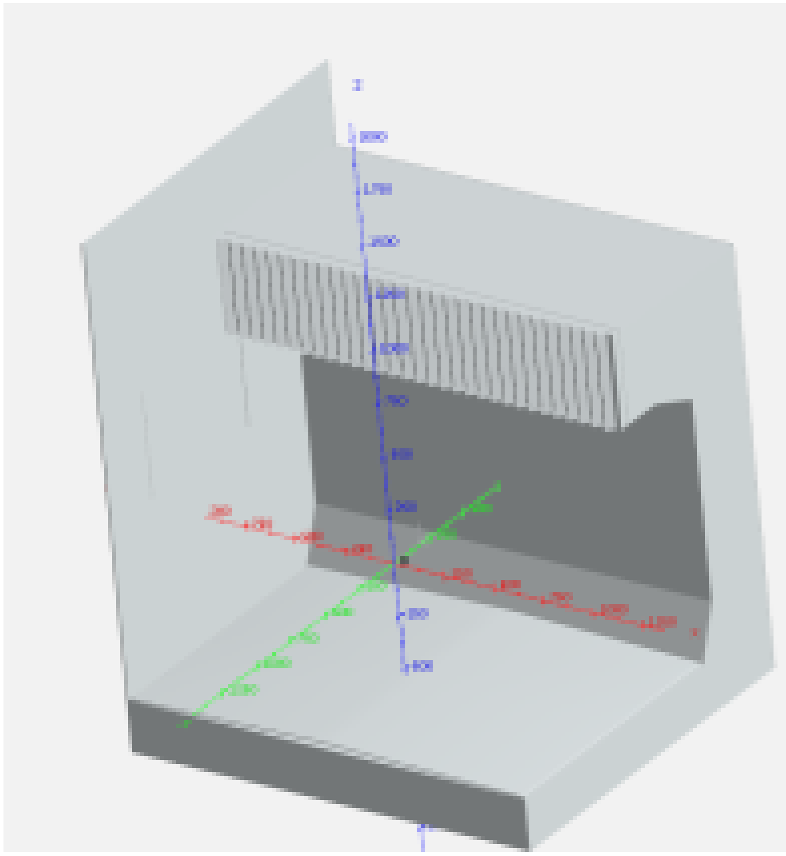


The "STL file" "base" is selected and accepted. Colors are selected at "Properties".





The "STL file" "base" has been accepted.

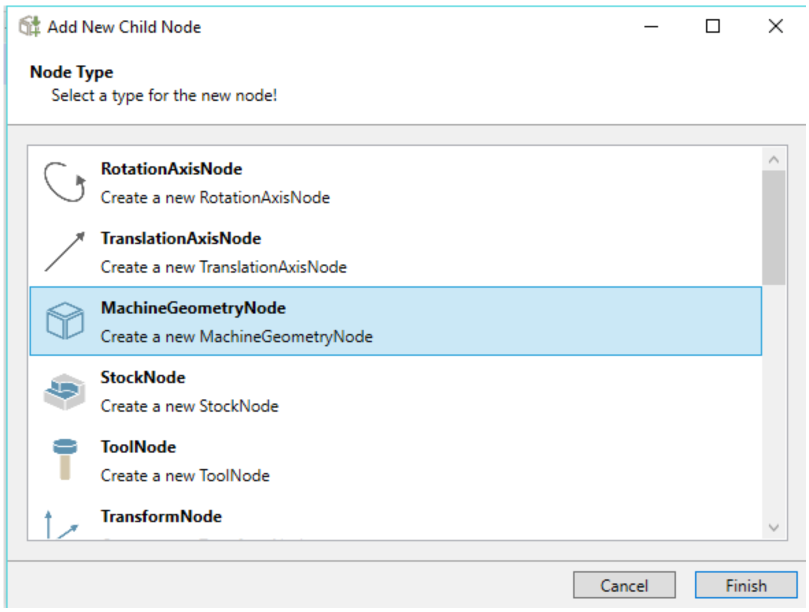


5.2.4 Integration Y axis

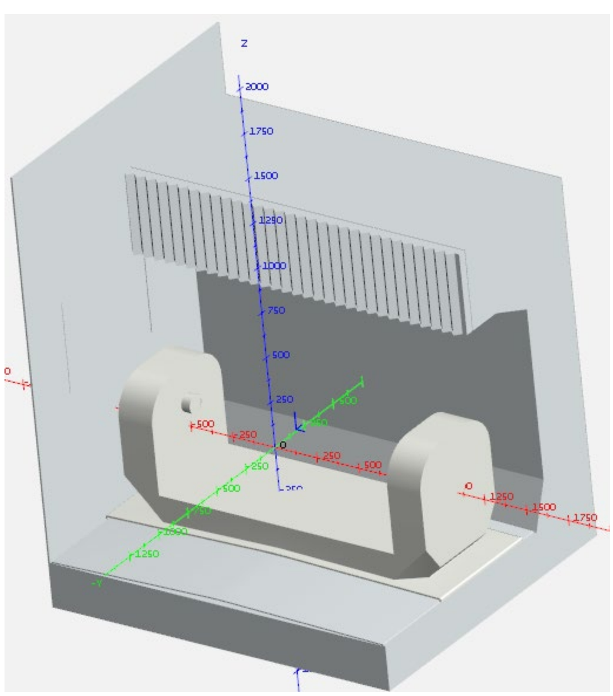
The "STL file" "xaxis" for the "Y axis" is integrated in the next step.

After selection of 

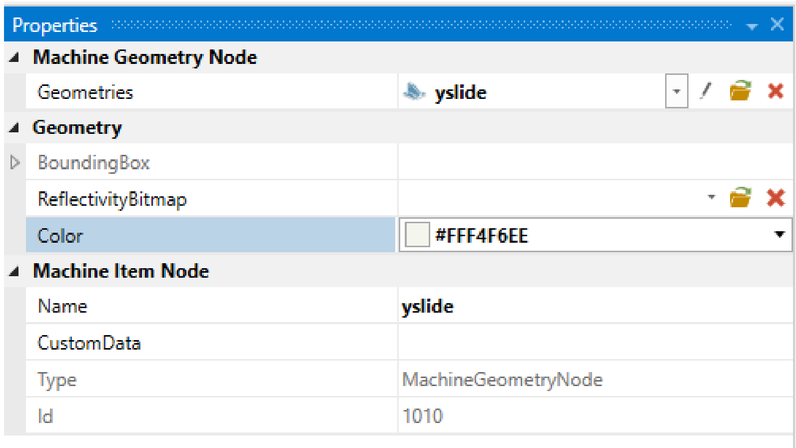
the "STL file" is integrated below the "Y axis".



"yslide" can be seen in the "Tree View".



The "STL file" with name "yslide" has been accepted.

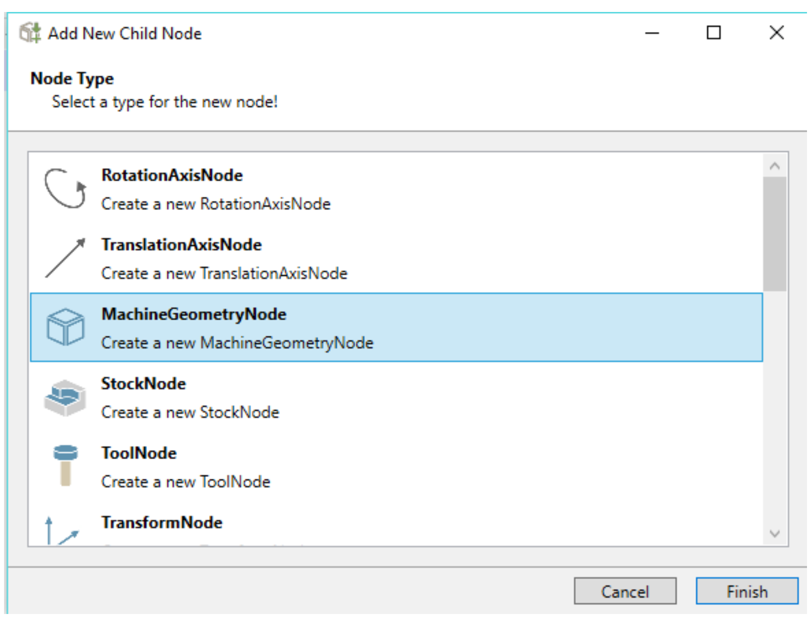


5.2.5 Integration A axis

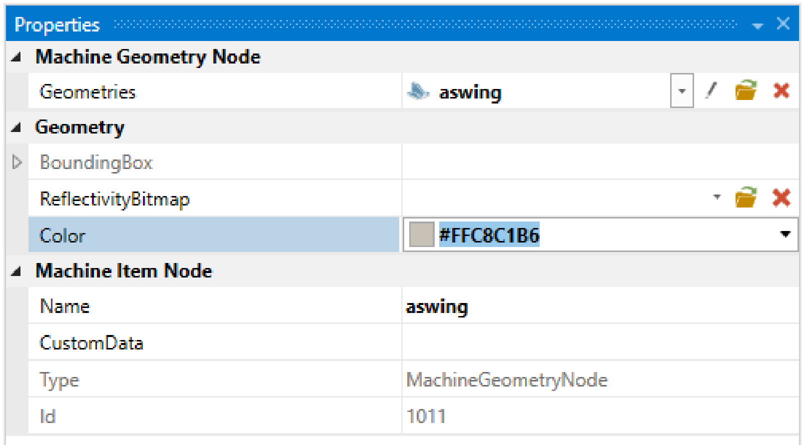
The "A axis" is integrated in the next step.

After selection of 

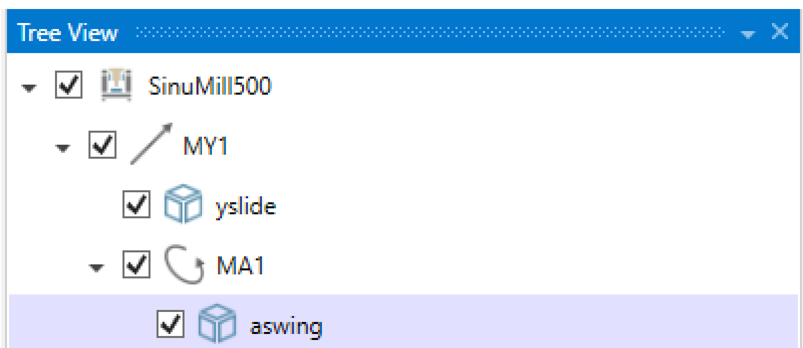
it is integrated below the "Y axis".

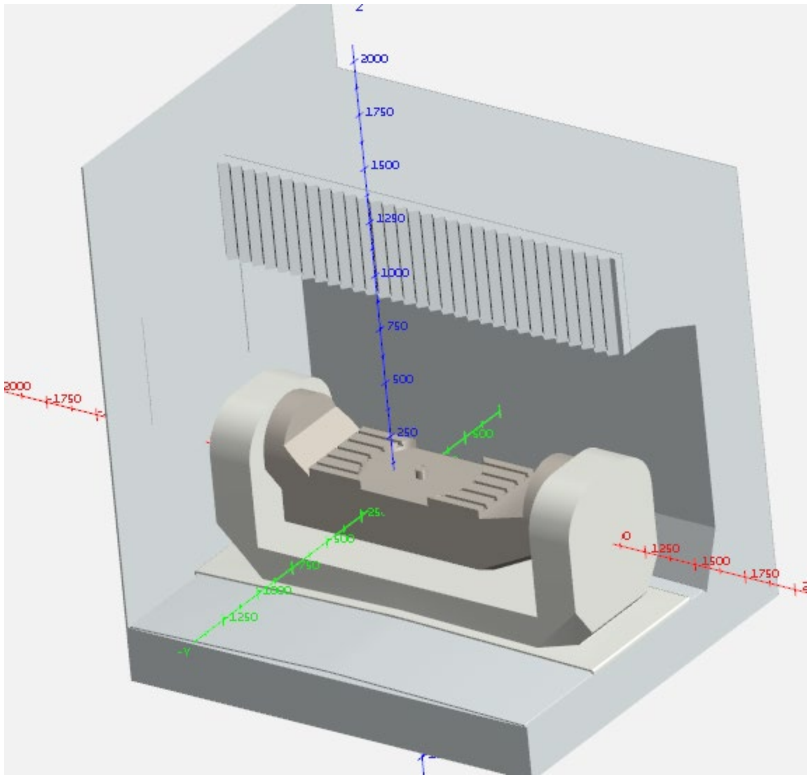


The "STL file" with name "aswing" has been accepted.



"aswing" can be seen in the "Tree View".

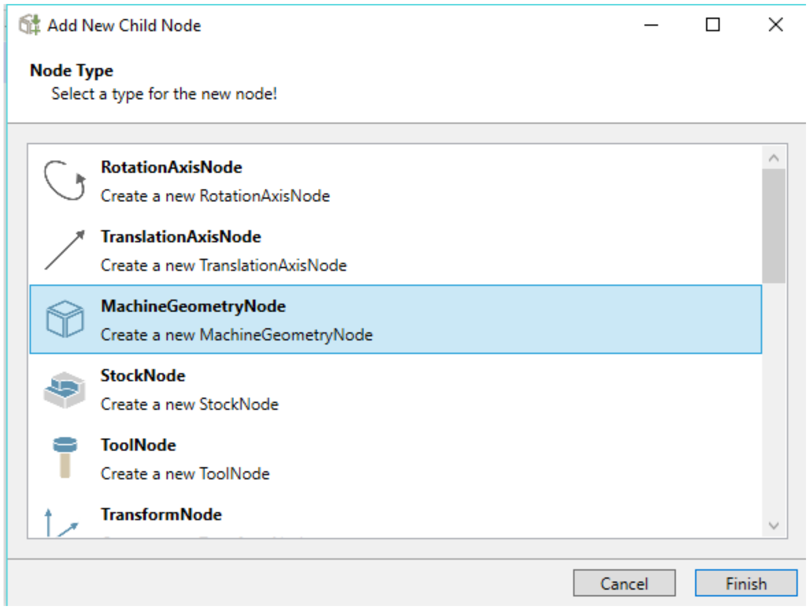




5.2.6 Integration C axis

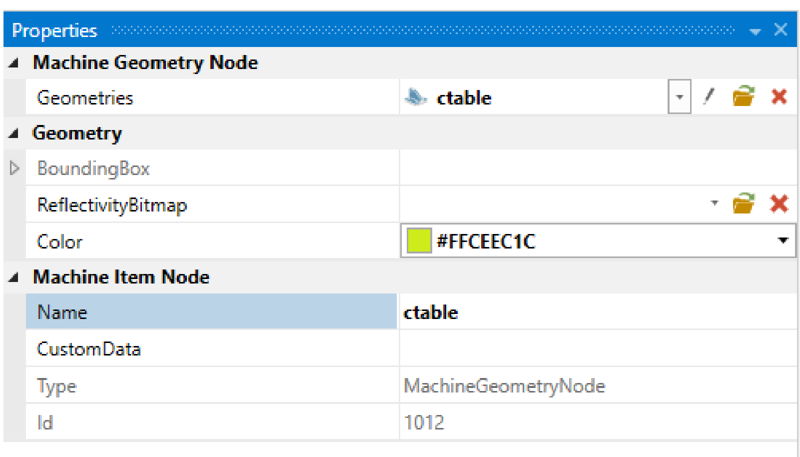
The "STL file" with the name "cslide" for the "C axis" is integrated in the next step.

After selection of 

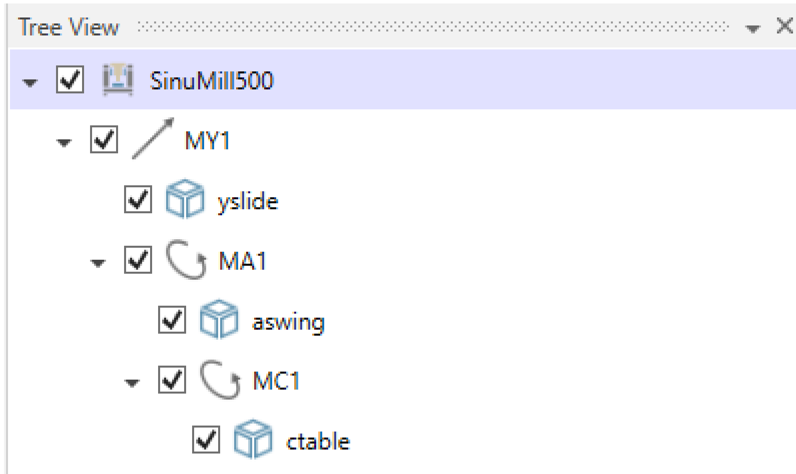


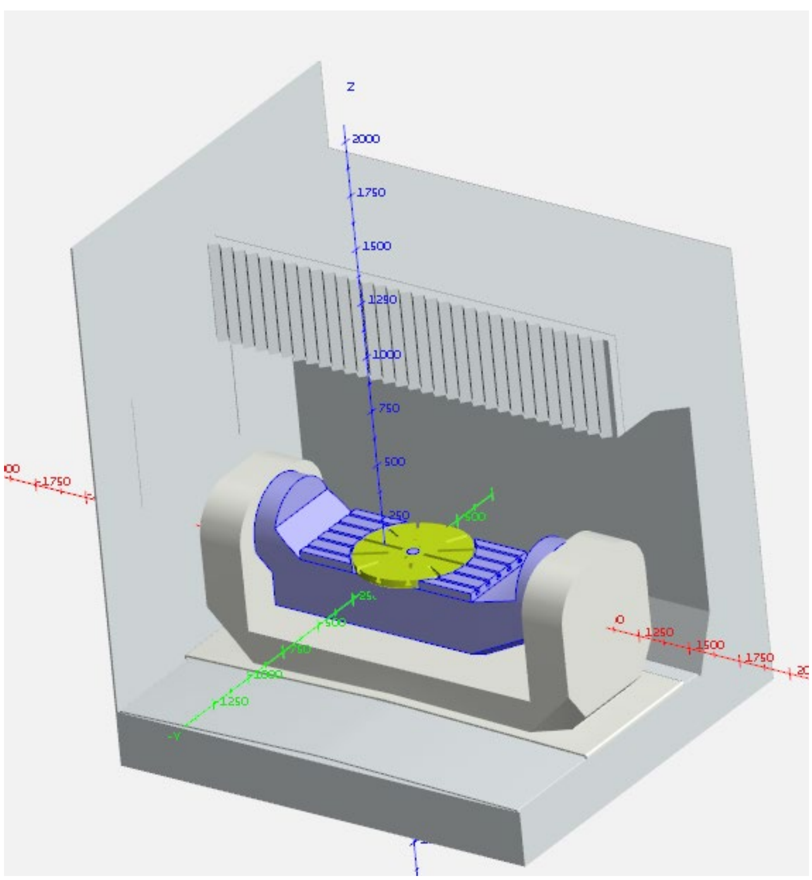
It is integrated below the "C axis".

The "STL file" with name "ctable" has been accepted.



"ctable" can be seen in the "Tree View".

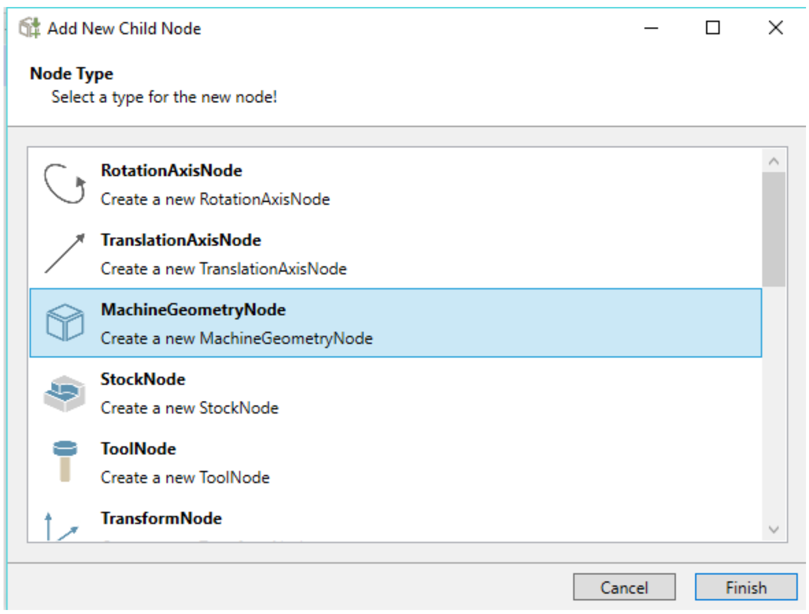




5.2.7 Integration X axis

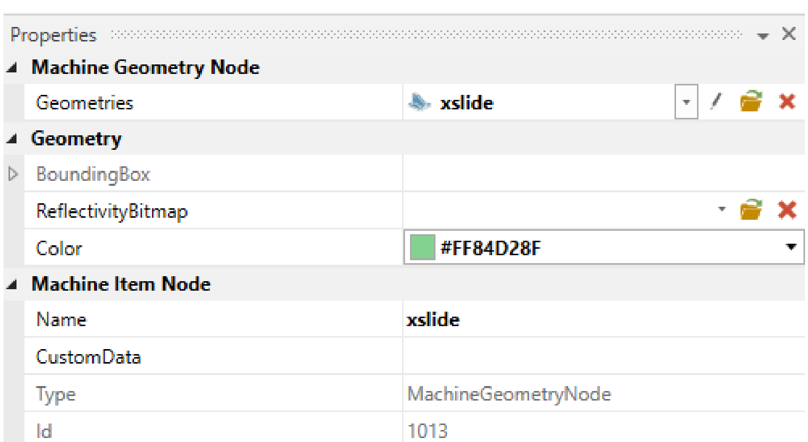
The "X axis" is integrated in the next step.

After selection of 

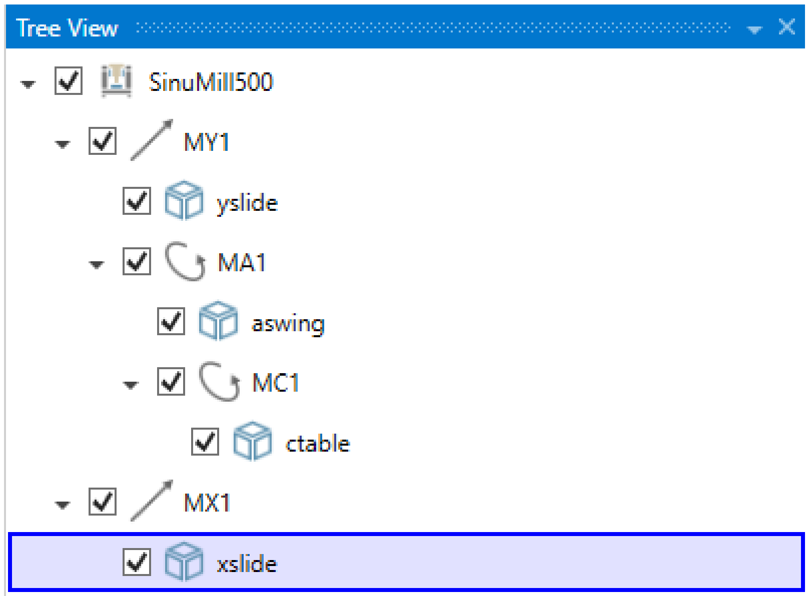


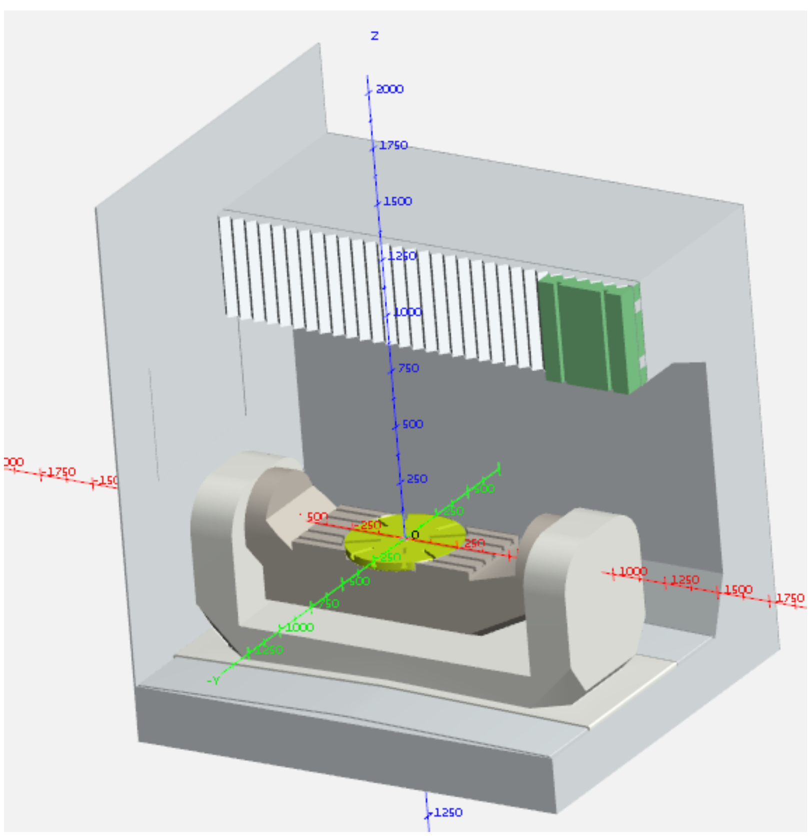
It is integrated below the "X axis".

The "STL file" with name "xslide" has been accepted.



"xslide" can be seen in the "Tree View".

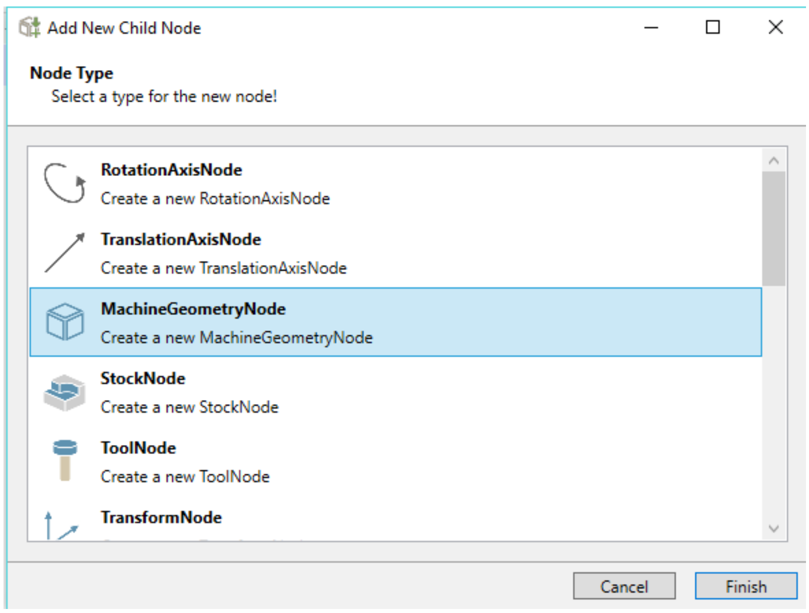




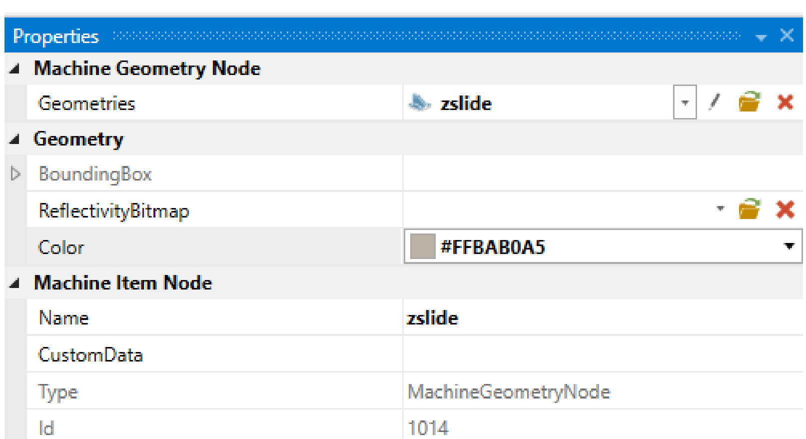
5.2.8 Integration Z axis

The "Z axis" is integrated in the next step.

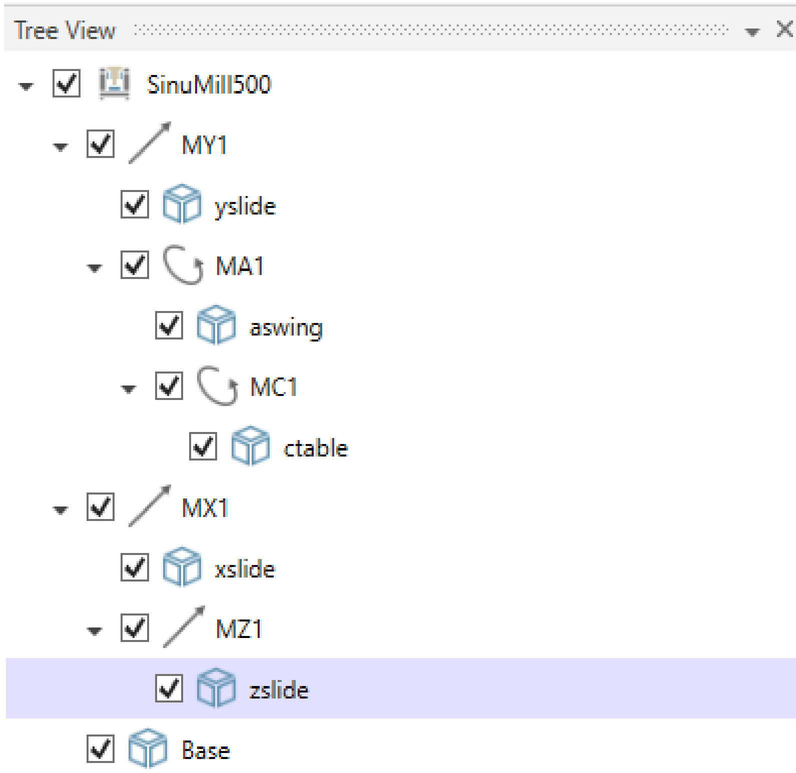
After selection of 



the "STL file" with name "zslide" is created.

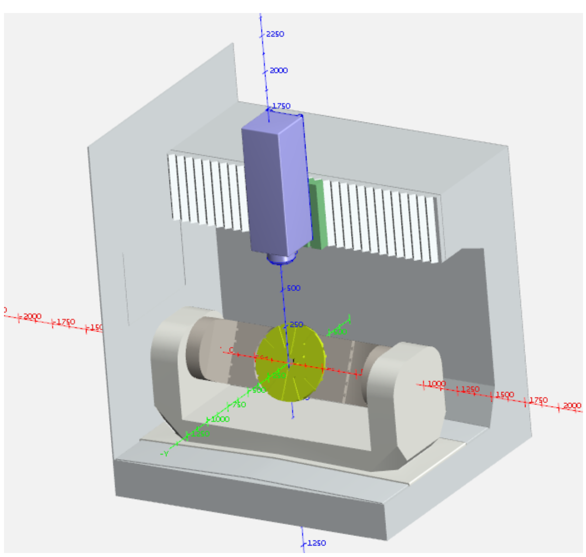


"zslide" can be seen in the "Tree View".



All machine axes are now linked with the associated "STL file".

The elements can be moved via the "Axis Control" function.

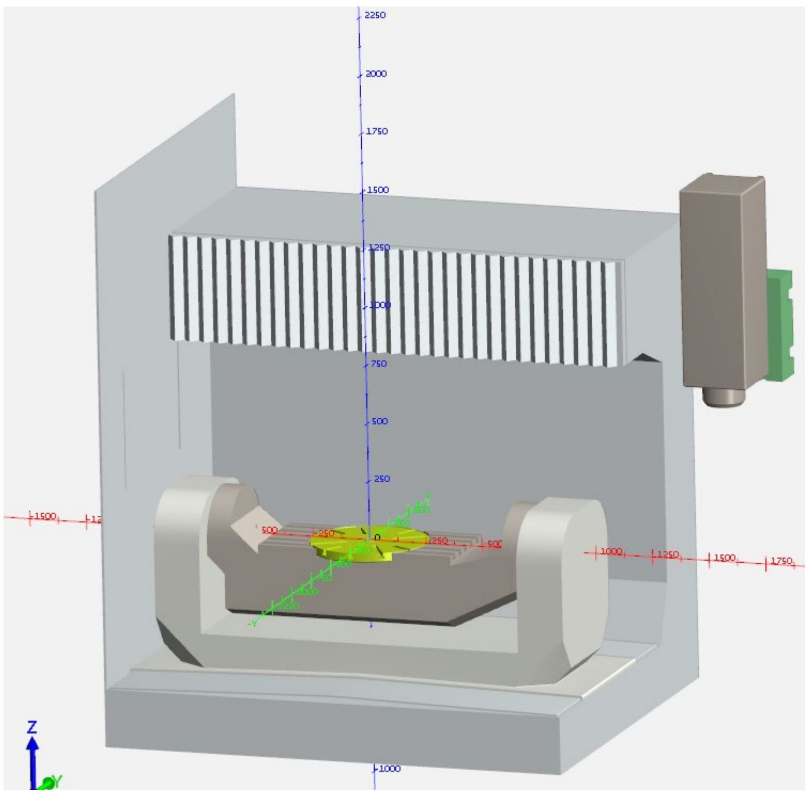


5.2.9 Optimize axis movements

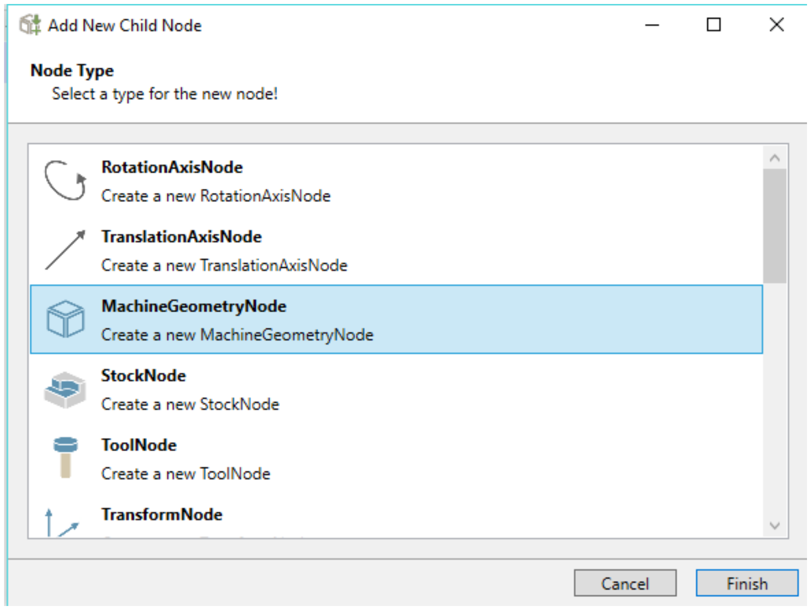
Whereby, the axes and axis movements are not yet optimized. This is done in the next steps.

X axis

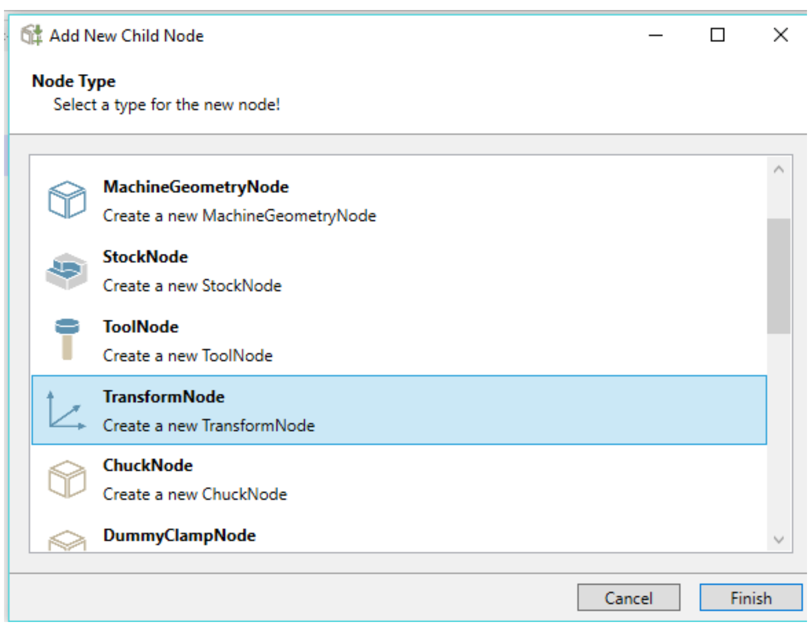
The traversal path of the "X axis" must still be optimized, because the traversal path does not yet match.



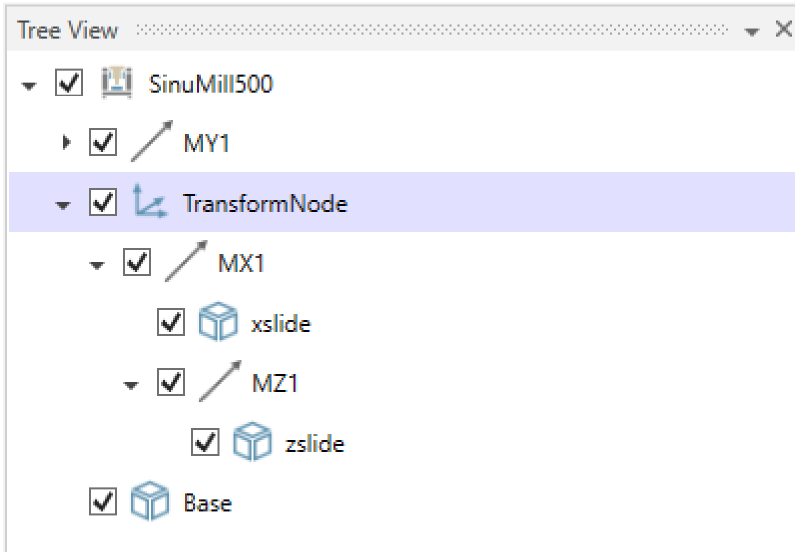
After selection of 



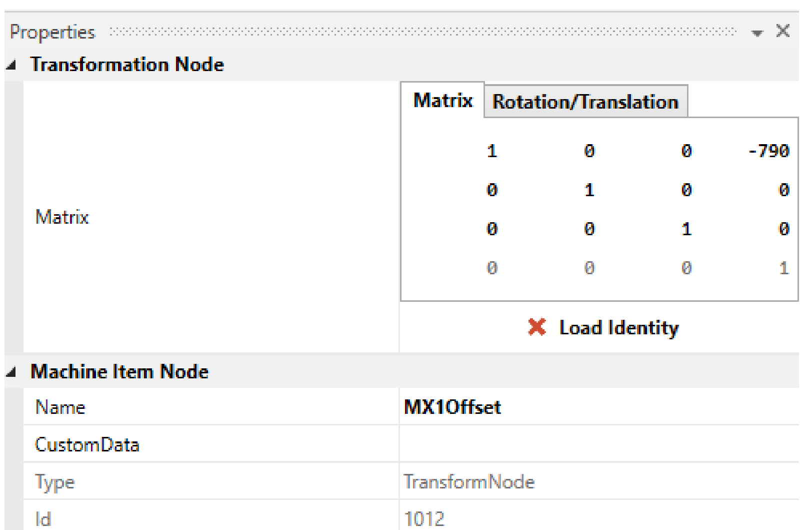
a "TransformNode" is integrated above the "X axis".



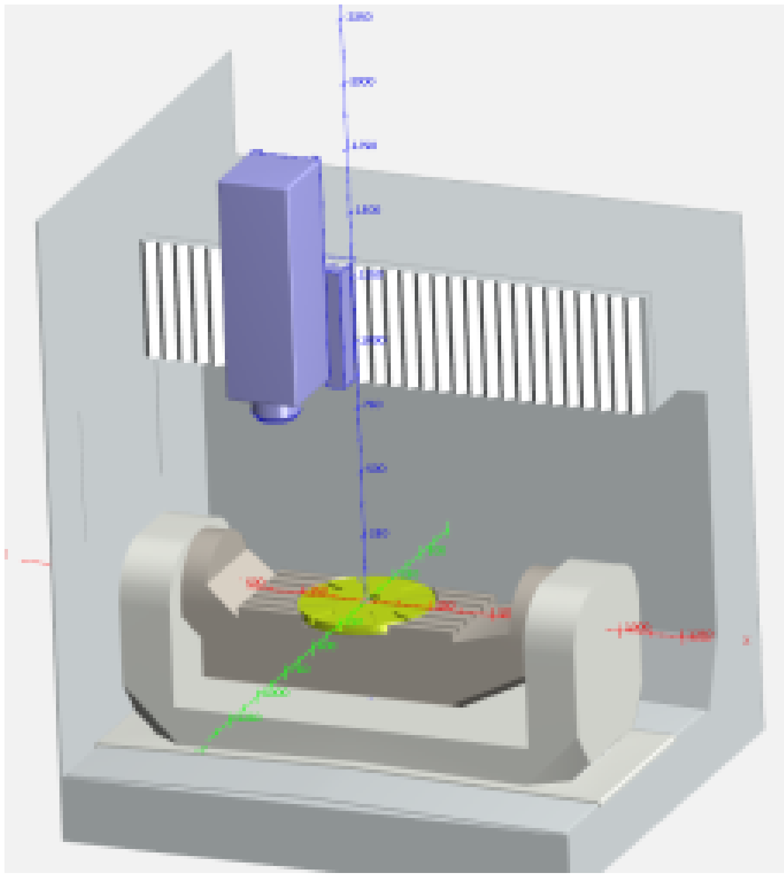
It is important that the "X axis" is located below the "TransformNode".



The value for the "X axis" offset and the name of the "TransformNode" are entered in the "Properties" input field of the "TransformNode".

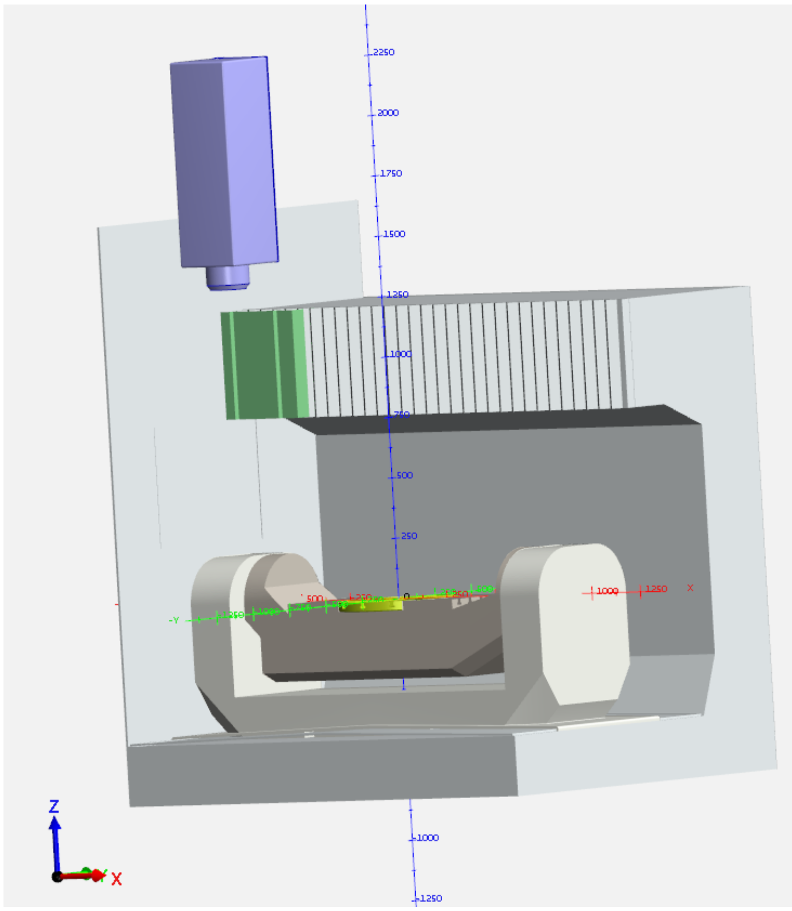


"X axis" is now correctly integrated with this offset.

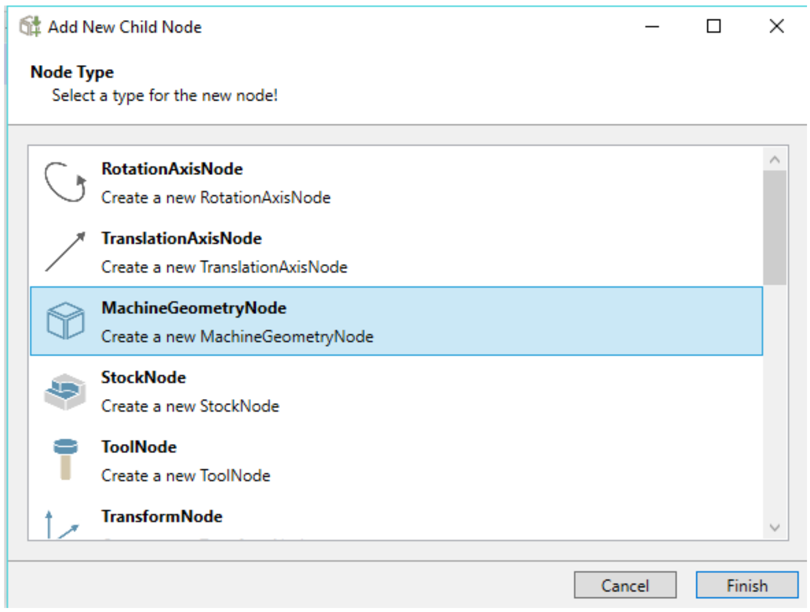


Z axis

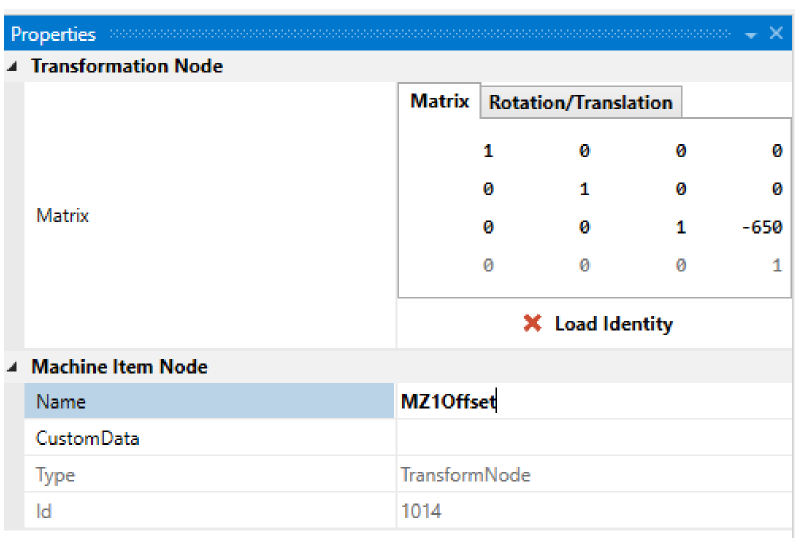
The "Z axis" is optimized in the next step. The traversal path also does not match here.

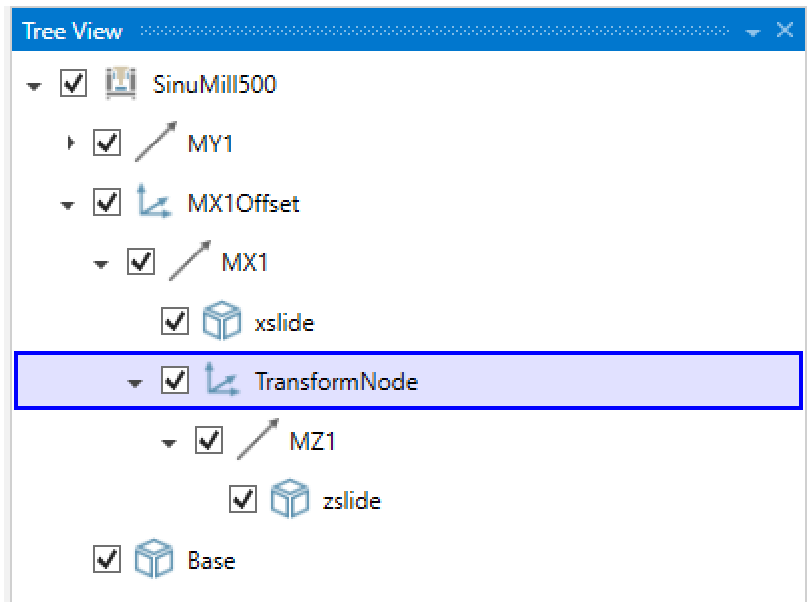


After selection of 

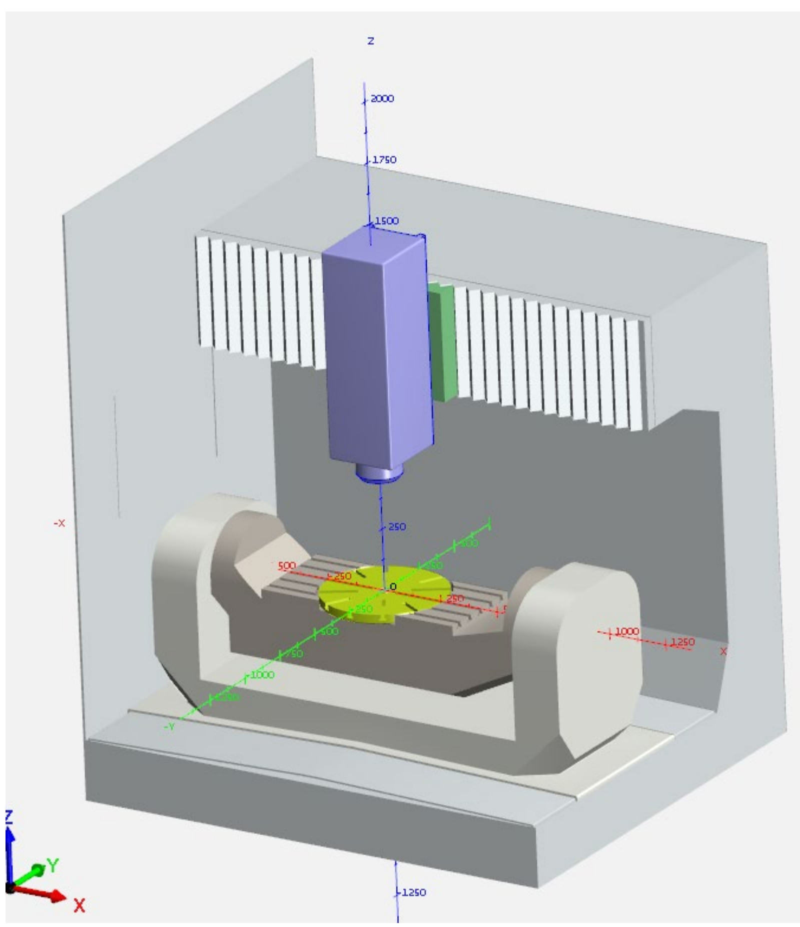


a "TransformNode" is added above the "Z axis", and the offset and name are entered there.





The "Z axis" is now correctly integrated with this offset.



5.2.10 Preparation for tools, fixture devices and stocks

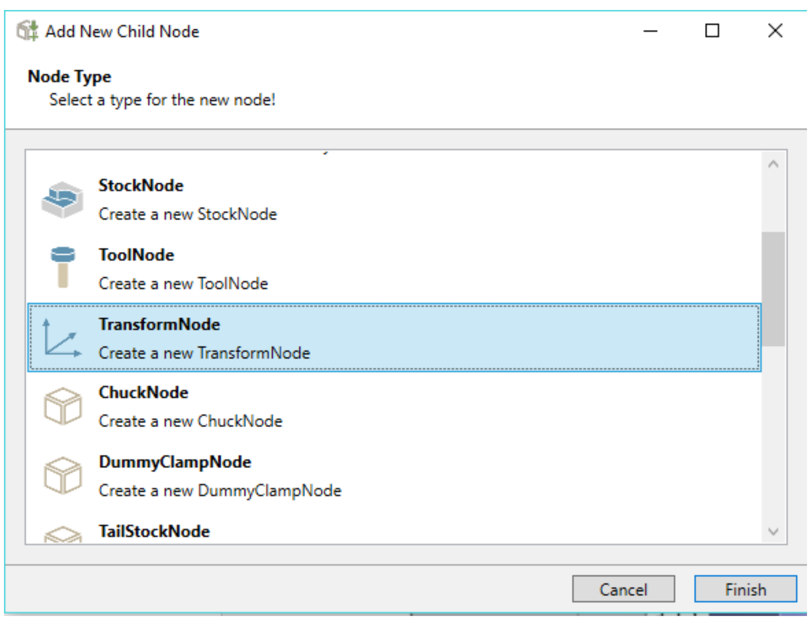
To later use the created 3D model, a routine must be performed to make the tool, the fixture devices and the stock visible in the "Protect MyMachine/ 3DTwin" or "Create MyVirtual Machine /3D" software.

This procedure must be observed. If this is not the case, the later representation of the machine does not work correctly.

Tools and fixture devices are not assigned directly to axes.

The steps listed here are not explained in detail, because they must be created for the system in this sequence and syntax.

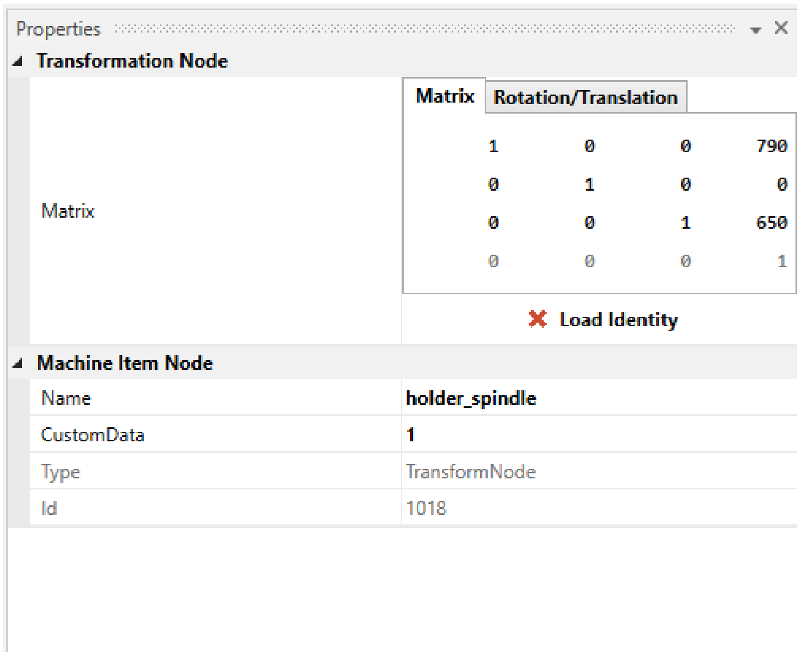
After selection of 

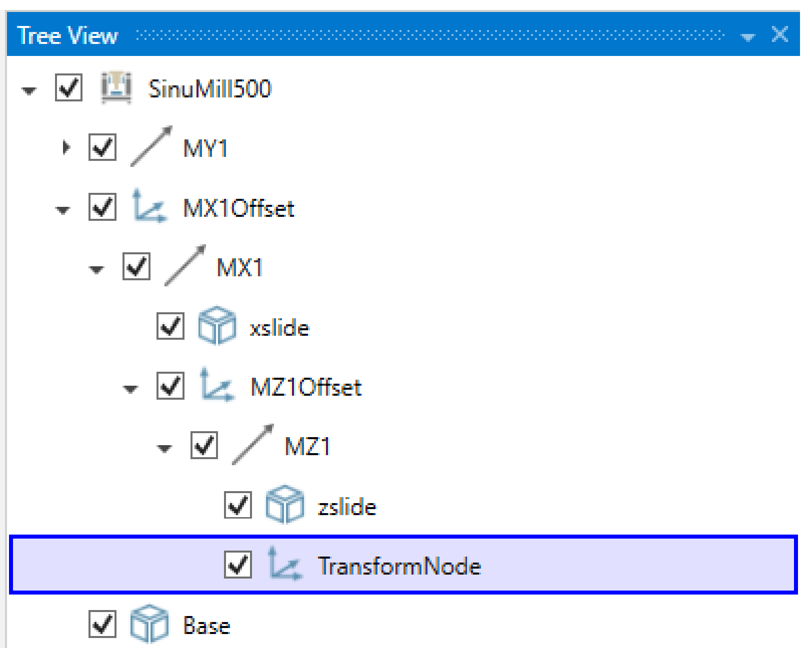


a "TransformNode" is added below the "Z axis".

Starting at the existing zero point, a "Dummy" offset must now be defined in which any fixture devices and stock may need to be considered later.

The values are entered after offset to the correct position.



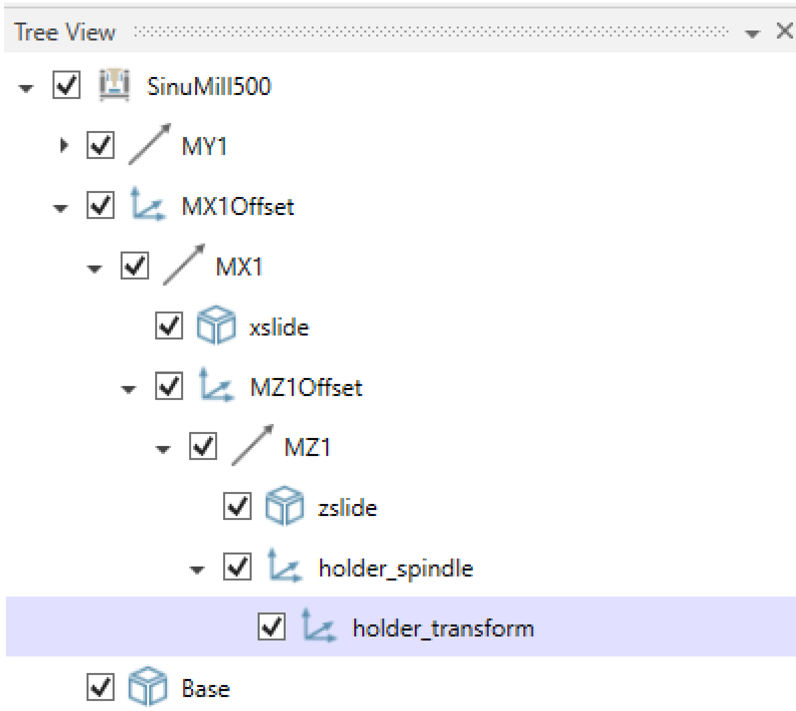


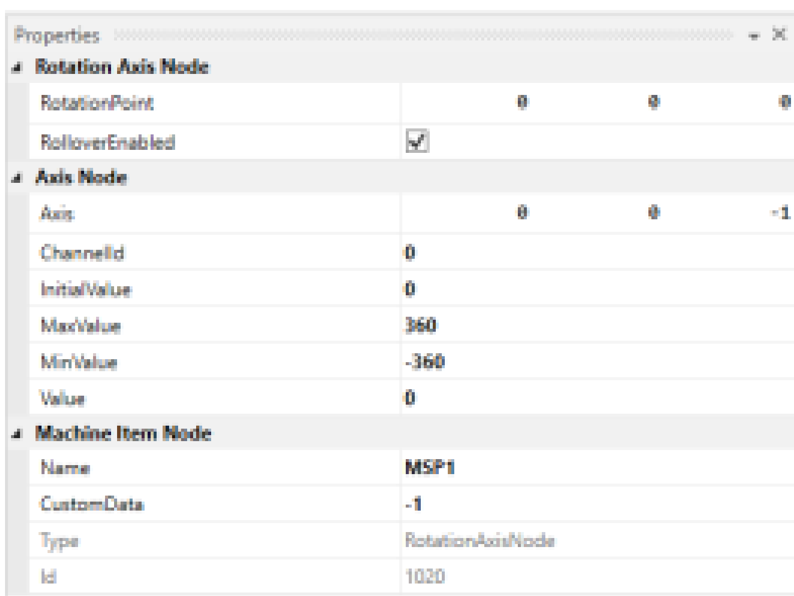
They describe the position of the spindle nose.

The designation "holder_spindle" is specified by the system and must be entered in this notation so it can be used later.

A "Dummy" offset must also be programmed here so that the spindle nose is later the reference point for the tool.

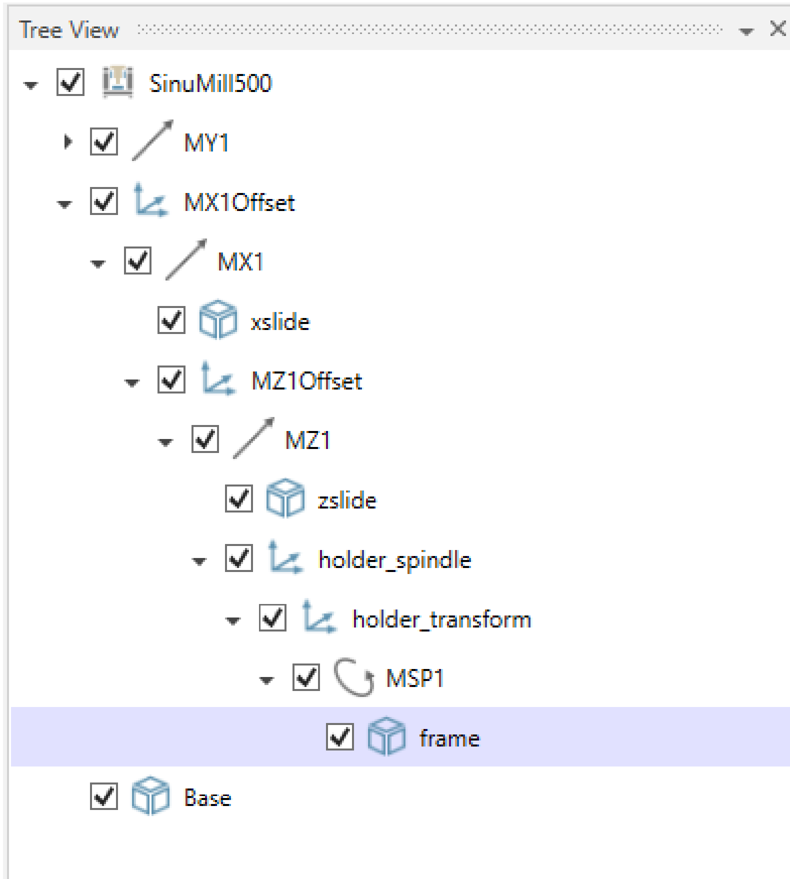
A "holder_transform" and the tool spindle "MSP1" must be added in the next step.

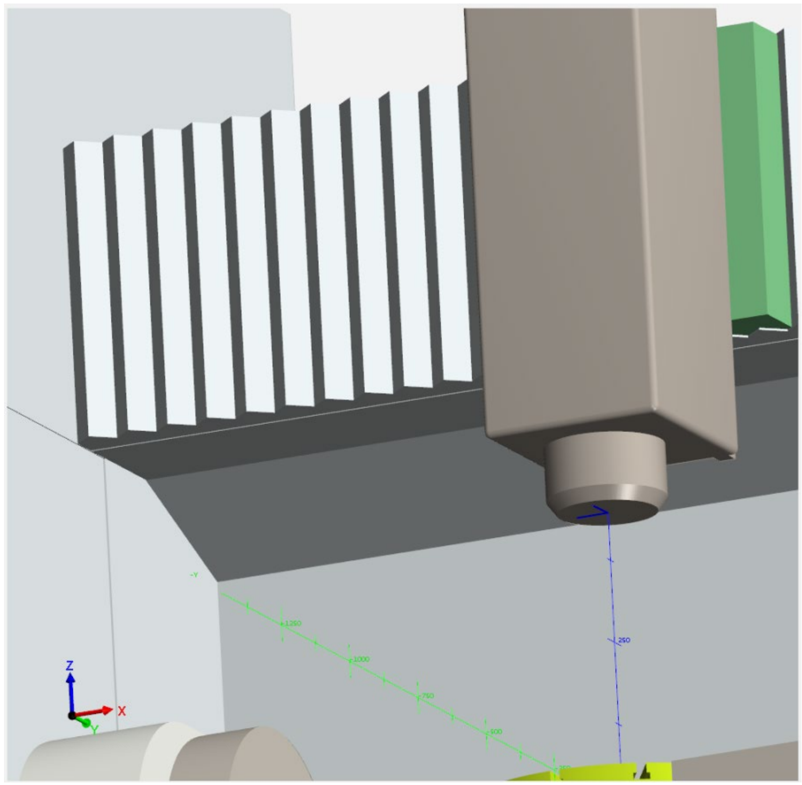




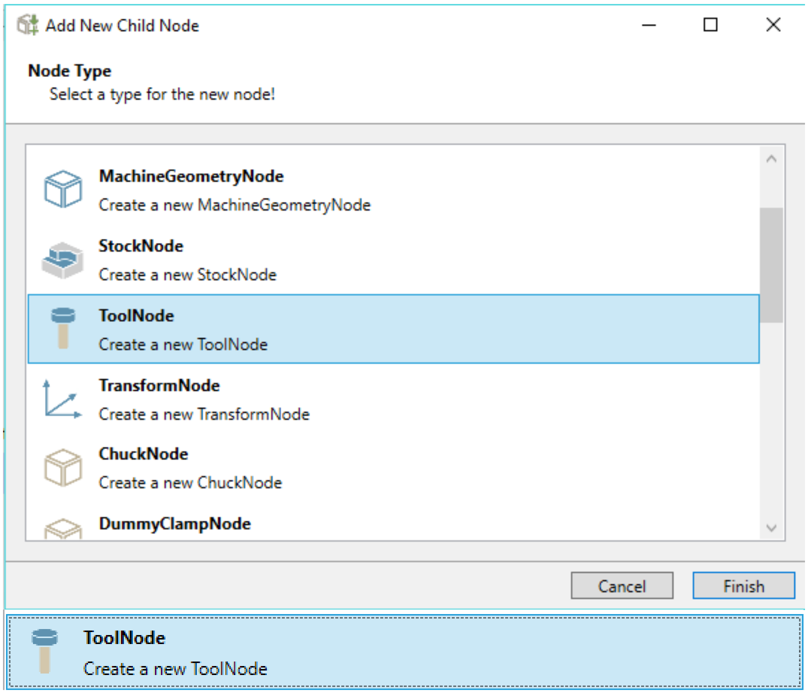
The spindle name of the later machine must be used here.

The correct rotation direction can be checked with "STL file" "frame".

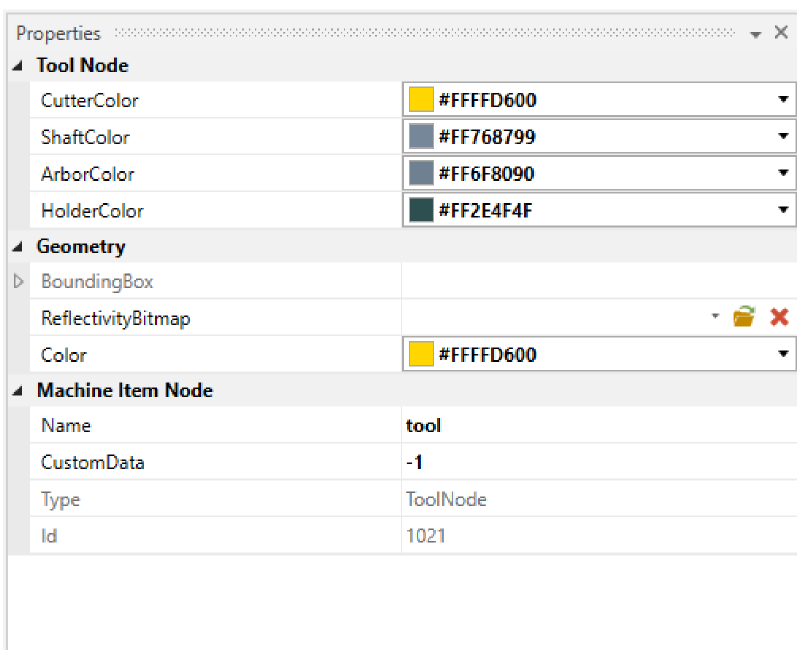


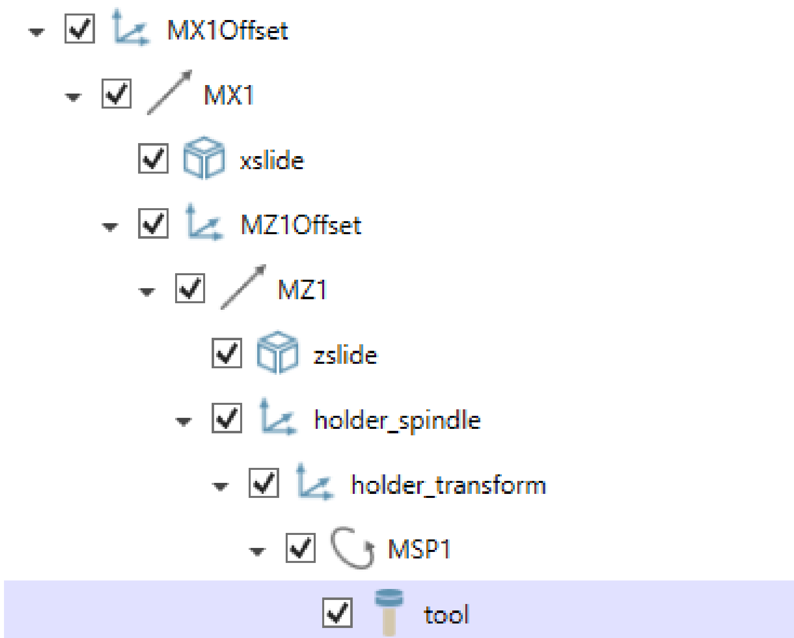


After selection of 



at "MSP1" spindle a tool is defined with "ToolNode" of the "placeholders". Caution: In this case, observe the "lowercase" notation.

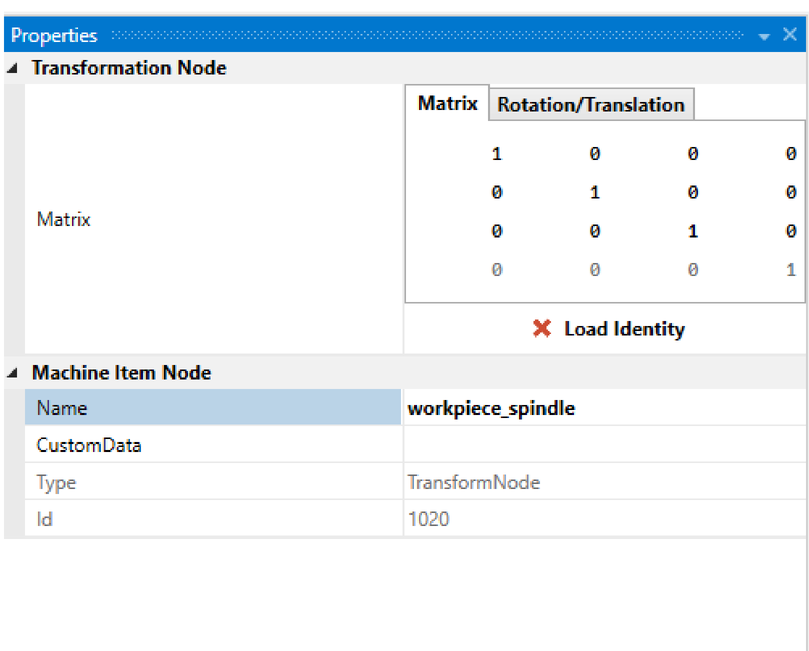


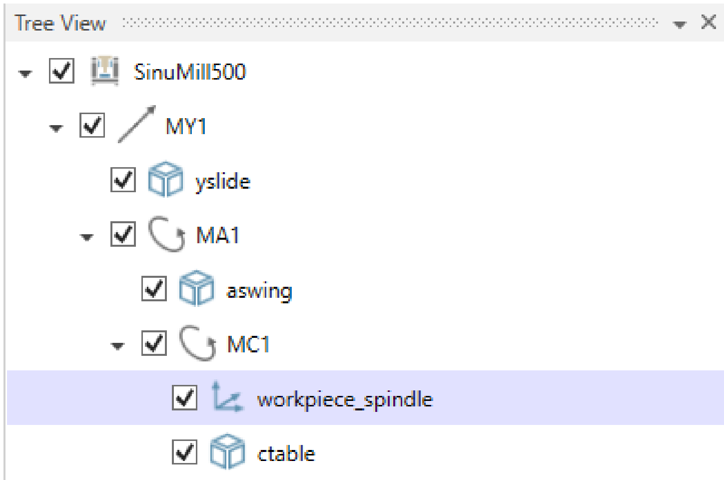


5.2.11 Defining workpiece area

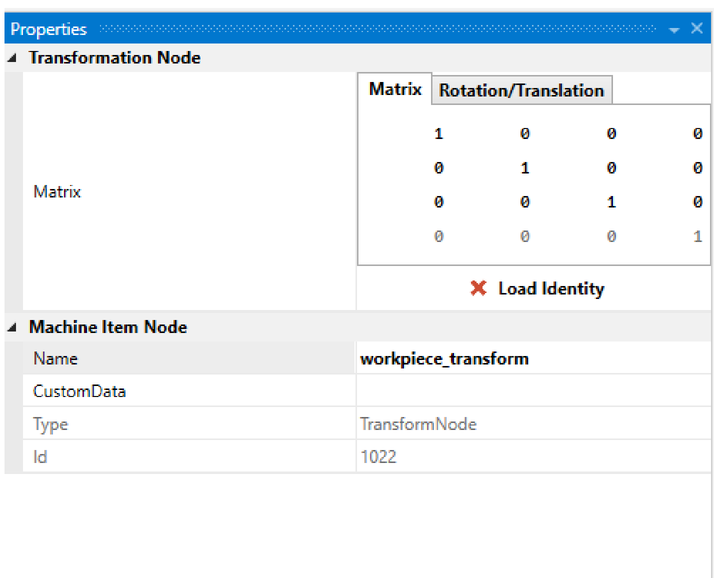
In the next steps, the area is defined for the workpiece.

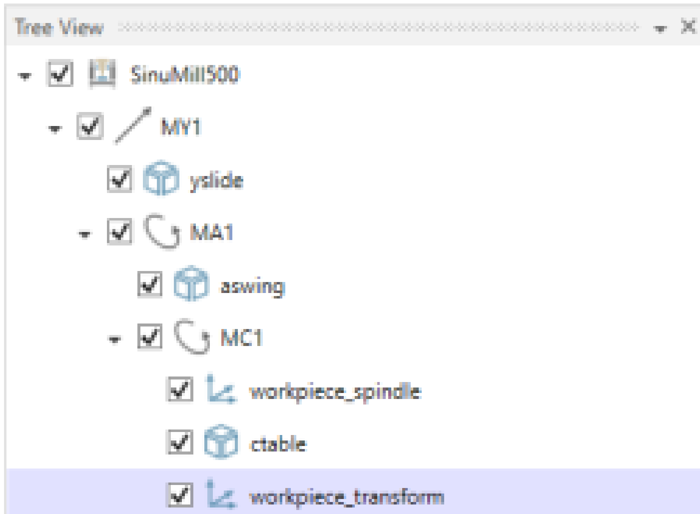
A "TransformNode" with the name "workpiece_spindle" is added below the "C axis".



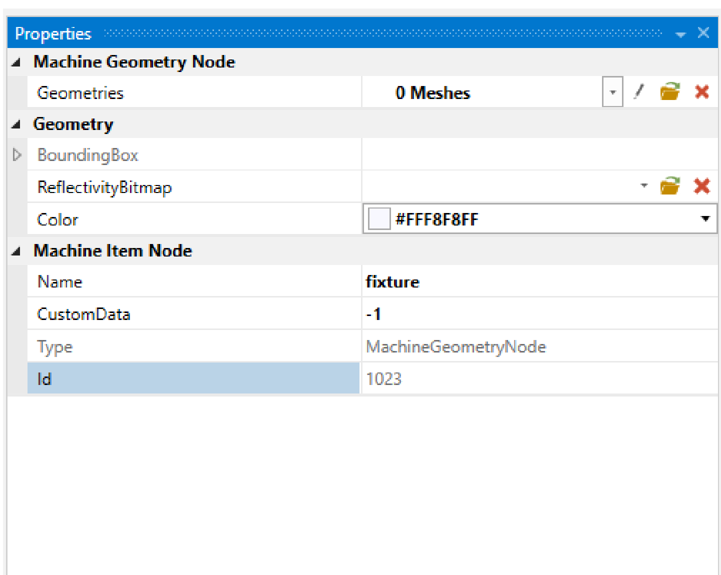


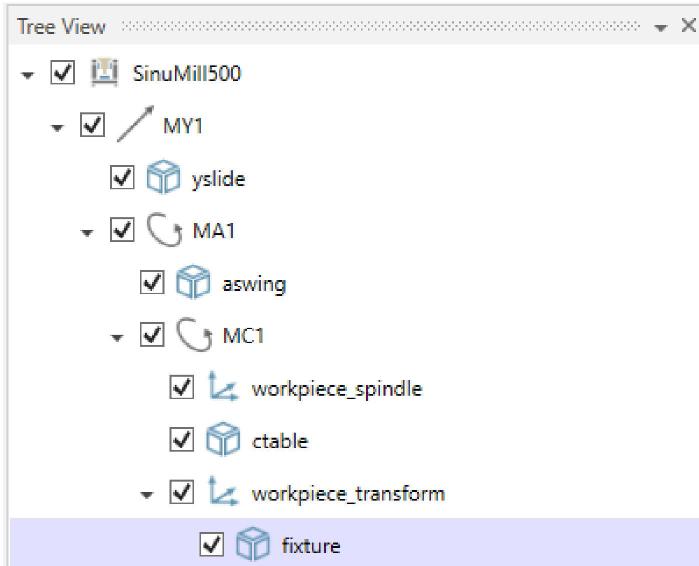
Another "TransformNode" with the name "workpiece_transform" is added below the STL file for the "ctable" C axis.



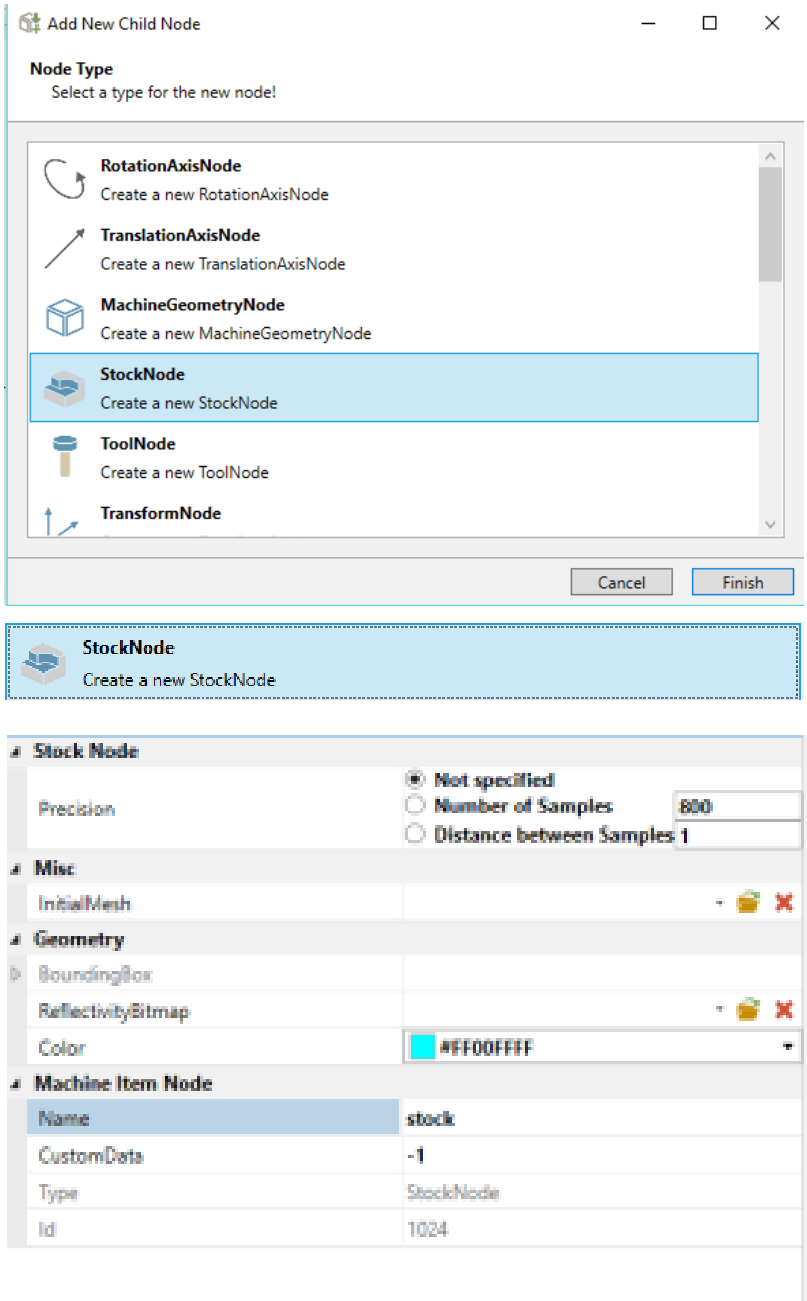


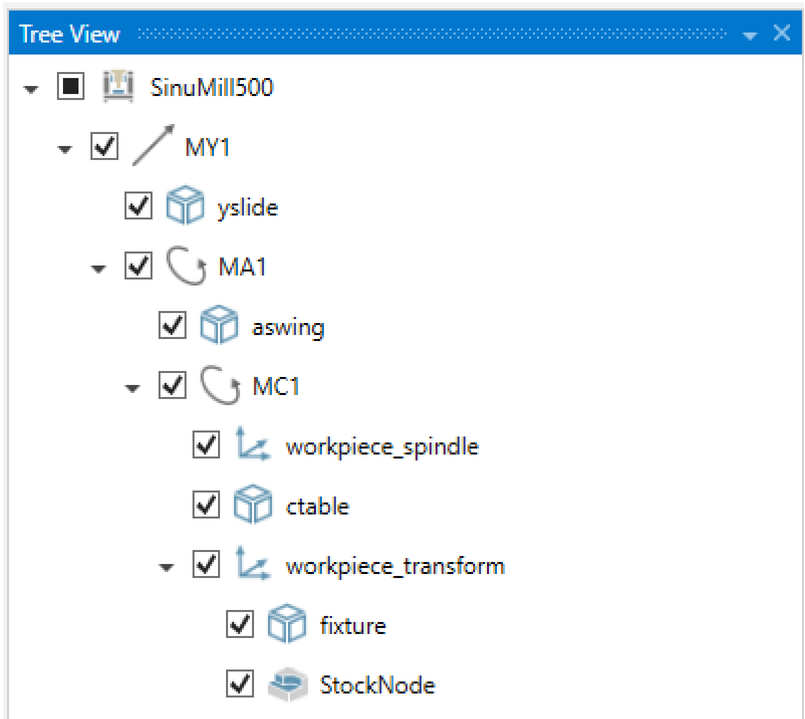
A "placeholder" for the fixture devices with the name "fixture" is added at "workpiece_transform".




Tree View
SinuMill500
MY1
yslide
MA1
aswing
MC1
workpiece_spindle
ctable
workpiece_transform
fixture

A placeholder for the stock with the name "stock" is integrated at "MachineGeometryNode" with the name "fixture".





5.2.12 Fast protection for workpiece setups

For this to work, the machine model must have a special machine geometry node with name "protection_area" below the workpiece transform. This node must contain a special crafted mesh, which is used to extract a contour that is then extruded with the given height value to create the initial protection volume.

The workpieces of the provided workpiece setup are then subtracted from this protection volume to create the final mesh. The resulting mesh is positioned in relation to the workpiece transform of the machine, so that it can be added as fixture geometry without additional positioning.

The mesh that is used to extract the contour must have the following properties:

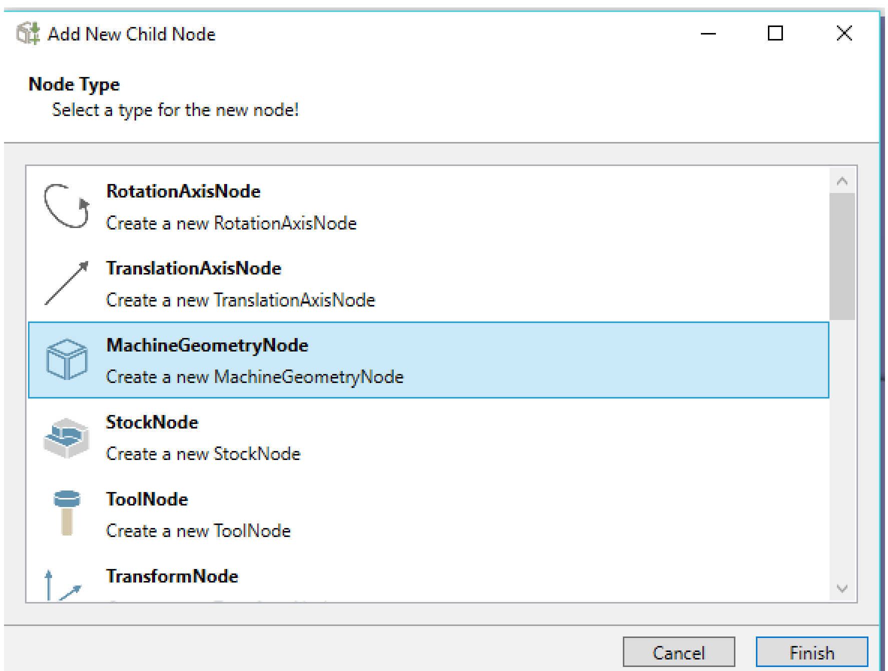
- The desired contour is generated from all mesh points in the XY plane at height Z = 0.
- There may not be mesh points in that plane that are not part of the desired contour.
- The extraction algorithm chooses a single starting point and then walks to the next nearest neighbor (euclidean distance) not yet part of the contour until all points are visited. If the desired contour is a circle or a rectangle, then this should automatically be true. If the contour is more complex, then this requires a bit of care when crafting the mesh.
- If the contour needs to be transformed to be placed on the table of the machine model, this can be achieved by adding a transform node between the workpiece transform and the "protection_area" node.

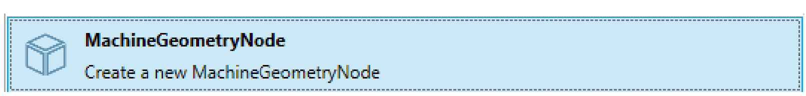
Such a mesh can be generated easily in a CAD program by drawing the contour in the XY plane and then extruding this contour by some arbitrary value e.g. 5 along the Z axis. The top and bottom need to be kept open. This final surface can then be exported as a mesh that should fulfill the requirements.

All necessary "nodes" to integrate this machine in "Protect MyMachine/ 3D Twin" or "Create MyVirtual Machine /3D" are created with it.

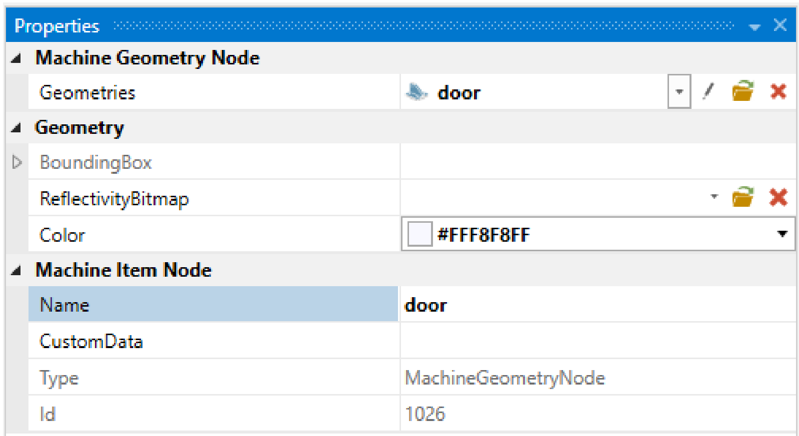
5.2.13 Integration *.stl door, housing and controller

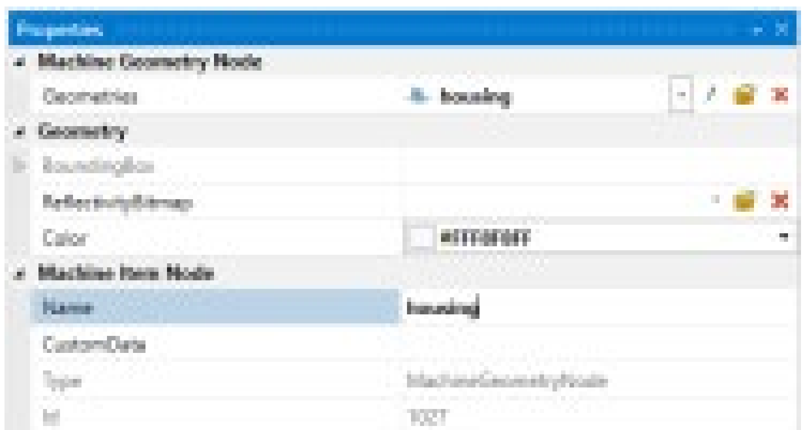
In the last step, after selection of 

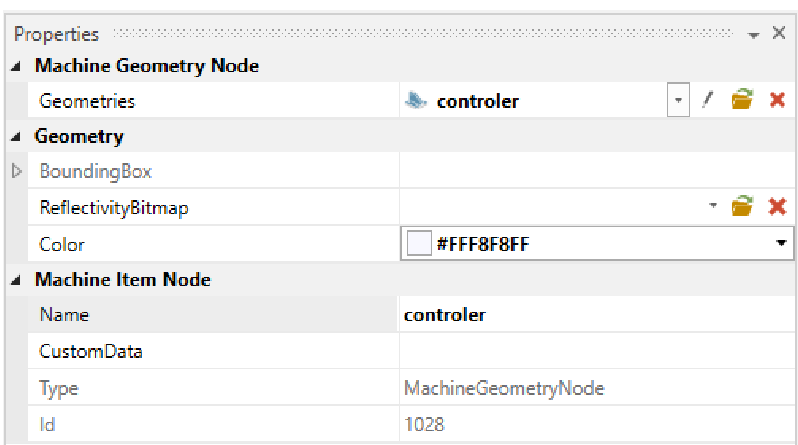




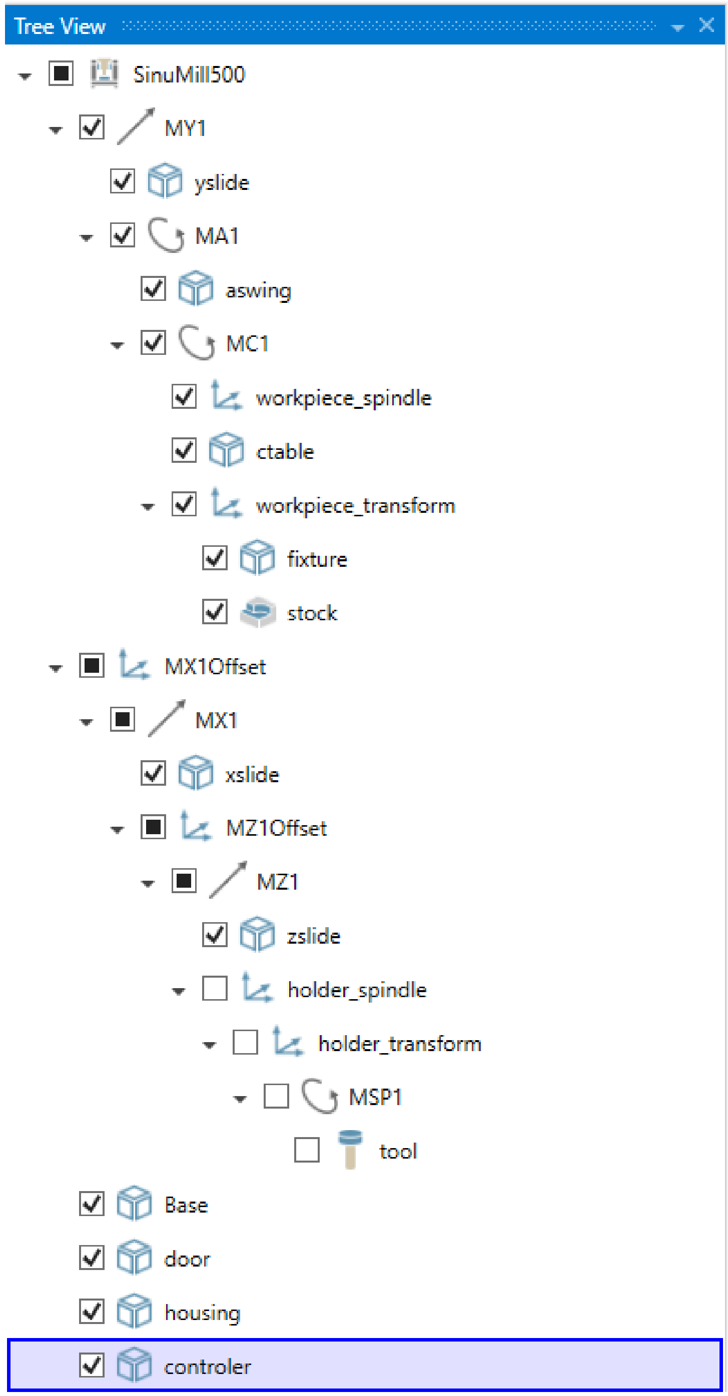
"MaschineGeometryNode", the "STL files" are integrated for the door, housing and controller.

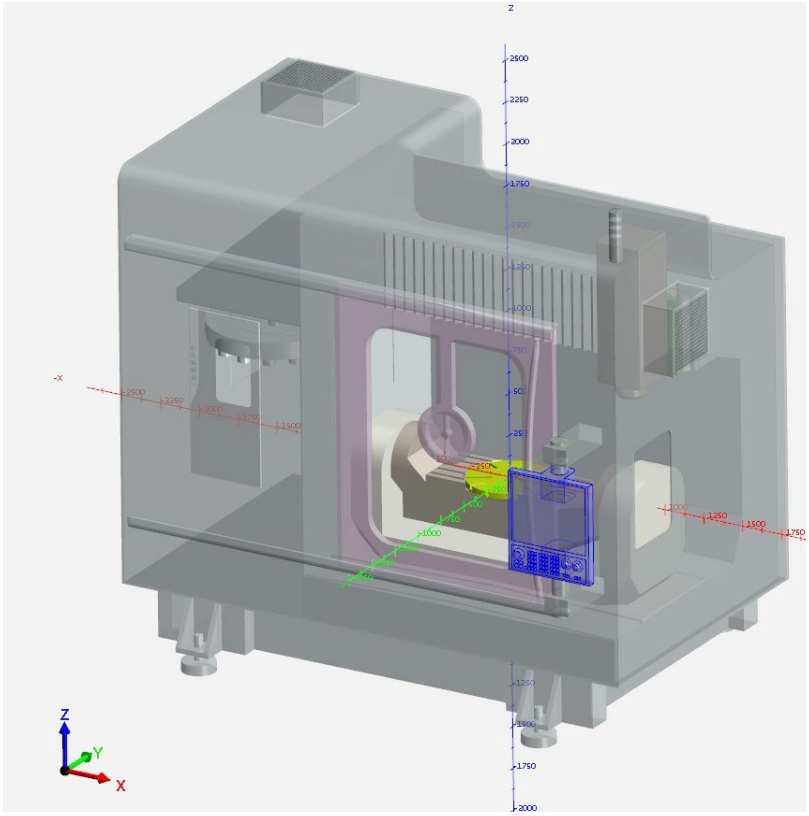

Properties
Machine Geometry Node
Geometries
door
Geometry
BoundingBox
ReflectivityBitmap
Color
#FFF8F8FF
Machine Item Node
Name
door
CustomData
Type
MachineGeometryNode
Id
1026


Machine Geometry Node
housing
Geometry
Machine Item Node
housing


Properties
Machine Geometry Node
Geometries
controler
Geometry
BoundingBox
ReflectivityBitmap
Color
#FFF8F8FF
Machine Item Node
Name
controler
CustomData
Type
MachineGeometryNode
Id
1028

The machine is completed in the "MachineBuilder" software.



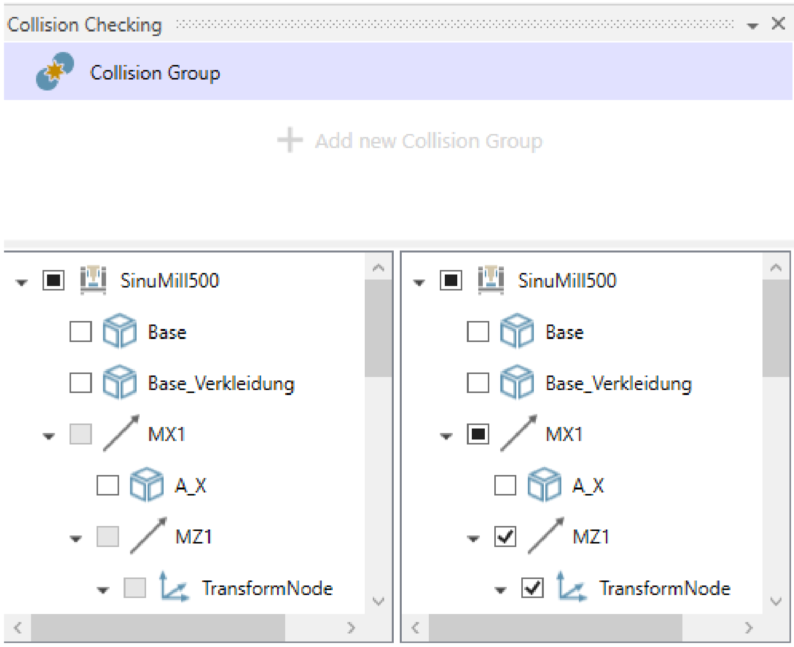


5.2.14 Adding Collision Pairs in Machine Builder

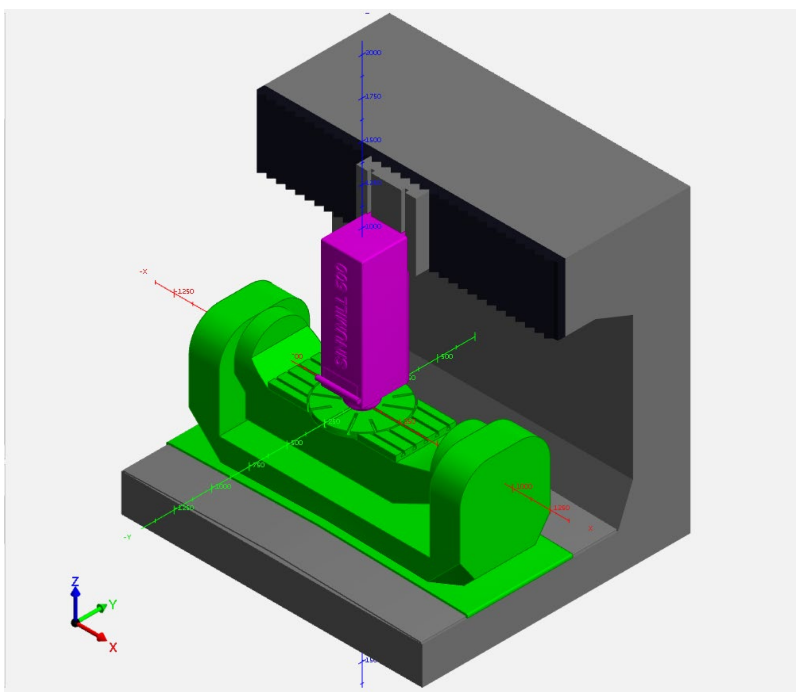
For the collision checking utility, the collision pairs must be added in the machine builder page.

After selection of Collision Checking 

in the Toolbox; Collision Checking page opens on the right down corner of the application.



By Pressing "Add New Collision Group" OEM can add new collision groups.



The safety distance of the collision group can be determined in the properties page.

Properties	
Collision-checked geometries	
Left	MY1
Right	MZ1
Misc	
Name	**Collision Group**
Custom	
SafetyDistance	**3**

Note

It is important to add Collision Groups in Machine Builder, because it is going to be used in "Protect MyMachine /3D Twin" and "Create MyVirtual Machine /3D" to make collision avoidance.

5.2.15 Adding Layers in Machine Builder

Layers are defined to make the machine parts visible/invisible in the runtime of 3D applications in order to observe the machining process better.

To define a layer, the "Custom Data" property of any node shall be edited with the name of the Layer. The geometries below that node will also be added to that Layer unless another Layer is defined in below geometries.

Machine Item Node	
Name	Housing_out
Type	MachineGeometryNode
Subtree size	1
Is collision checked	
In collision groups	
Collision subtree size	0
Custom data	Housing

5.3 Kinematic Head Change and Scripting Support

Note

Only available for:

Protect MyMachine /3D Twin from V1.3.1

Create MyVirtual Machine /3D from V1.3.2

It is possible to define kinematic changes by adding special script to the machine model xml file manually.

Via this script, it is possible to:

- Support kinematic head changes
- PLC driven/activated components

Collisions caused by appearing geometry/linked axis will be reported with the next position simulation result. This requires certain synchronization to make sure actions are applied before new lookahead is checked. This can be probably achieved with a combination of "G4" (dwell) and "STOPRE". Otherwise, it is undefined when the geometry changes are taken into account.

In the script, you should:

- Define a signal, which will trigger an action on a value change.
- Define an action for each value of the Signal source.
- Define the list of commands inside an action.

5.3.1 Structure of the script

signals

signal attributes:

- `name`
  Unique name among signals. Used by actions to reference this signal.
- `source`
  BTSS variable path defining where the signal value should be read from. The variable needs to be directly convertible to an integer.
- `default`
  The default value to be used until values from the control arrive.
- `(Optionally) alias`
  Alternative for source in the CMVM use-case.

Multiple signals can be defined.

Source has to have a default value, which will be triggered on application startup.

actions

actions define a list of commands that will be executed when the referenced signal takes on the specified value. Define an action block for each value of the signal source.

The commands are executed in the defined order. This might be relevant if components are attached or detached, e. g. 'linking' commands need to follow 'attaching' commands.

- sub element `trigger`
  - `signal`
    Has to match a previously defined signal name.
  - `value`
    Defines the signal value for which the commands get executed once.
- sub element `commands`
  With at least one `command` child. Otherwise, the action is ignored.

commands

commands define what work is done when an action is triggered.

command attributes:

- `name`
  Specifies the type of command / what kind of work is to be done.
- `parameters`
  Contains the list of parameters. The required or possible parameter names are depending on the used name.
- `parameter`
  - `name`
    Name of the parameter. Meaning depending on specific command. Might not need to be unique for a command list.
  - `value`
    The parameter value.

There are predefined commands and parameters that can be defined in an action, as below:

- `AttachComponent` command
  Moves a kinematic node below a new parent which is a specific other kinematic node.

  Parameters:
  - `component`
    The component to move. Refers to a unique node name of the kinematic tree.
  - `target`
    The new parent for the component. Refers to a unique node name of the kinematic tree.
- `DetachComponent` command

  Moves a kinematic node outside of the active kinematic tree. Afterwards it doesn't have a parent node. The entire component is excluded from visibility and collision checking until it gets re-attached.

Parameters:

– `component`
  The component to detach. Refers to a unique node name of the kinematic tree.

  This can be specified multiple times to be done for multiple components.

- `LinkAxis` command

  Links the value of a 'follow' axis node to a 'lead' axis node's value with an optional translation ratio.

  Parameters:

  – `Axis`
    The name of the axis that should follow. Refers to a unique node name of the kinematic tree.

  – `LeadingAxis`
    The name of the axis that should lead. Refers to a unique node name of the kinematic tree.

  – `Ratio`
    Optional. The ratio between lead and follow axis value `followValue = ratio x leadValue`.

- `UnlinkAxis` command

  Removes all links where this axis node is the follower. Doesn't change the current axis value. Future changes of the 'leading axis' value are not followed.

  Parameters:

  – Axis
    The name of the axis to unlink. Refers to a unique node name of the kinematic tree.
    This can be specified multiple times to unlink multiple axes.

- `LinkToolNode` command

  Links a tool mountstation to a specific tool magazine place.

  Parameters:

  – `ToolNode`
    The name of the tool mountstation to link Refers to a unique node name of the kinematic tree.

  – `TargetToolNode`
    The tool magazine location to use. Needs to be a name conforming to normal tool mountstation naming, see also chapter *7.3 Naming conventions for mount stations*.

- `UnlinkToolNode` command

  Removes the links of a tool mountstation to any magazine place. Also removes the currently active tool geometry.

  Parameters:

  – `ToolNode`
    The name of the tool mountstation to unlink. Refers to a unique node name of the kinematic tree.
    This can be specified multiple times to unlink multiple tool mountstations.

5.3.2 Script structure

```
<?xml version="1.0" encoding="UTF-8"?>
<machine_definition>
  [...]
  <extension>
    <signals version="1.0">
      <signal name="SignalUniqueName" source="/Some/Btss/variablePath[1]" ali-
as="$SomeEquivalentNCVariable[0]" default="0"/>
    </signals>
    <actions version="1.0">
      <action>
        <trigger signal="SignalUniqueName" value="0"/>
        <commands>
        </commands>
      </action>
    </actions>
  </extension>
</machine_definition>
```

5.3.3 Example script

```
<extension>
  <signals version="1.0">
    <signal name="HeadSwitch" source="/Nck/ProtectedArea/kinSwitch[1]" ali-
as="$NK_SWITCH[0]" default="0"/>
  </signals>
  <actions version="1.0">
    <action>
      <trigger signal="HeadSwitch" value="0"/>
      <commands>
        <command name="DetachComponent">
          <parameters>
            <parameter name="Component" value="head2"/>
            <parameter name="Component" value="Head4"/>
          </parameters>
        </command>
        <command name="AttachComponent">
          <parameters>
            <parameter name="Component" value="head2"/>
            <parameter name="Target" value="head_mount"/>
          </parameters>
        </command>
        <command name="LinkAxis">
          <parameters>
            <parameter name="Axis" value="MA11_head1"/>
            <parameter name="LeadingAxis" value="MA11"/>
            <parameter name="Ratio" value="1.0"/>
```

```
        </parameters>
      </command>
      <command name="LinkToolNode">
        <parameters>
          <parameter name="ToolNode" value="tool_head1"/>
          <parameter name="TargetToolNode" value="tool_9998_1"/>
        </parameters>
      </command>      </commands>
    </action>
  </actions>
</extension>
```

5.3.4 Limitations

- Only mount transformations of tool mountstations which are initially linked to a magazine location are considered. Also the relation to the respective magazine location is fixed afterwards. That means linking or unlinking of tool mountstations does not change or update the mount transformations.
- Actions are triggered in the present (can not be predicted/happen in lookahead)
- Spindles that drive tools must be in the same component as the respective tools if they are to be attached and detached by commands. Outside of these limitations the behavior is unspecified.
- activeToolLocation is only applied to initially (possibly statically) linked tool mountstations and their magazine locations.

Note

The script should be added at the end of the machine model xml file, and then the model can be exported as "*.mkc" for the use in 3D applications.

For the full support of <extension> in XML and "*.mkc" usage, please contact with customer support. The support of <extension> in the MachineBuilder when used as "*.mkc" is only supported with a certain version. As a rule, you can use the machine model as a "*.zip" file with the individual files "*.stl" and "*.xml".

5.4 Turrets and attributes

To model turrets, there are three special custom attributes available on MachineBuilder.

5.4.1 activeToolLocation

Of all the tools mounted on a turret, only the active one (in the NCK) should be enabled for cutting operation. Interferences by other tools should be recognized as collisions. Therefore, It is possible to limit the "active" tool to a tool location, for enabling the cutting operation.

The tool mount point name (e.g. tool_9998_1) should be set for "activeToolLocation" custom attribute on the common parent turret transform node.

Only the tool that resides in this place with its current location (non-home) will be able to cut and will be rotated potentially by a turret spindle.

5.4.2 turretToolSpindle

It is possible to have a driven tool (by NC Spindle axis) in the turret. The Custom attribute "turretToolSpindle" should be set with the NC axis name of a spindle axis (e.g. MSC3) on the common parent turret transform node.

Notes/Restrictions to consider:

- This spindle axis must not be present as a RotationAxisNode.
- When using this attribute, "activeToolLocation" must be used.
- Only one tool (the active tool) can spin at a time.
- Configuring these attributes in a way that multiple tools are active when the linked spindle spins, leads to undefined behavior.
- Turning tools will not be rotated in any case.
- If multiple ancestors of a tool mount station have this attribute configured, this is undefined behavior.

5.4.3 mountTransformation

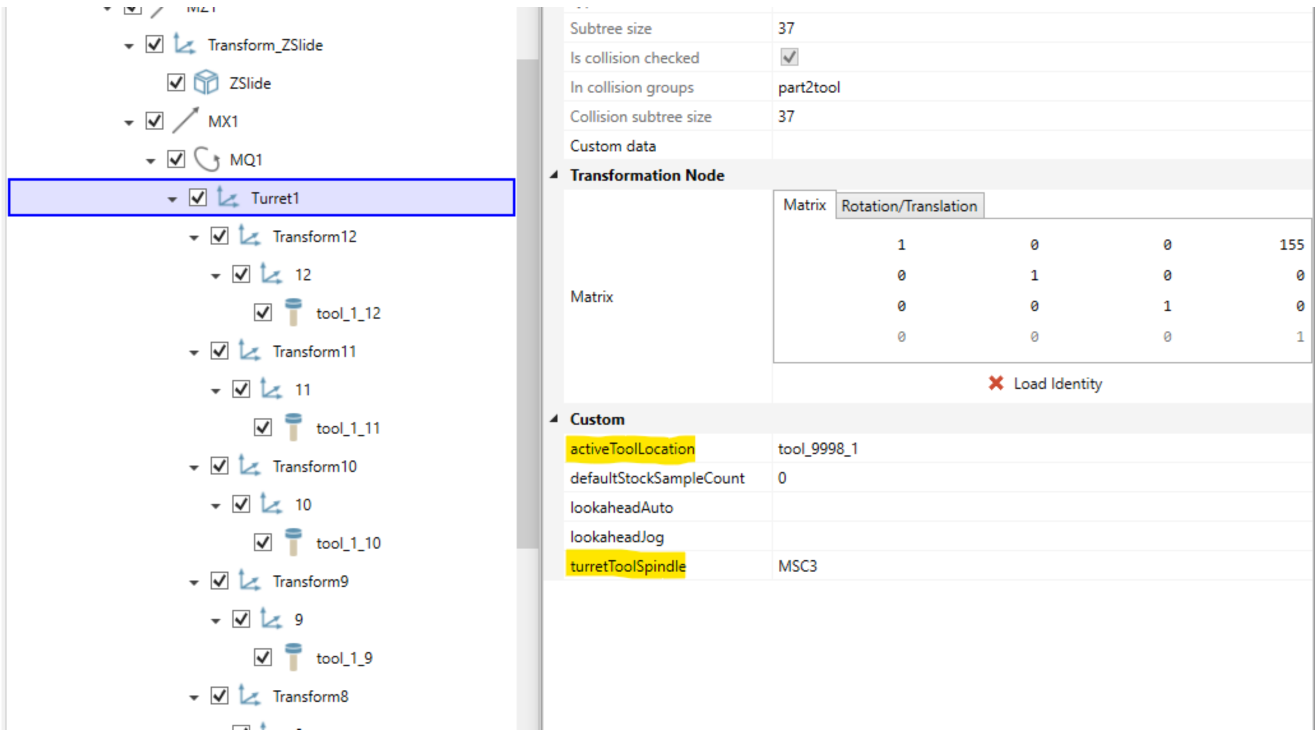
It is possible to define a separate mount coordinate system for a toolNode. This means the 3D holder/adapter geometry with their Mounting Points will be placed based on this coordinate system.

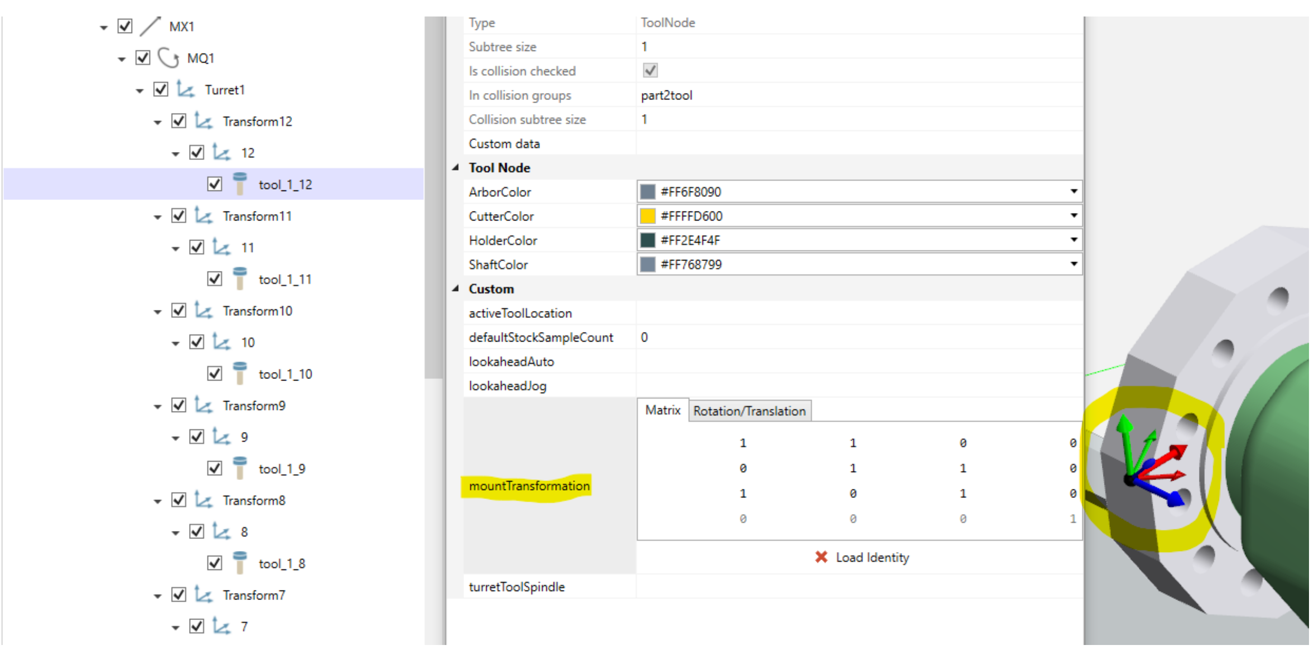
Notes/Restrictions to consider:

- Function is supported starting with /3D v1.3.1.
- Below mountTransformation extension should be added to CustomExtensions.xml

```
<extension base="ToolNode">
<attribute name="mountTransformation" type="matrix"
show_coordinate_system="true"
description="Separate tool mount coordinate system, based on nodes
local coordinate system"/>
 </extension>
```

- mountTransformation information is only supported for turret tool nodes and not for spindle tool nodes.
- Placement of parametric geometries or Reference points for binding to tool edges will still be evaluated in existing toolNode transform.
- mountTransformation will be effective on tools based on their home location (their home location has to match to the location name where this attribute is defined on).

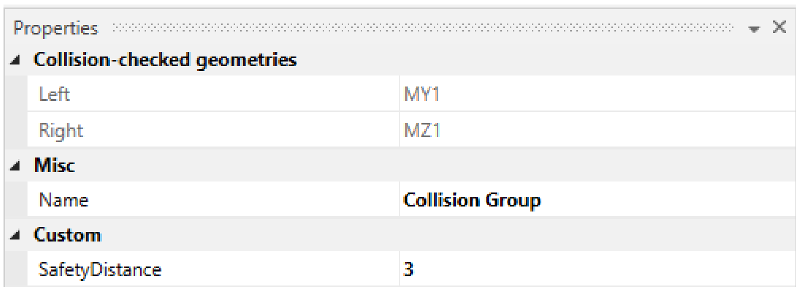

Transform_ZSlide
ZSlide
MX1
MQ1
Turret1
Transform12
12
tool_1_12
Transform11
11
tool_1_11
Transform10
10
tool_1_10
Transform9
9
tool_1_9
Transform8
Subtree size
37
Is collision checked
In collision groups
part2tool
Collision subtree size
37
Custom data
Transformation Node
Matrix
Rotation/Translation
Matrix
1 0 0 155
0 1 0 0
0 0 1 0
0 0 0 1
Load Identity
Custom
activeToolLocation
tool_9998_1
defaultStockSampleCount
0
lookaheadAuto
lookaheadJog
turretToolSpindle
MSC3


MX1
MQ1
Turret1
Transform12
12
tool_1_12
Transform11
11
tool_1_11
Transform10
10
tool_1_10
Transform9
9
tool_1_9
Transform8
8
tool_1_8
Transform7
7
Type
ToolNode
Subtree size
1
Is collision checked
In collision groups
part2tool
Collision subtree size
1
Custom data
Tool Node
ArborColor
#FF6F8090
CutterColor
#FFFFD600
HolderColor
#FF2E4F4F
ShaftColor
#FF768799
Custom
activeToolLocation
defaultStockSampleCount
0
lookaheadAuto
lookaheadJog
mountTransformation
Matrix
Rotation/Translation
1 1 0 0
0 1 1 0
1 0 1 0
0 0 0 1
Load Identity
turretToolSpindle

5.5 Safety distances (only Protect MyMachine /3D Twin)

Collision checking is configured via collision groups. The safety distance of the resulting collision pairs can be configured via a special attribute "safetyDistance". The default safety distance is 3mm.

If multiple groups apply to a pair the maximum distance is used.



5.6 Look ahead visualization (only Protect MyMachine /3D Twin)

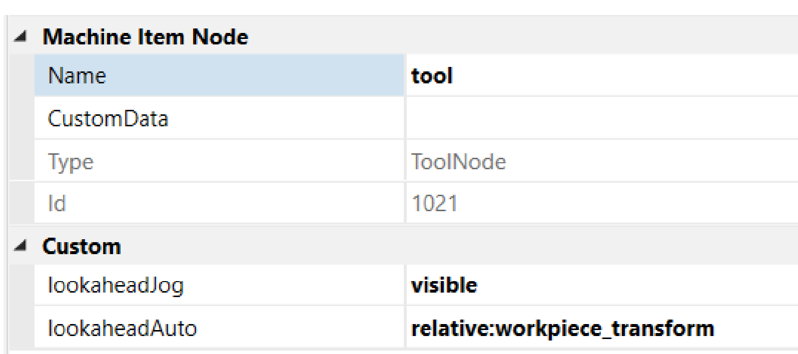
Following attributes are available to configure the look ahead visualization.

- `lookaheadJog` (will apply for JOG mode)
- `lookaheadAuto` (will apply for AUTO mode)

The value can be "" (empty string = inherit, applies also if attribute is not specified), `hidden`, `visible`, `relative`:{`nodename`}.

The values are inherited from parent nodes, the default to inherit is `hidden`. But only geometry nodes will be visualized in the end.

All nodes that get a look ahead visualization need to have a unique name.



5.7 Collision Detection

The collision pairs for a machine are defined in the machine model.

Collision checking is always performed between component pairs (if groups are given, collisions are checked between all possible pairs in the two different groups).

Only machine components are defined in the machine model.

Possible collision pairs are:

- Machine component <-> Machine component
- Machine component <-> Fixture
- Machine component <-> Stock
- Machine component <-> Tool
- Fixture <-> Fixture
- Fixture <-> Stock
- Fixture <-> Tool
- Stock <-> Stock
- Stock <-> Tool
- Tool <-> Tool

Safety distances between components default to 3 mm. Violations of the safety distance are treated in the same way as actual collisions between components if not specified otherwise below.

Safety distance violations are indicated by orange coloring of the objects in the corresponding collision pairs. Actual collisions are indicated by red coloring.

For tools vs stock collision pairs there can only be one safety distance in the end which is effective for all of these pairs. This one safety distance is the maximum distance of all configured tool vs stock pairs.

5.8 Axis couplings

Note

Only available for:

PMM /3D Twin from V1.3.2

It is possible to define axes in machine model, which are not defined on NC, but coupled with NC axes.

In the machine model as a child of <machine_definition> node, <axis_couplings> element should be added to define coupled axes.

XML tag	Attribute	Values	Details
axis_couplings	-	-	
linear_coupling	driving_axis	String ID of the driving axis node.	required
	dependant_axis	String ID of the driven axis node	required
	direction	"FORWARD", "REVERSE", or "BIDIRECTIONAL" (Only "FORWARD" is supported)	required
	gear_ratio	Float The quotient of driven axis speed divided by the driving axis speed.	required

Example

```
<?xml version="1.0" encoding="UTF-8" ?>
<machine_definition>
  <machine_data name="5AxHeadHead" version="1.8" units="metric" controller="">
    <view_transform
      initialvalue="1,0,0,0,0,1,0,0,0,0,1,0,0,0,0,1" />
  </machine_data>
  <axis id="X" type="translation" x="1.0" y="0.0" z="0.0"
      minvalue="-100000.0" maxvalue="100000.0"
      valuetype="cont" initial_value="0.0">
    <axis id="Y" type="translation" x="0.0" y="1.0" z="0.0"
        minvalue="-100000.0" maxvalue="100000.0"
        valuetype="cont" initial_value="0.0">
      <axis id="Z" type="translation" x="0.0" y="0.0" z="1.0"
          minvalue="-100000.0" maxvalue="100000.0"
          valuetype="cont" initial_value="150.0">
        <axis id="C" type="rotation" x="0.0" y="0.0" z="1.0"
             minvalue="-100000.0" maxvalue="100000.0" valuetype="cont"
             initial_value="0.0" rzx="0.0" rzy="0.0" rzz="0.0">
          <axis id="B" type="rotation" x="0.0" y="1.0" z="0.0"
               minvalue="-100000.0" maxvalue="100000.0" valuetype="cont"
               initial_value="0.0" rzx="0.0" rzy="0.0" rzz="0.0">
            <transform id="holder_transform"
                initialvalue="1,0,0,0,0,1,0,0,0,0,1,0,0,0,0,1">
```

```
              <geometry name="tool" geo="tool.stl" alpha="0.0" reflectivity="0.10"
                   reflectivityBitmapFileName="tablereflection.bmp" objtype="tool"
                   cuttr="0.827" cuttg="0.780" cuttb="0.674"
                   noncuttr="0.501" noncuttg="0.5011" noncuttb="0.501"
                   arborr="0.39215687" arborg="0.392" arborb="0.392"
                   holderr="0.368" holderg="0.572" holderb="0.690" />
            </transform>
          </axis>
        </axis>
      </axis>
    </axis>
  </axis>
  <transform id="workpiece_transform"
     initialvalue="1.0,0.0,0.0,0.0,0.0,1.0,0.0,0.0,0.0,0.0,1.0,0.0,0.0,0.0,0.0,1.0">
    <geometry name="fixture" geo="fixture.stl" clrr="0.501" clrg="0.501" clrb="0.501"
       alpha="0.0" reflectivity="0.0" reflectivityBitmapFileName="" objtype="fixture" />
    <geometry name="initialstock" geo="stock.stl" clrr="0.752" clrg="0.752" clrb="0.752"
       alpha="0.80" reflectivity="0.0" reflectivityBitmapFileName="" objtype="initialstock" />
    <geometry name="stock" geo="stock.stl" clrr="0.752" clrg="0.752" clrb="0.752"
       alpha="0.0" reflectivity="0.40" reflectivityBitmapFileName="tablereflection.bmp"
       objtype="stock" />

    <geometry name="toolpath" geo="toolpath.asc" clrr="0.0" clrg="0.0" clrb="0.0"
       alpha="0.0" reflectivity="0.0" reflectivityBitmapFileName="" objtype="toolpath" />
    <geometry name="workpiece" geo="workpiece.stl" clrr="0.752" clrg="0.752" clrb="0.752"
       alpha="0.0" reflectivity="0.40" reflectivityBitmapFileName="tablereflection.bmp"
       objtype="workpiece" />
  </transform>
  <collcheck id="cc1" name="cc1" group1="tool" group2="workpiece" />
  <axis_couplings>
    <linear_coupling driving_axis="X" dependant_axis="Y" direction="BIDIRECTIONAL" gear_ratio="1" />
    <linear_coupling driving_axis="Y" dependant_axis="C" direction="BIDIRECTIONAL" gear_ratio="2.25" />
  </axis_couplings>
</machine_definition>
```

Integrating 3D model into target system 6

6.1 Import machine model into Protect MyMachine /3D Twin

After completing 3D machine, the saved file must be uploaded to Microbox with Samba Folder.

- The data (*.stl, *.xml) must be zipped directly without a parent folder. Otherwise you can not import the model.
- The data (*.mkc) can be used directly by moving it to samba folder. No "*.zip" required additionally.

When the application is started, go to settings page and click "Import Machine Model" button to import the machine model data. And 3D machine model will be imported to the application.

For further information, the installation manual of the application Protect MyMachine /3D Twin can be examined.

6.2 Import machine model into Create MyVirtual Machine /3D

Create MyVirtual Machine /3D uses a storage folder on the virtual memory card. You create the machine model (kinematics description and geometries) in the MachineBuilder of ModulWorks and export it as "machine container" in the "*.mkc" format or use the "*.stl" and "*.xml" directly.

You can find the folder on the virtual memory card in your user directory.

- "C:\Users\<username>\AppData\Local\Siemens\Automation\SINUMERIK ONE\ncu\card\oem\sinumerik\3d\model\"

For further information, user manual of the application can be examined.

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7.1 Naming conventions for tools and turrets

Tool nodes in the machine model need to follow a naming convention to establish a relation between the tool locations used in the simulation and the tooling information defined on the NC. Tool nodes with a name that cannot be mapped to NC data remain empty. Tooling information defined on the NC which cannot be matched to an appropriately named tool node will not be used in the simulation. For milling machines, this means that the main spindle needs to be present in the simulation, which typically relates to magazine 9998 and slot 1 on the NC. For turning machines with a turret, each turret location needs to be defined in the machine model by a dedicated tool node with matching naming.

The full definition for a tool name contains the elements "tool organization unit" (to), "magazine" (mag) and "slot"/"pocket" (slot) and each of these is associated with a number.

Naming of a tool is defined as:
tool_to{tool_organization_unit_id}_mag{magazine_id}_slot{slot_id}

That form can be simplified in the following ways

- The abbreviations "to", "mag", "slot" can be omitted.
- A tool organization unit of "1" is assumed, if its definition is left out and the tool name only contains two numbers.
- If all elements are left out and only the name "tool" is used, then this is equivalent *to tool_to1_mag9998_slo*t1, which is the main tool spindle.

Examples

Name	To	Mag	Slot
tool_2_3	1	2	3
tool_2_mag1_2	2	1	2
tool_to1_mag1_slot2	1	1	2
tool	1	9998	1

7.1.1 Multi spindle

Defining multiple spindles for the same NC tool information is also possible, by adding a suffix to the tool name as "tool_1_9998_1_*".

Examples

Name	To	Mag	Slot
tool_1_9998_1_1	1	9998	1
tool_1_9998_1_2	1	9998	1
tool_1_9998_1_abc	1	9998	1

7.2 Naming conventions for mount stations

For the machines with one mount station, the mount point for stocks and fixtures can be called "workpiece_transform".

For the machines with multiple mount station, the mount point for stocks and fixtures has to be named as one of the options below:

- setup_{station_number}
- setup_{area_number}_{station_number}
- setup_ms{station_number}
- setup_wa{area_number}_ms{station_number}
- setup_workarea{area_number}_mountstation{station_number}

{area_number} and {station_number} placeholders should be replaced with unique integers.

Below this placeholder nodes for stocks (StockNode called "stock") and fixtures (MachineGeometryNode called "fixture") can be placed to define collision pairs and lookahead visualization for them.

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